LMGB · SESSION COMPANION · SINGAPORE

Location analytics, from intuition to evidence

Every idea is built on the board first, from something you already know, and then run on real data from Singapore in the browser workbench.

Tool: **quantng.github.io/location-lab** · country guide: **quantng.github.io/location-lab/singapore/**

Chrome or Edge on a laptop. Nothing to install. Files you upload never leave your laptop.

THE PLACE

Singapore in one slide

Singapore is a city-state of about 6 million people on 735 km². About 77% of resident households live in HDB flats, public housing built by the Housing & Development Board and sold on 99-year leases. Home ownership is above 90%, and flats are traded on an open resale market whose every transaction is published.

The island is divided into 55 URA planning areas. HDB towns are planned around MRT stations, hawker centres (cooked-food markets), schools and parks, which makes Singapore a clean laboratory for asking what drives price.

Currency	Singapore dollar (S$); prices quoted per m² (psm)
Admin units	55 URA planning areas (Master Plan 2019)
Business core	Raffles Place / Downtown Core (the CBD)
Transit	MRT (heavy rail) and LRT (light rail feeders)
Why it matters here	Planned, complete and public data: ideal for spatial regression

THE DATA

What is in the catalogue

HDB resale blocks (8,047, data.gov.sg + OneMap): price_psm (S$ per m², 12-month median), n_sales, floor area, town, lease start.

MRT & LRT stations, **hawker centres** (NEA), **55 planning areas** (URA MP2019), **Botanic Gardens** outline, and **NDVI** from Sentinel-2, 5 Dec 2025.

Every vector layer ships as GeoJSON (plain text), GeoParquet (compressed columns) and FlatGeobuf (with a built-in spatial index). Pick the format in the Add data window and compare the load size.

Layer	GeoJSON	GeoParquet	FlatGeobuf
Botanic Gardens	4 KB	18 KB	3 KB
Hawker centres	35 KB	19 KB	29 KB
HDB resale blocks	2.8 MB	318 KB	1.9 MB
MRT & LRT stations	32 KB	15 KB	28 KB
Planning areas	138 KB	54 KB	59 KB

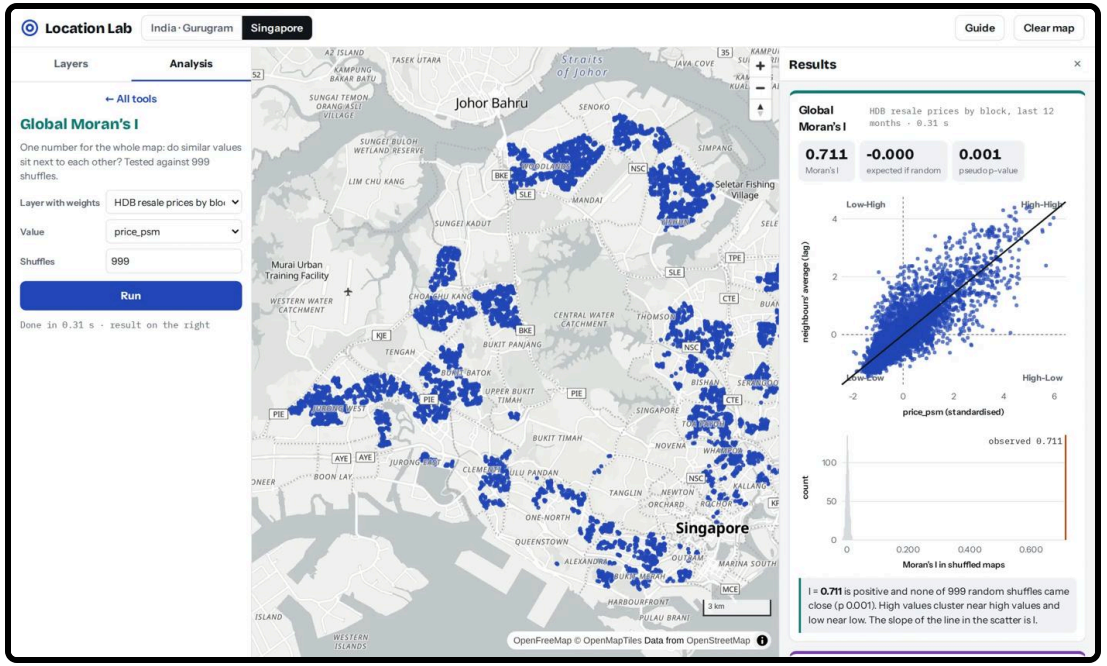
THE TOOL

Three moves: add, drop, read

1 · Add data. Layers tab → **+ Add data** → choose the file format → click a dataset. It appears as a layer card; use the card to colour by any field, zoom, open the table or download.

2 · Drop an analysis. Analysis tab → drag a tool onto the map or onto a layer card (or just click it). Check the settings, press **Run**.

3 · Read the result. The map restyles, the legend updates, and the Results drawer on the right holds the numbers, the chart and a one-line reading. Click any feature to see its values.



THE SESSION IN ONE LINE

Is “where” random? If not, why not?

Part	Question	Tools
1 · Spatial data recap	Load, measure, filter, join, bin, smooth, read a satellite image	Nearest · SQL · Box→distance · Join · H3 · Heatmap · NDVI
2 · Neighbours and lag	Who counts as “near”, and how are they doing?	Weights · Spatial lag
3 · Point patterns	Clustered, dispersed or random?	Point pattern test
4 · Autocorrelation	Do similar values sit together, and where?	Moran’s I · LISA · Gi*
5 · Clusters	Which points form one market?	DBSCAN
6 · Surfaces	What is the value between samples?	Sample · IDW · Kriging
7 · Explain	Which factors, and do they vary by place?	Distance · OLS · Lag · Error · GWR

PART 1

Spatial data recap

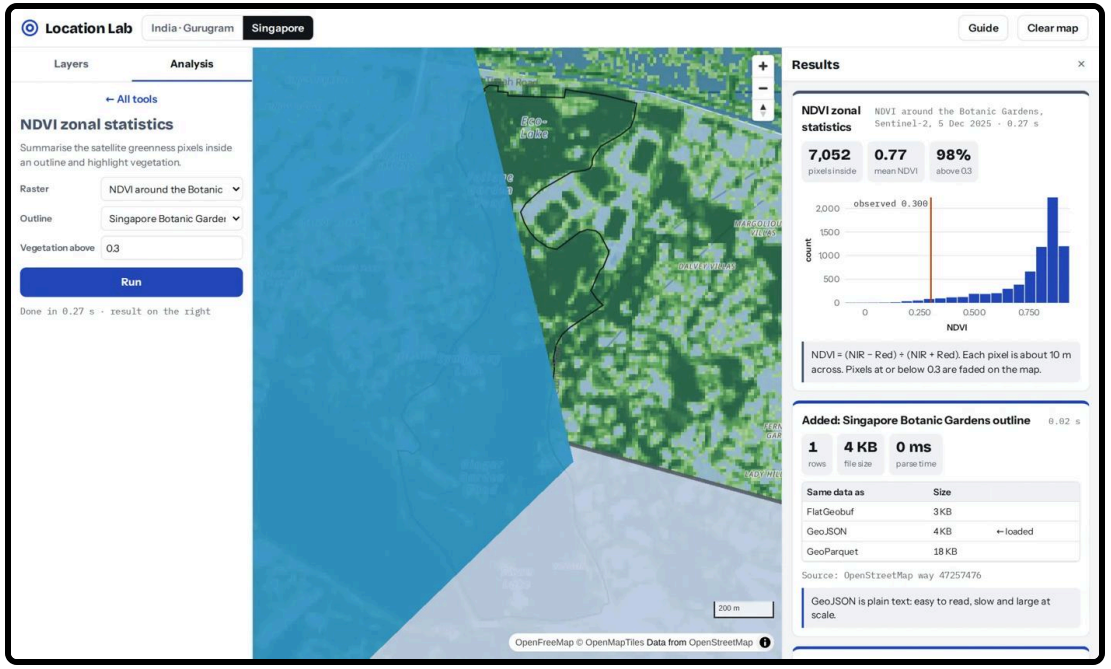
The path we take on the board:

Load → **Measure** → **Filter** → **Join** → **Bin** → **Smooth** → **Pixels**

Seven moves you already know

Last session you learned the grammar of spatial data. A file carries **geometry** (where) and **attributes** (what). You **measured** distance on a curved Earth, **filtered** cheaply with a bounding box before the exact test, **joined** points to the polygon they fall in, **binned** points into H3 hexagons, **smoothed** them into a heatmap and **summarised** satellite pixels inside an outline.

Today's question is new: once the points are on the map, is their arrangement telling us something, or could it be chance?



Where business already uses this

IN THE NEWS · AMAZON WEB SERVICES · 31 JUL 2026

AWS adds “Search Nearby” on GrabMaps for Singapore and Malaysia

Find places within a radius, ranked by distance: the “Nearest to a point” block, sold as a cloud service built on Grab’s own map data.

https://aws.amazon.com/about-aws/whats-new/2026/07/amazon-location-service-search-nearby-grabmaps

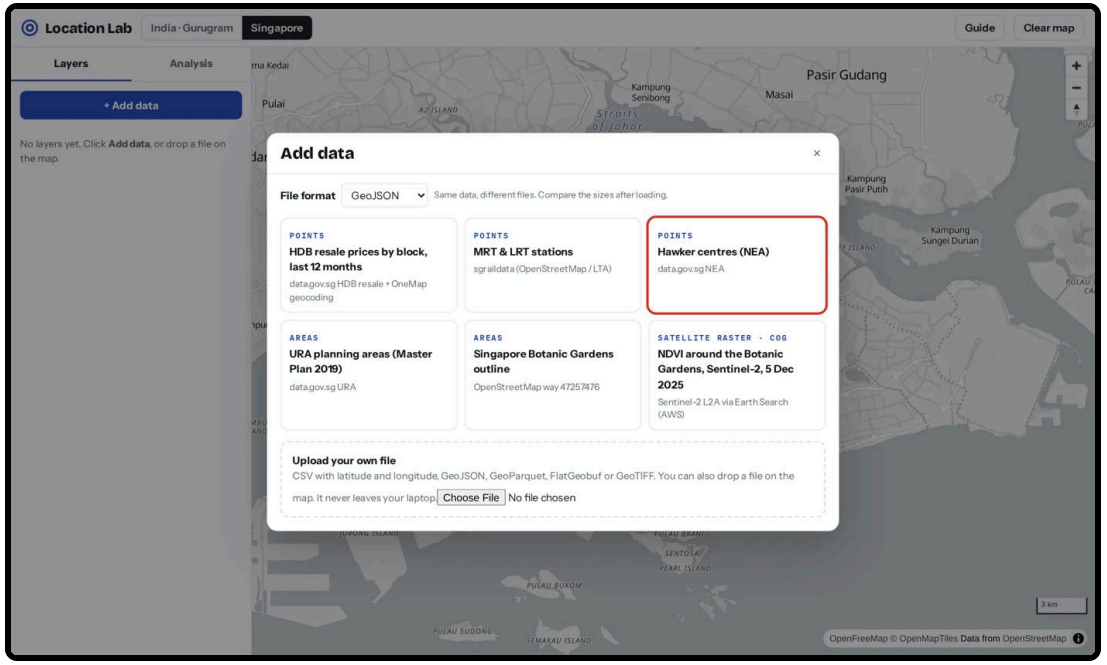
LAB 1 · SINGAPORE

1 · Spatial data in seven moves

The same seven moves on Singapore's hawker centres, URA planning areas and a Sentinel-2 image of the Botanic Gardens.

Open **quantng.github.io/location-lab**, choose **Singapore** in the top bar, then follow the steps.

1. Load the hawker centres.
2. Explore the table.
3. Nearest hawker centres to Raffles Place.
4. Ask the same question in SQL.
5. Box filter, then exact 2 km around Orchard.
6. Load the 55 URA planning areas.
7. Spatial join: planning area of each hawker centre.
8. H3 hexagons in 3D.
9. Heatmap.
10. Load the NDVI raster (COG).
11. Load the Botanic Gardens outline.
12. Zonal statistics for the Gardens.



Load the hawker centres.

DO THIS

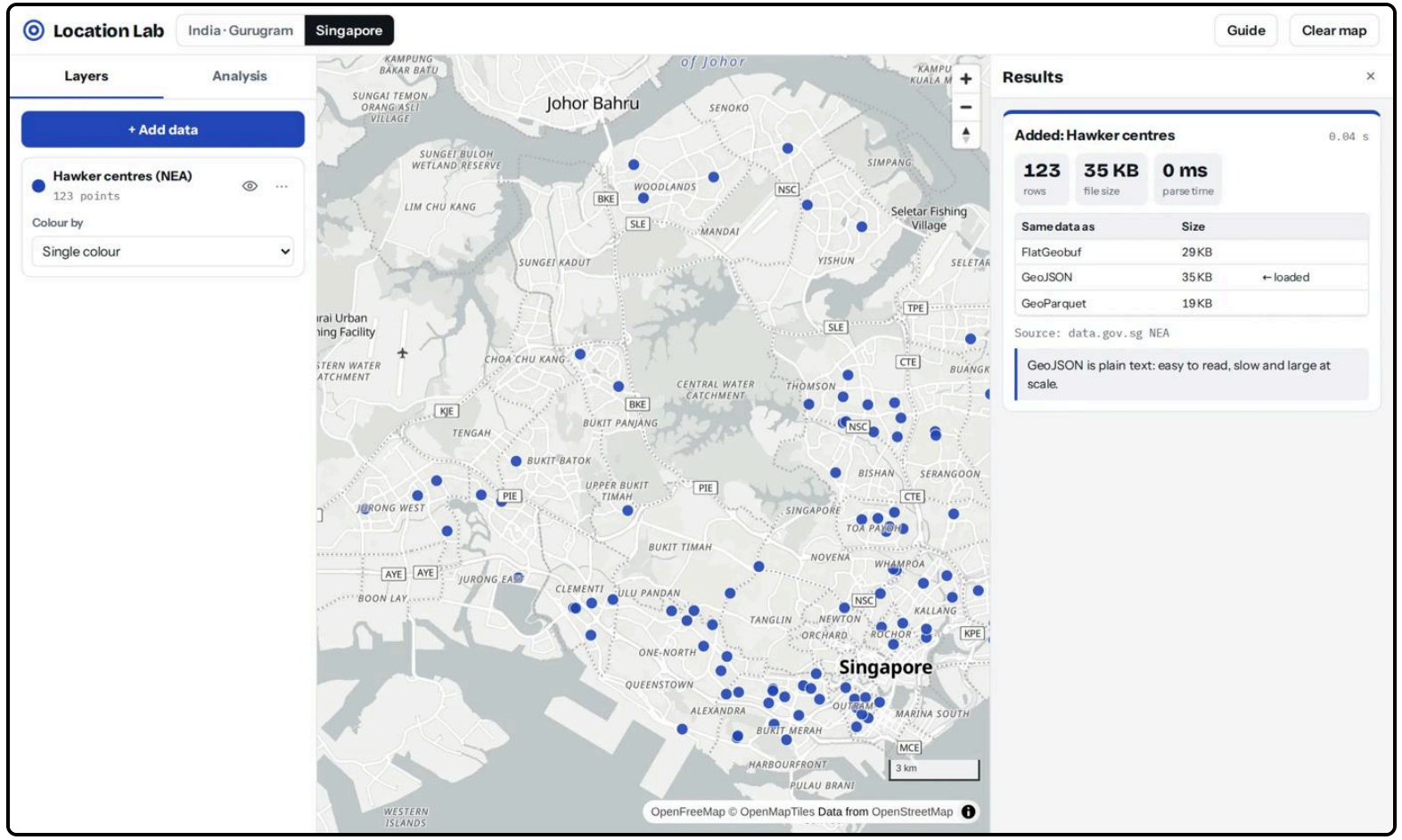
Layers tab → **+ Add data** → pick the format → click the dataset.

Dataset	Hawker centres (NEA)
File format	GeoJSON

YOU SHOULD SEE

123 rows | **35 KB** file size | **0 ms** parse time

> GeoJSON is plain text: easy to read, slow and large at scale.



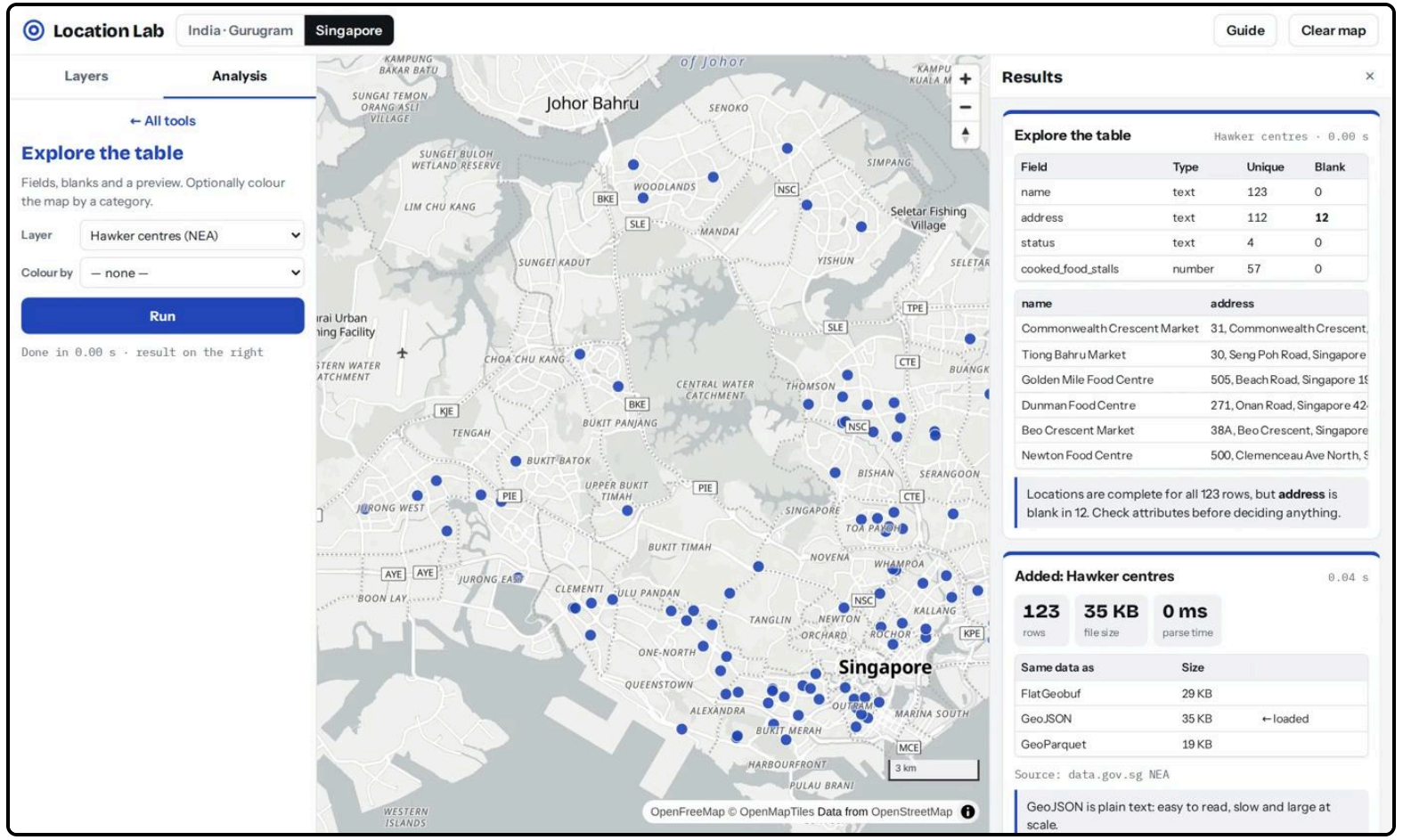
Explore the table.

DO THIS

Analysis tab → drag **Explore the table** onto the map (or click it), check the settings, press **Run**.

Layer	Hawker centres (NEA)
Colour by	— none —

> Locations are complete for all 123 rows, but address is blank in 12. Check attributes before deciding anything.



Nearest hawker centres to Raffles Place.

DO THIS

Analysis tab → drag **Nearest to a point** onto the map (or click it), check the settings, press **Run**.

Layer	Hawker centres (NEA)
From	Raffles Place MRT (CBD)
Show nearest	5

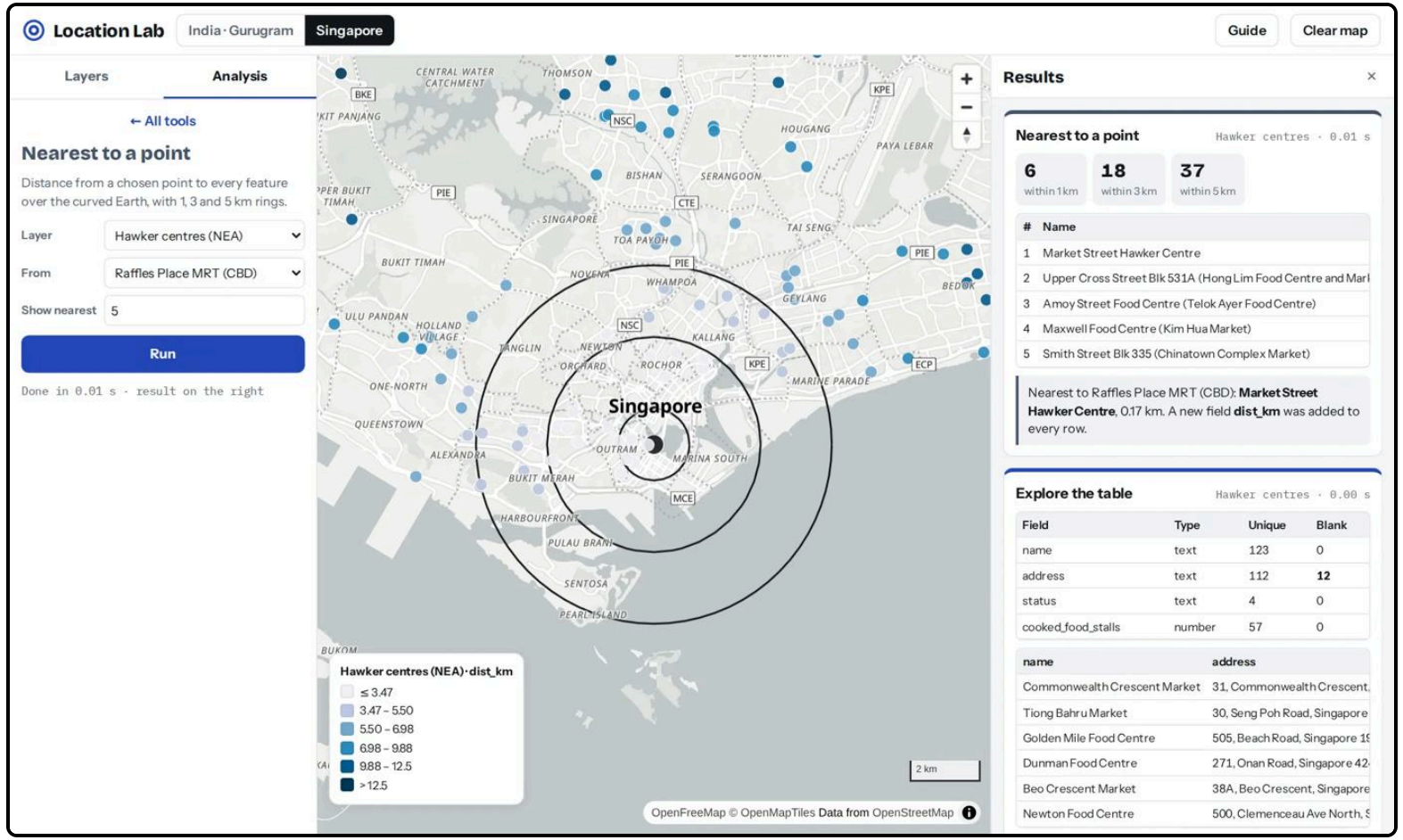
YOU SHOULD SEE

6 within 1 km

18 within 3 km

37 within 5 km

> Nearest to Raffles Place MRT (CBD): Market Street Hawker Centre, 0.17 km. A new field dist_km was added to every row.

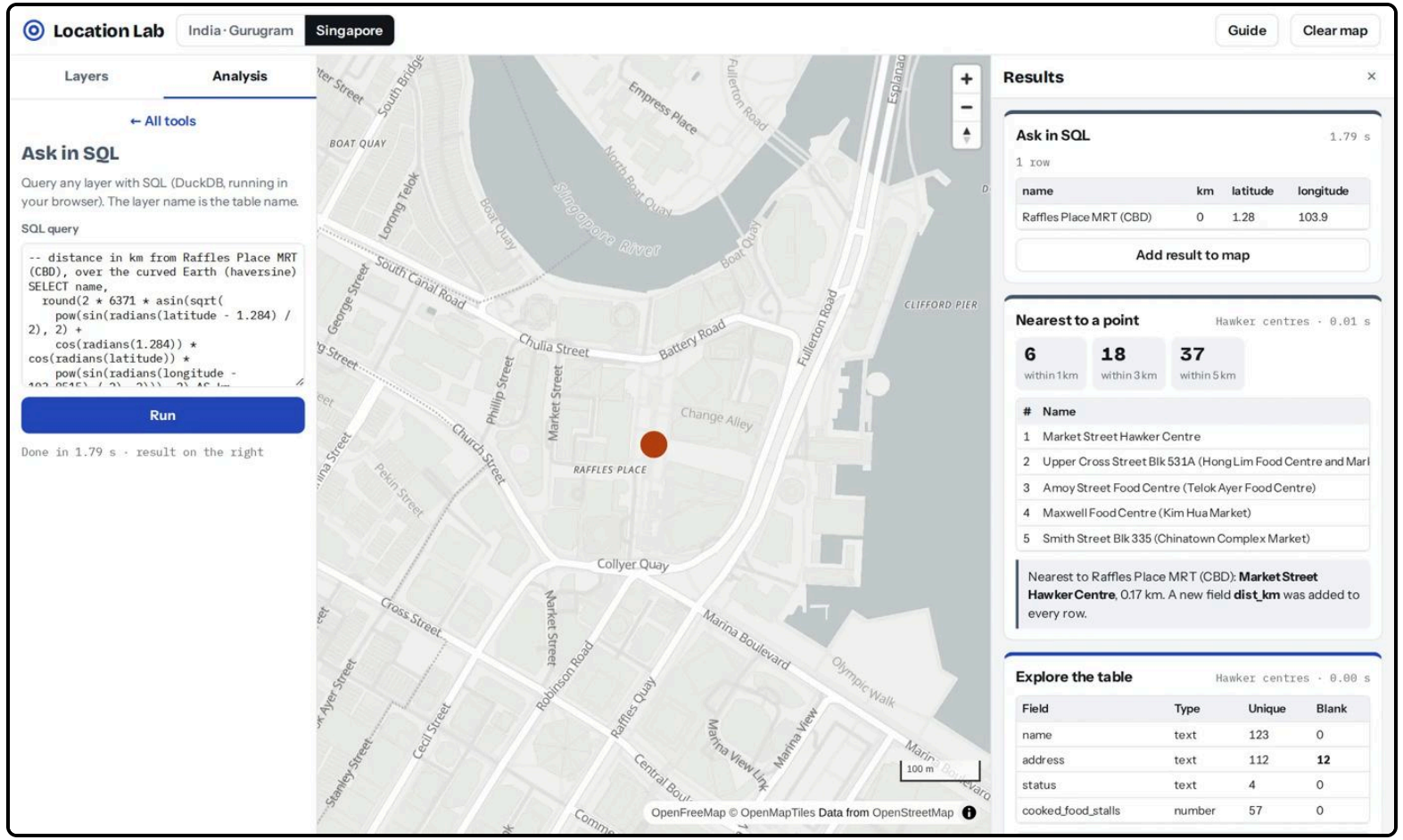


Ask the same question in SQL.

DO THIS

Analysis tab → drag **Ask in SQL** onto the map (or click it), check the settings, press **Run**.

SQL query	(query in the box)



Box filter, then exact 2 km around Orchard.

DO THIS

Analysis tab → drag **Box filter, then distance** onto the map (or click it), check the settings, press **Run**.

Layer	Hawker centres (NEA)
Centre	Orchard MRT
Radius (km)	2

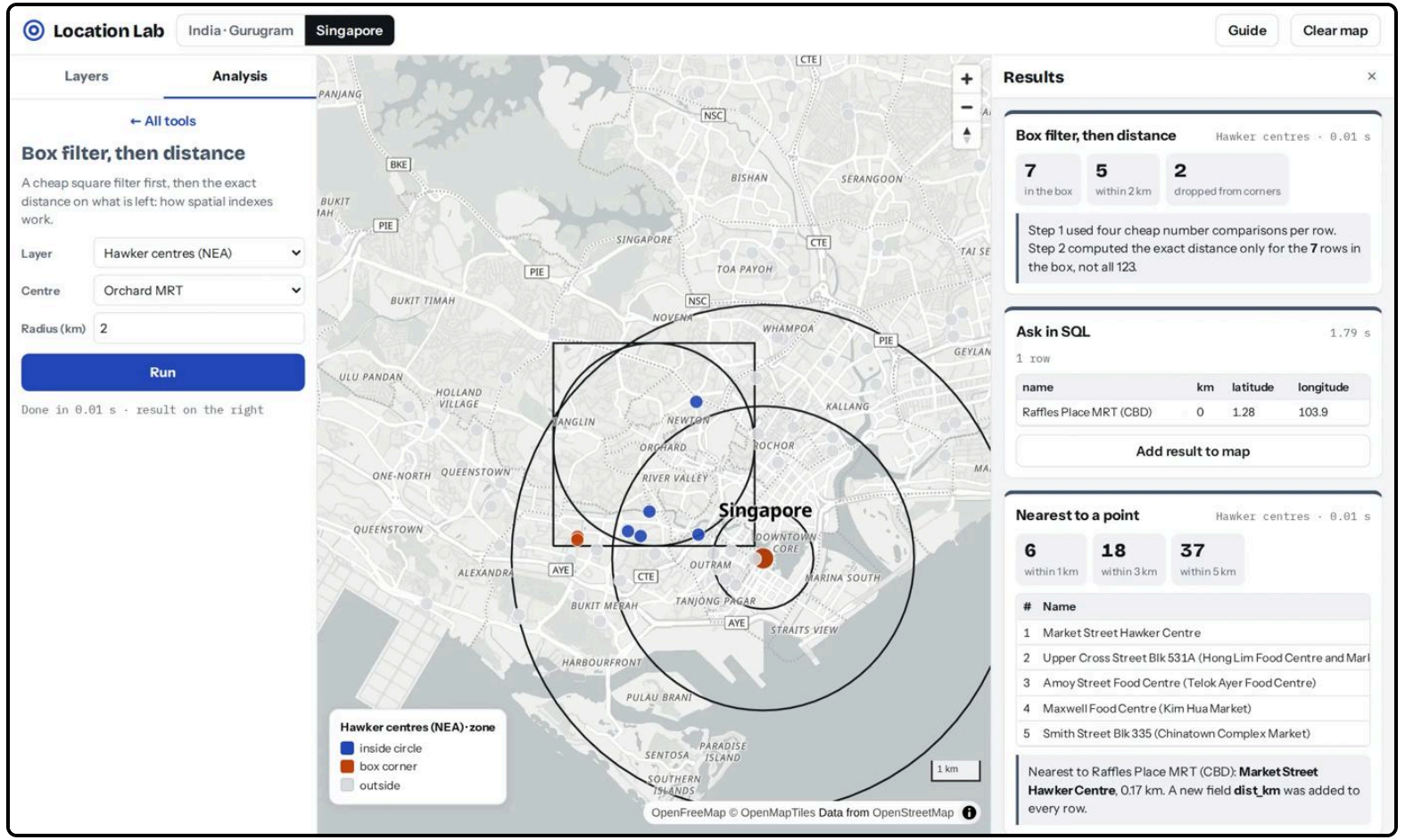
YOU SHOULD SEE

7
in the box

5
within 2 km

2
dropped from corners

> Step 1 used four cheap number comparisons per row. Step 2 computed the exact distance only for the 7 rows in the box, not all 123.



Load the 55 URA planning areas.

DO THIS

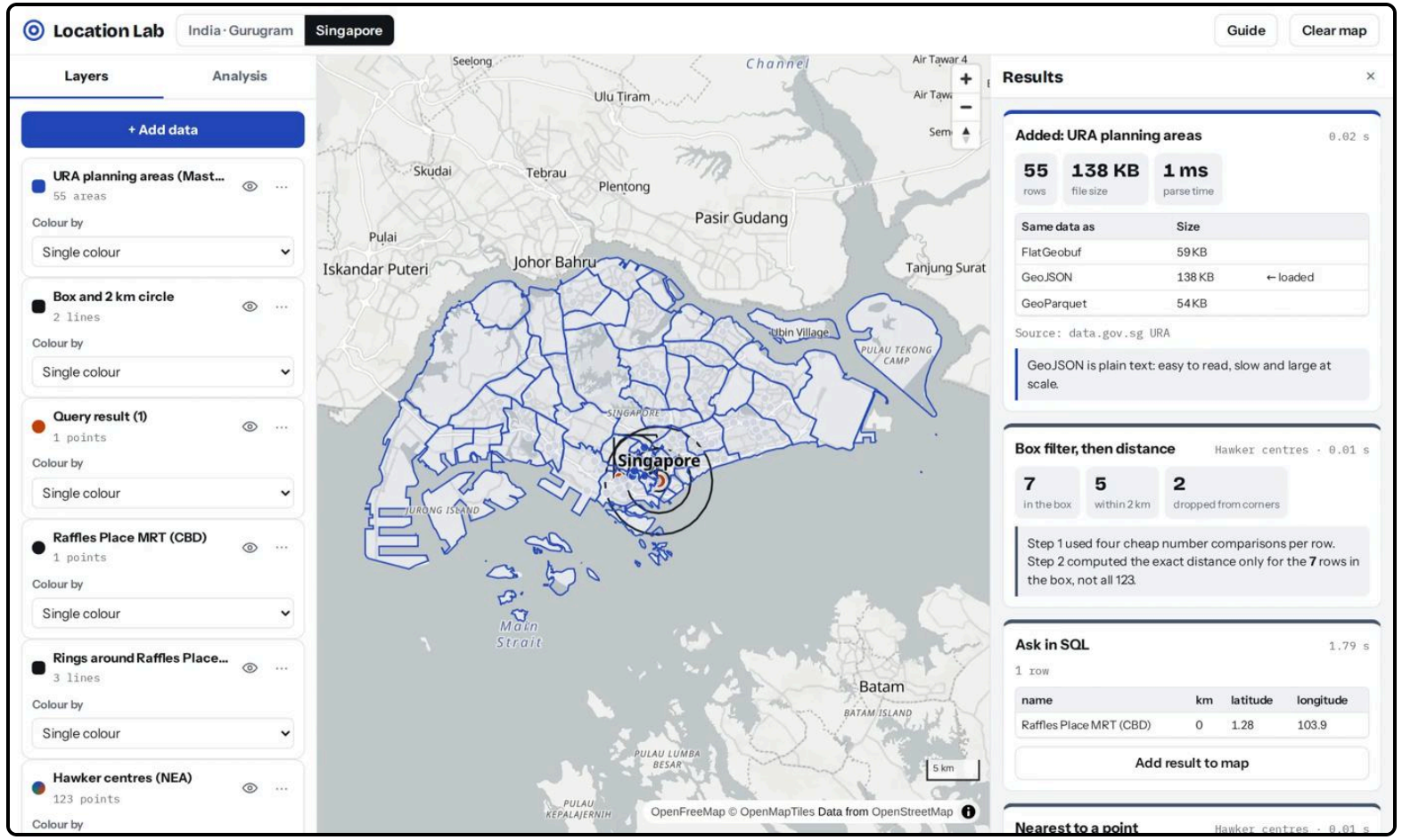
Layers tab → **+ Add data** → pick the format → click the dataset.

Dataset	URA planning areas (Master Plan 2019)
File format	GeoJSON

YOU SHOULD SEE

55 rows | **138 KB** file size | **1 ms** parse time

> GeoJSON is plain text: easy to read, slow and large at scale.



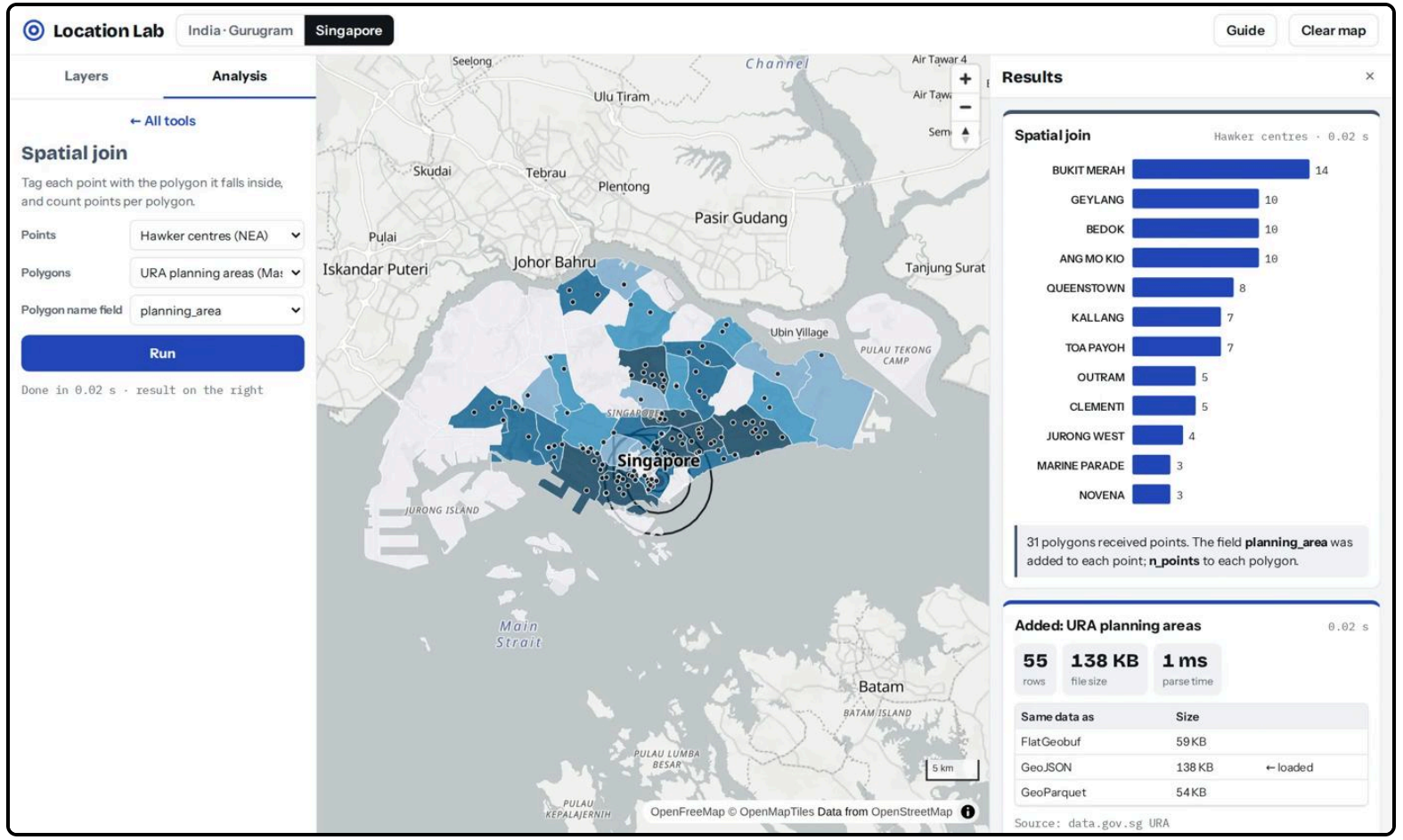
Spatial join: planning area of each hawker centre.

DO THIS

Analysis tab → drag **Spatial join** onto the map (or click it), check the settings, press **Run**.

Points	Hawker centres (NEA)
Polygons	URA planning areas (Master Plan 2019)
Polygon name field	planning_area

> 31 polygons received points. The field planning_area was added to each point; n_points to each polygon.



H3 hexagons in 3D.

DO THIS

Analysis tab → drag **H3 hexagon bins** onto the map (or click it), check the settings, press **Run**.

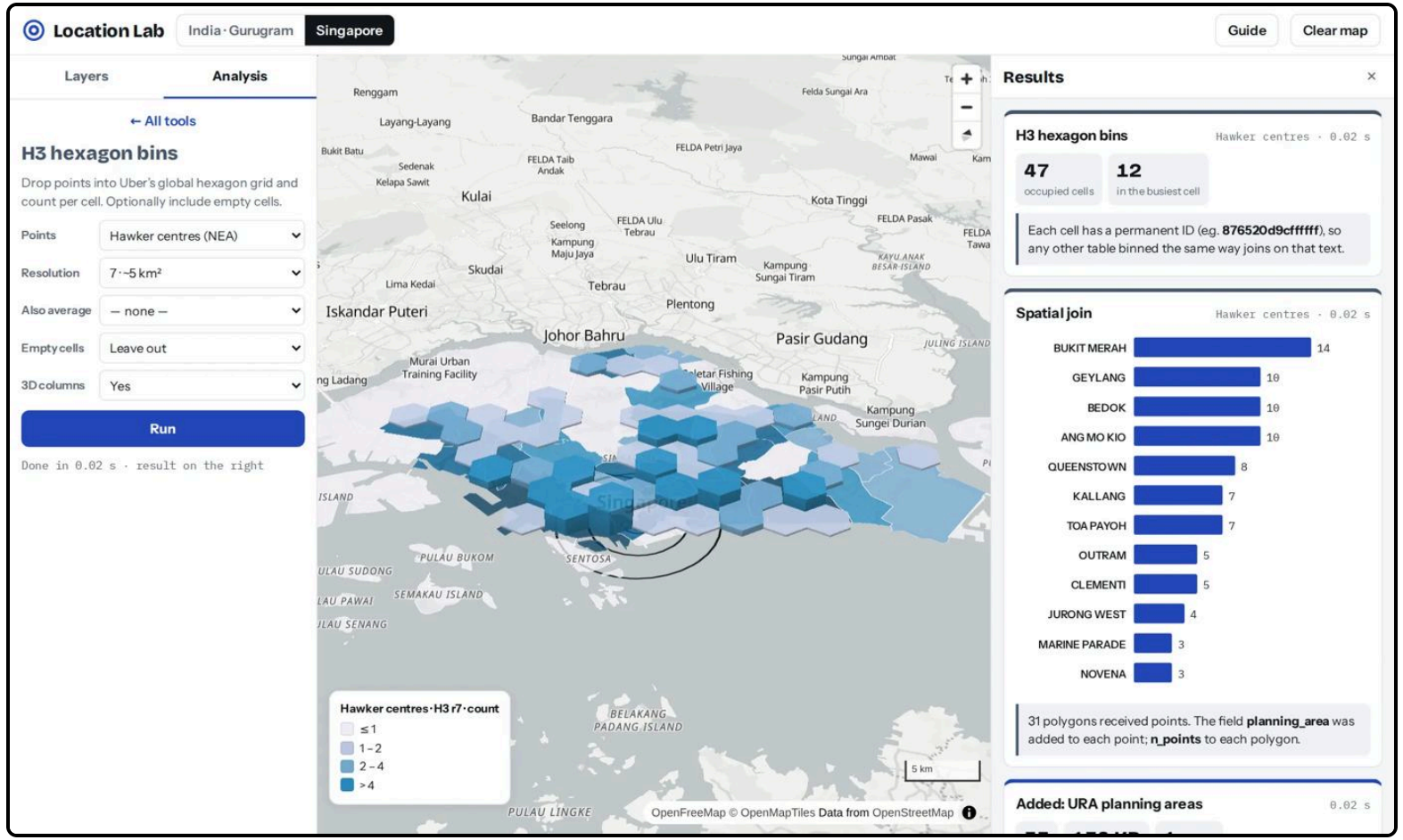
Points	Hawker centres (NEA)
Resolution	7 · ~5 km²
Also average	— none —
Empty cells	Leave out
3D columns	Yes

YOU SHOULD SEE

47
occupied cells

12
in the busiest cell

> Each cell has a permanent ID (e.g. 876520d9cfffffff), so any other table binned the same way joins on that text.



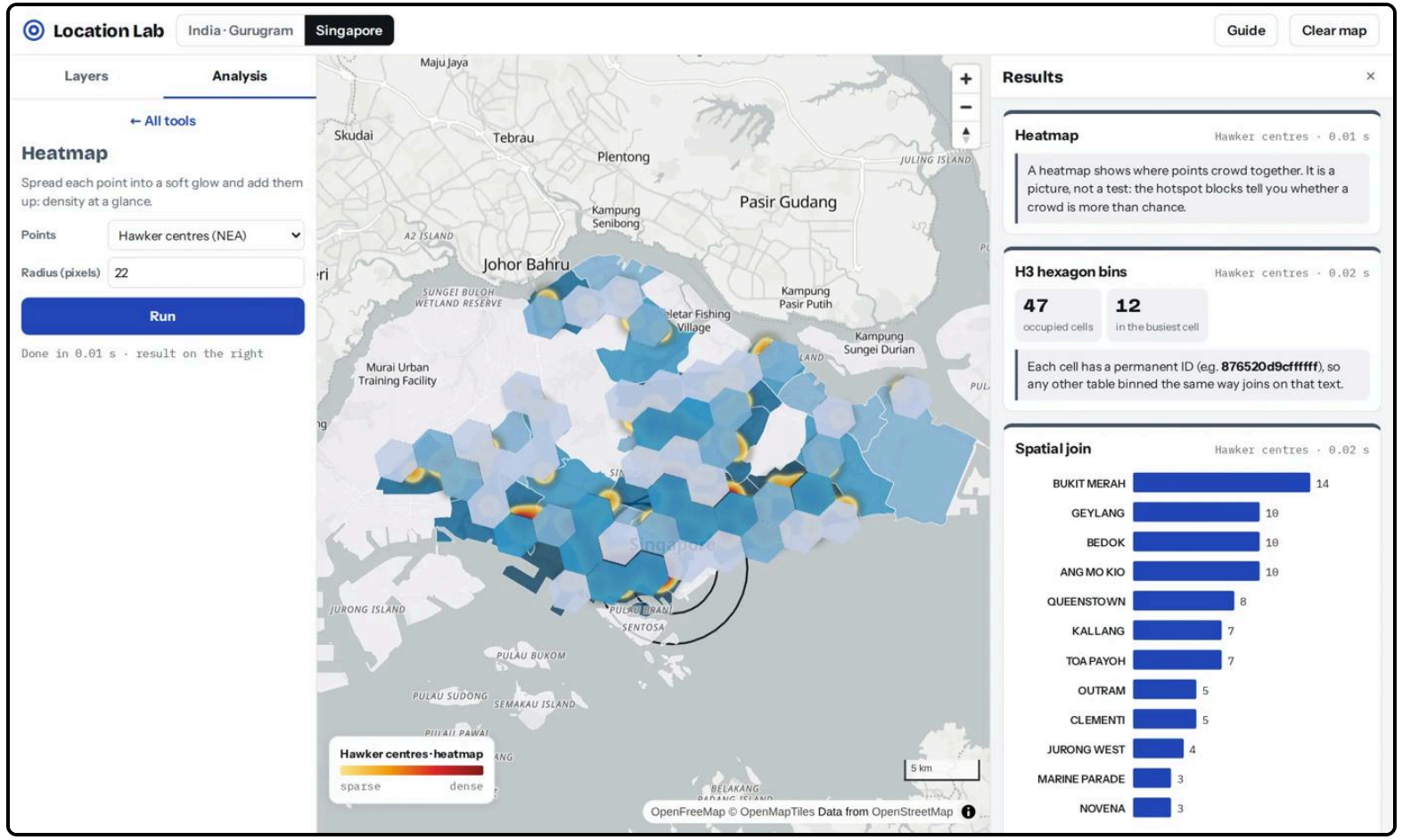
Heatmap.

DO THIS

Analysis tab → drag **Heatmap** onto the map (or click it), check the settings, press **Run**.

Points	Hawker centres (NEA)
Radius (pixels)	22

> A heatmap shows where points crowd together. It is a picture, not a test: the hotspot blocks tell you whether a crowd is more than chance.



Load the NDVI raster (COG).

DO THIS

Layers tab → **+ Add data** → pick the format → click the dataset.

Dataset	NDVI around the Botanic Gardens, Sentinel-2, 5 Dec 2025
File format	Cloud-Optimised GeoTIFF

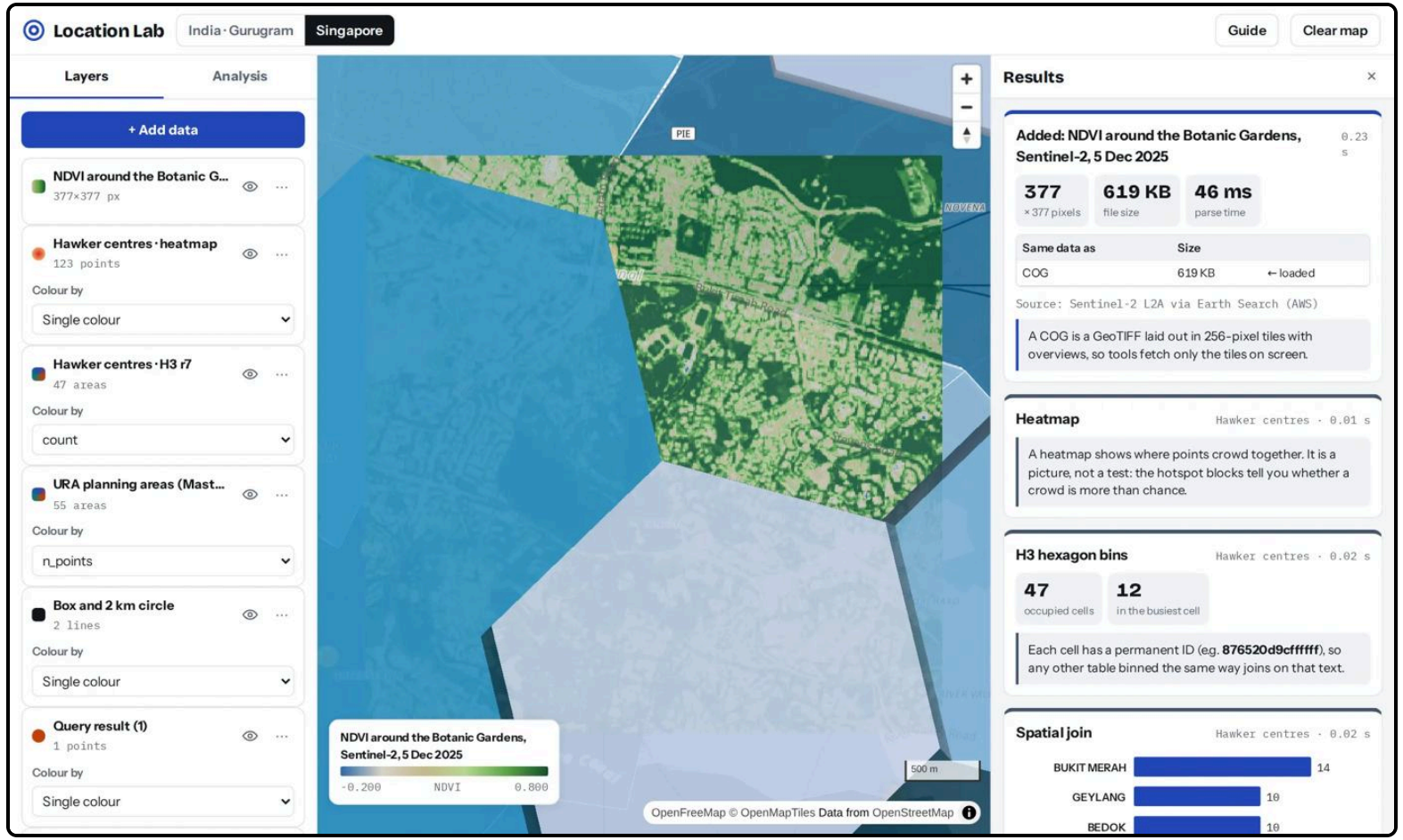
YOU SHOULD SEE

377
× 377 pixels

619 KB
file size

46 ms
parse time

> A COG is a GeoTIFF laid out in 256-pixel tiles with overviews, so tools fetch only the tiles on screen.



Load the Botanic Gardens outline.

DO THIS

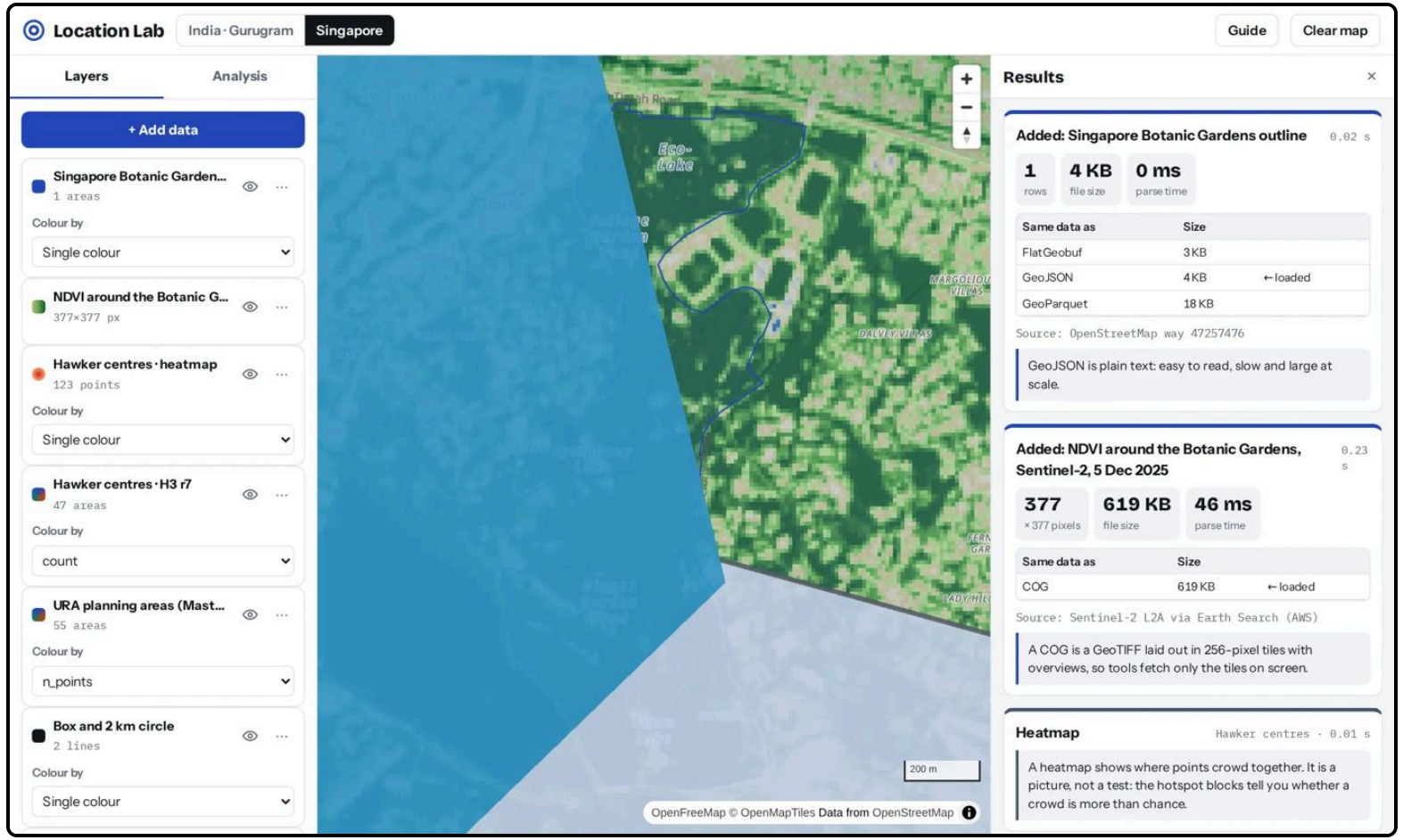
Layers tab → **+ Add data** → pick the format → click the dataset.

Dataset	Singapore Botanic Gardens outline
File format	GeoJSON

YOU SHOULD SEE

1 rows | **4 KB** file size | **0 ms** parse time

> GeoJSON is plain text: easy to read, slow and large at scale.



Zonal statistics for the Gardens.

DO THIS

Analysis tab → drag **NDVI zonal statistics** onto the map (or click it), check the settings, press **Run**.

Raster	NDVI around the Botanic Gardens, Sentinel-2, 5 Dec 2025
Outline	Singapore Botanic Gardens outline
Vegetation above	0.3

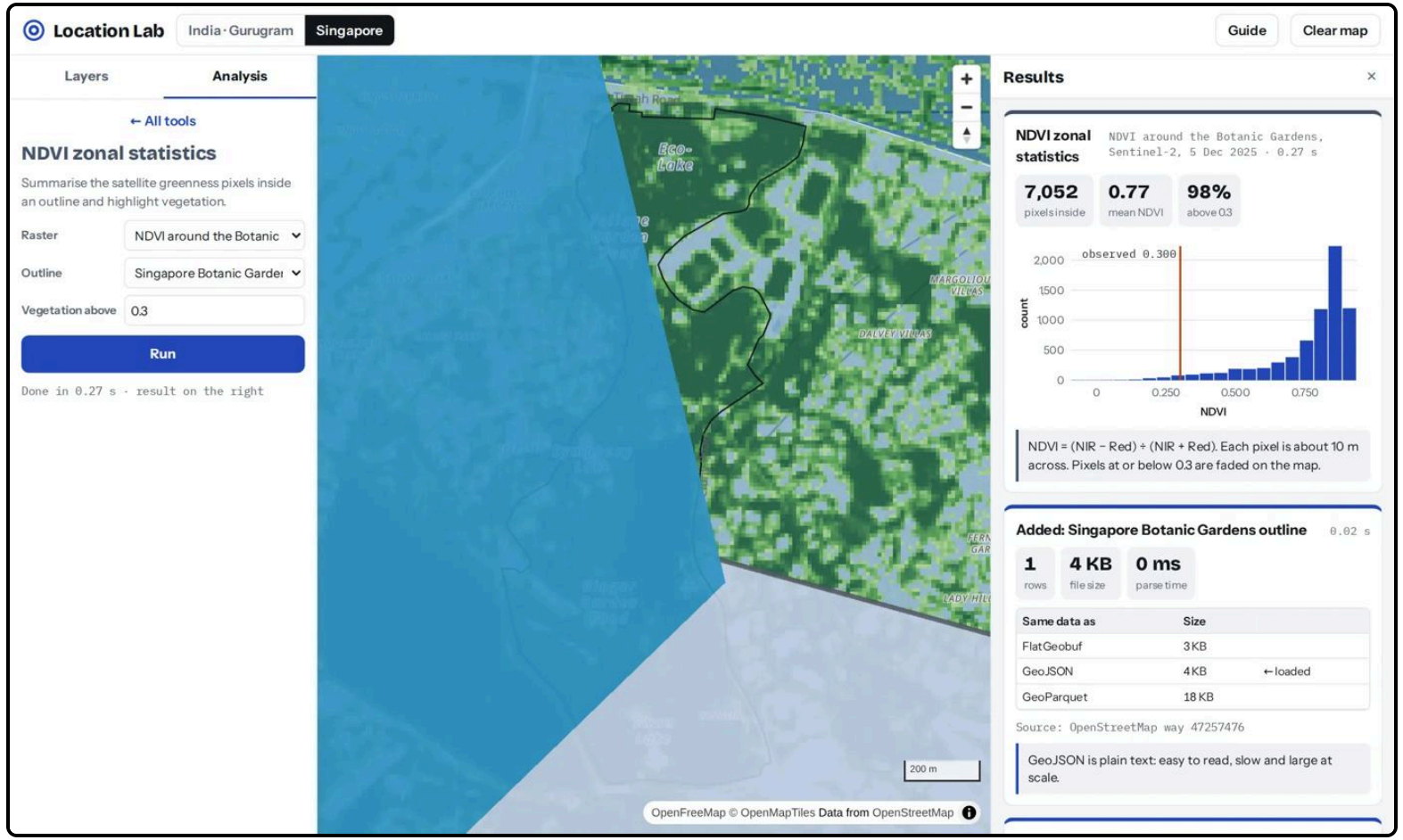
YOU SHOULD SEE

7,052 pixels inside

0.77 mean NDVI

98% above 0.3

> NDVI = (NIR − Red) ÷ (NIR + Red). Each pixel is about 10 m across. Pixels at or below 0.3 are faded on the map.



Three questions

1. A box filter returns more blocks than the exact distance test. Which is right?
2. How much smaller is the HDB file as GeoParquet than as GeoJSON?
3. Why bin HDB blocks into H3 hexagons?

Answers

1. **A box filter returns more blocks than the exact distance test. Which is right?**
   The exact test; the box over-includes its corners.
2. **How much smaller is the HDB file as GeoParquet than as GeoJSON?**
   About 2.8 MB versus about 0.3 MB, roughly nine times smaller.
3. **Why bin HDB blocks into H3 hexagons?**
   Equal-area cells with permanent IDs make counts comparable and joinable across datasets.

PART 2

Neighbours and spatial lag

The path we take on the board:

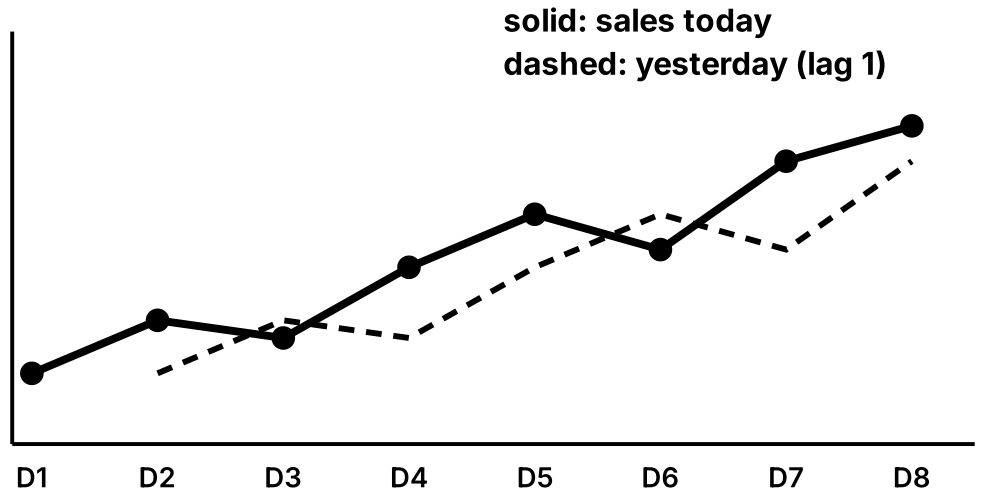
Temporal lag → **Space has no direction** → **Rook · queen · hex** → **Band · k nearest** →
Weights W → **Spatial lag Wy**

Start with time: what is a “lag”?

Take daily sales. Shift the series by one day and lay it beside the original: today’s value next to yesterday’s. That shifted copy is the **lag**, written y_{t-1}.

If today tends to look like yesterday, the two series move together and their correlation is high. Economists call this **autocorrelation**: a variable correlated with its own past.

Time is easy because it has one direction and one obvious neighbour: the day before.



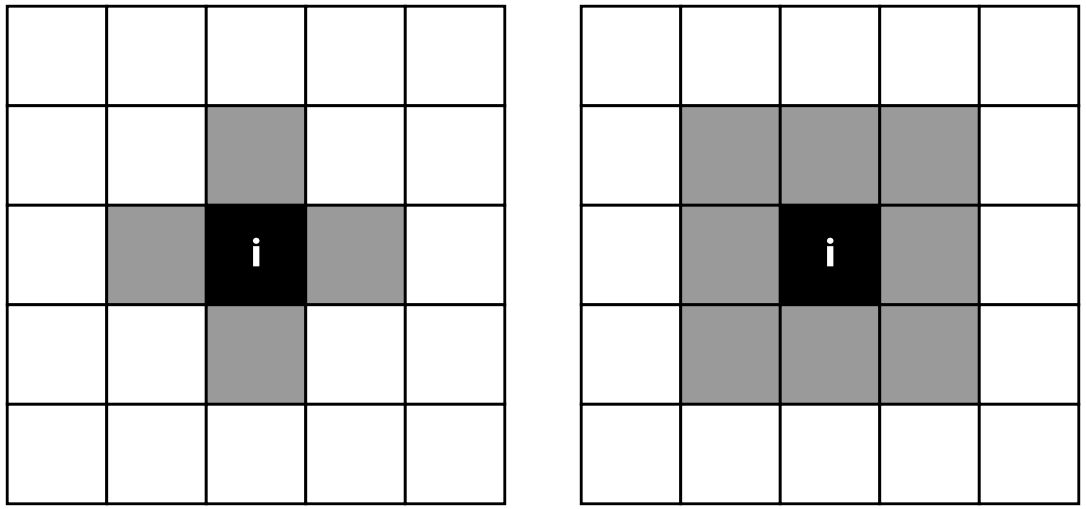
Sales over 8 days and the same series shifted by one day

Space has no “yesterday”

A café on Golf Course Road has neighbours to the north, south, east and west, near and far. There is no single “previous” location. So before we can lag anything in space, we must **decide who counts as a neighbour**.

With a grid, chess gives two answers. The **rook** moves along rows and columns: four neighbours that share an edge. The **queen** also moves diagonally: eight neighbours that share an edge or a corner.

The same rules apply to districts or planning areas: rook means sharing a border, queen means touching even at a single point.



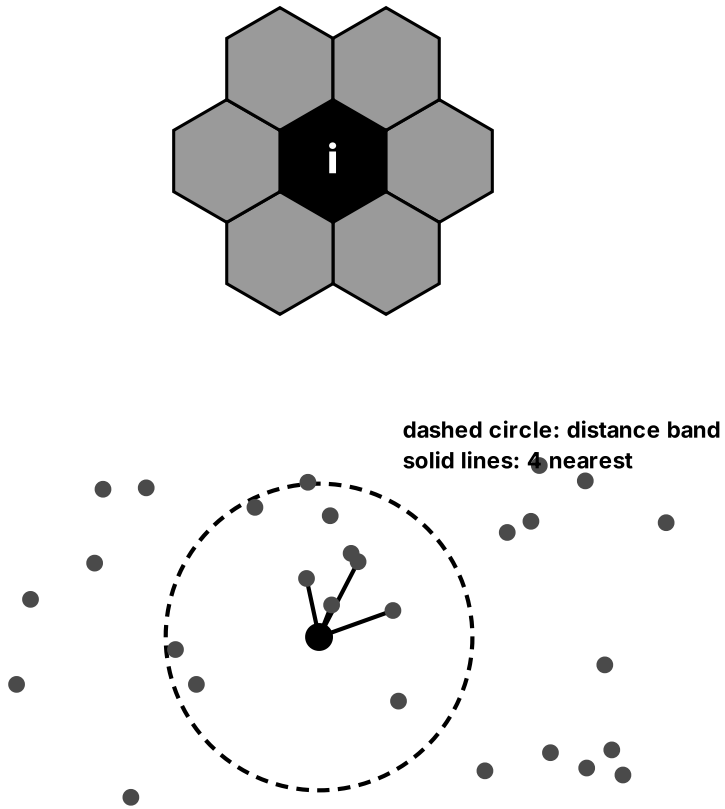
Rook (left): 4 neighbours · Queen (right): 8 neighbours

Hexagons, distance bands, and k nearest

Hexagons remove the rook/queen ambiguity: every H3 cell has exactly six neighbours, all at the same distance, all sharing a full edge.

For **points** there are no borders, so we use distance. A **distance band** makes everyone within d metres a neighbour; in sparse areas some points get none (islands). **k nearest** gives each point exactly k neighbours, however crowded or empty its area.

HDB blocks are dense in towns and sparse near parks, so we use the 8 nearest blocks.



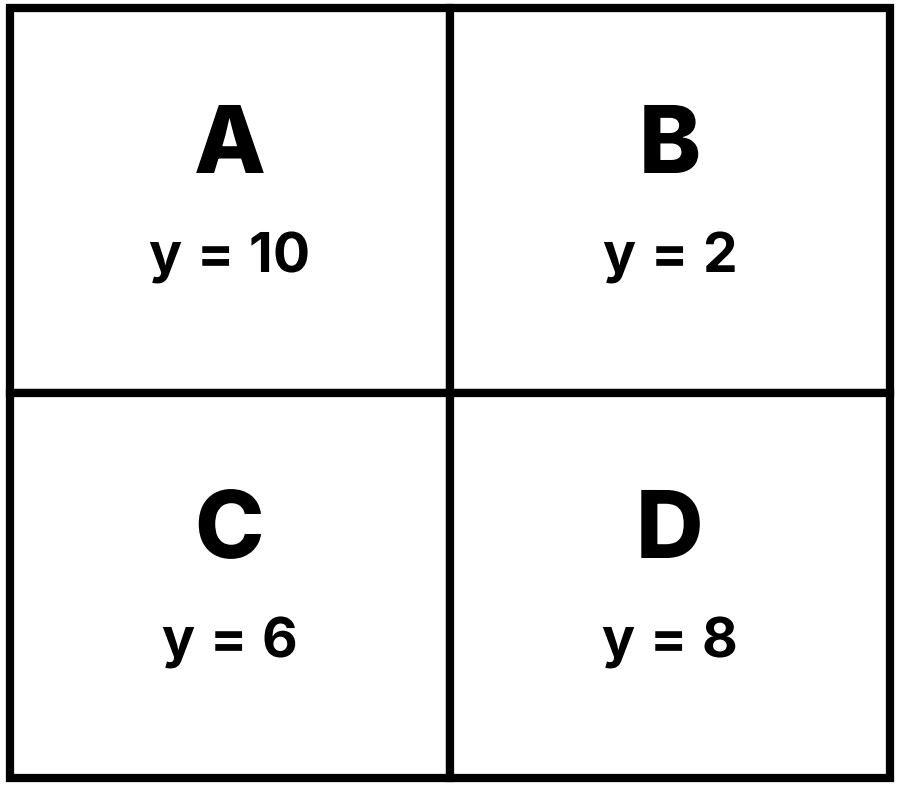
Top: six hexagon neighbours · Bottom: band vs 4 nearest

Write it down: the weights matrix W

Four areas in a 2 × 2 grid with values A = 10, B = 2, C = 6, D = 8. Under the rook rule, A touches B and C; B touches A and D; and so on. Put a 1 where two areas are neighbours and 0 elsewhere: that table is **W**.

	A	B	C	D
A	0	½	½	0
B	½	0	0	½
C	½	0	0	½
D	0	½	½	0

Dividing each row by its number of neighbours (**row-standardising**) makes every row sum to 1, so multiplying by W gives an **average**.



Rook neighbours: A–B, A–C, B–D, C–D

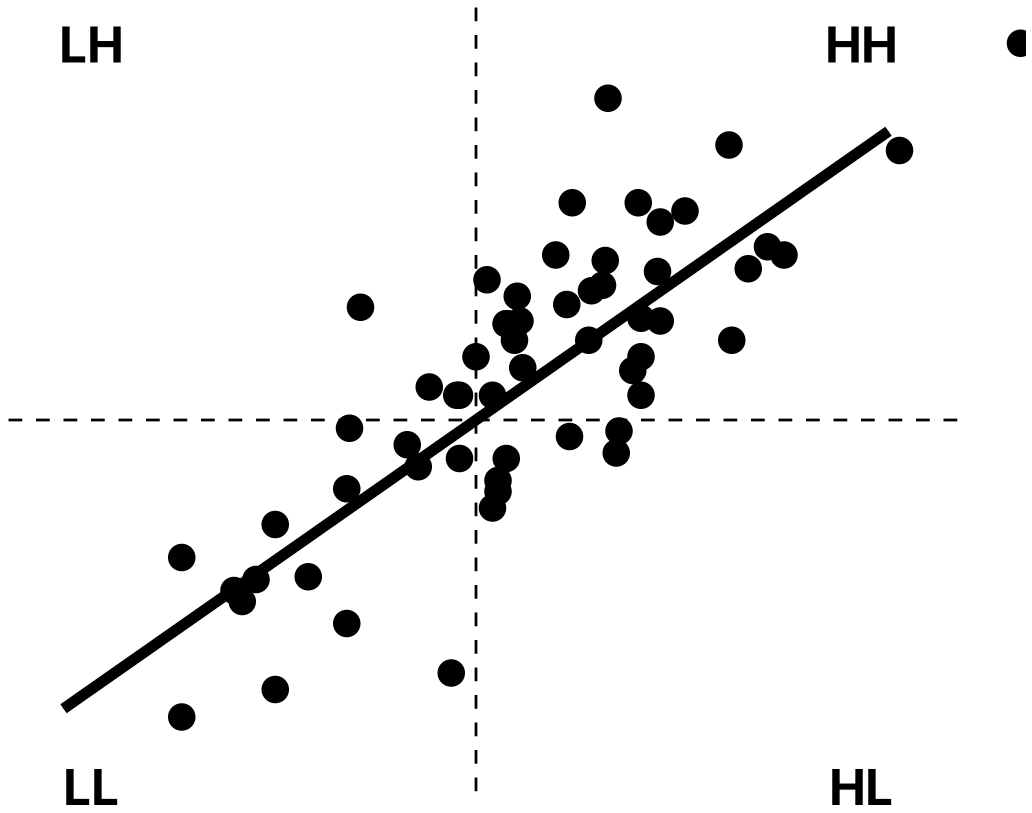
The spatial lag: “how is my neighbourhood doing?”

The spatial lag is the space version of yesterday’s value: the weighted average of a feature’s neighbours.

$$\text{lag}(y)_i = \Sigma_j \, w_{ij} \, y_j = (Wy)_i$$

A: (2 + 6) ÷ 2 = **4** · B: (10 + 8) ÷ 2 = **9** · C: (10 + 8) ÷ 2 = **9** · D: (2 + 6) ÷ 2 = **4**.

A is high (10) but its neighbourhood is low (4); B is low (2) but surrounded by high values (9). Plot each value against its lag: a rising cloud means similar values sit together; a falling one, like our 2 × 2 example, is a checkerboard.



Rising cloud: neighbours resemble each other

Where business already uses this

IN THE NEWS · PROPUBLICA · 29 OCT 2015

Uber prices each city zone separately; a block’s walk can halve the fare

Neighbouring zones can surge very differently. Defining “neighbour” and watching the lag across zones is a pricing question.

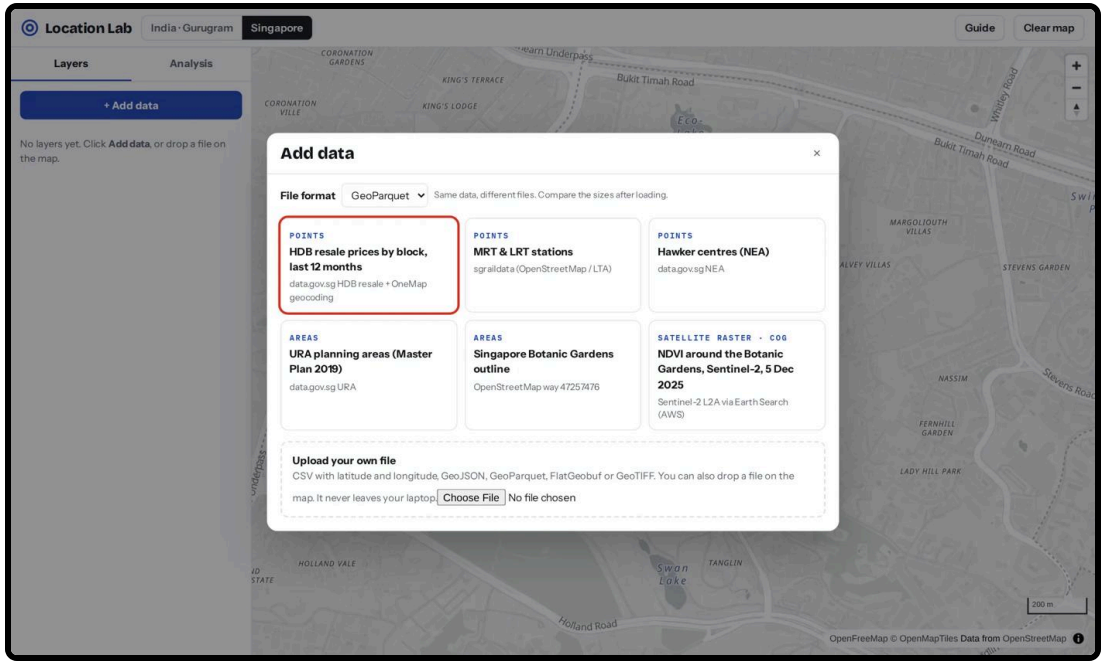
https://www.propublica.org/article/uber-surge-pricing-may-not-lead-to-a-surge-in-drivers

2 · Neighbours and spatial lag

Each HDB block’s neighbours are its 8 nearest blocks. Lag = their average price.

Open **quantng.github.io/location-lab**, choose **Singapore** in the top bar, then follow the steps.

1. Load HDB resale blocks (GeoParquet).
2. 8-nearest-neighbour weights. Click a block to see its neighbours.
3. Spatial lag of price per m².



LAB 2 · STEP 1 OF 3

Load HDB resale blocks (GeoParquet).

DO THIS

Layers tab → **+ Add data** → pick the format → click the dataset.

Dataset	HDB resale prices by block, last 12 months
File format	GeoParquet

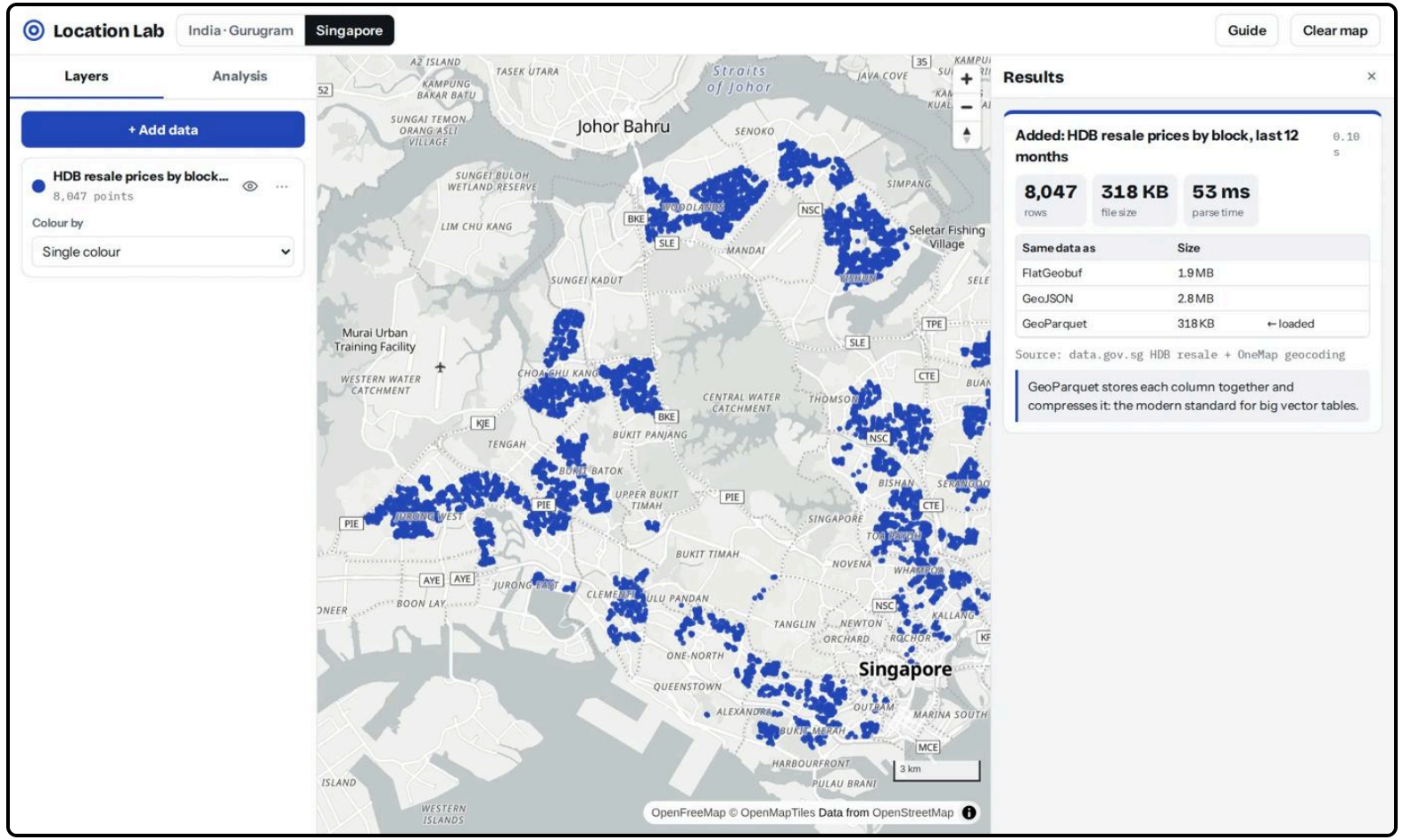
YOU SHOULD SEE

8,047 rows

318 KB file size

53 ms parse time

> GeoParquet stores each column together and compresses it: the modern standard for big vector tables.



8-nearest-neighbour weights. Click a block to see its neighbours.

DO THIS

Analysis tab → drag **Spatial weights** onto the map (or click it), check the settings, press **Run**.

Layer	HDB resale prices by block, last 12 months
Neighbours are	the k nearest
k	8

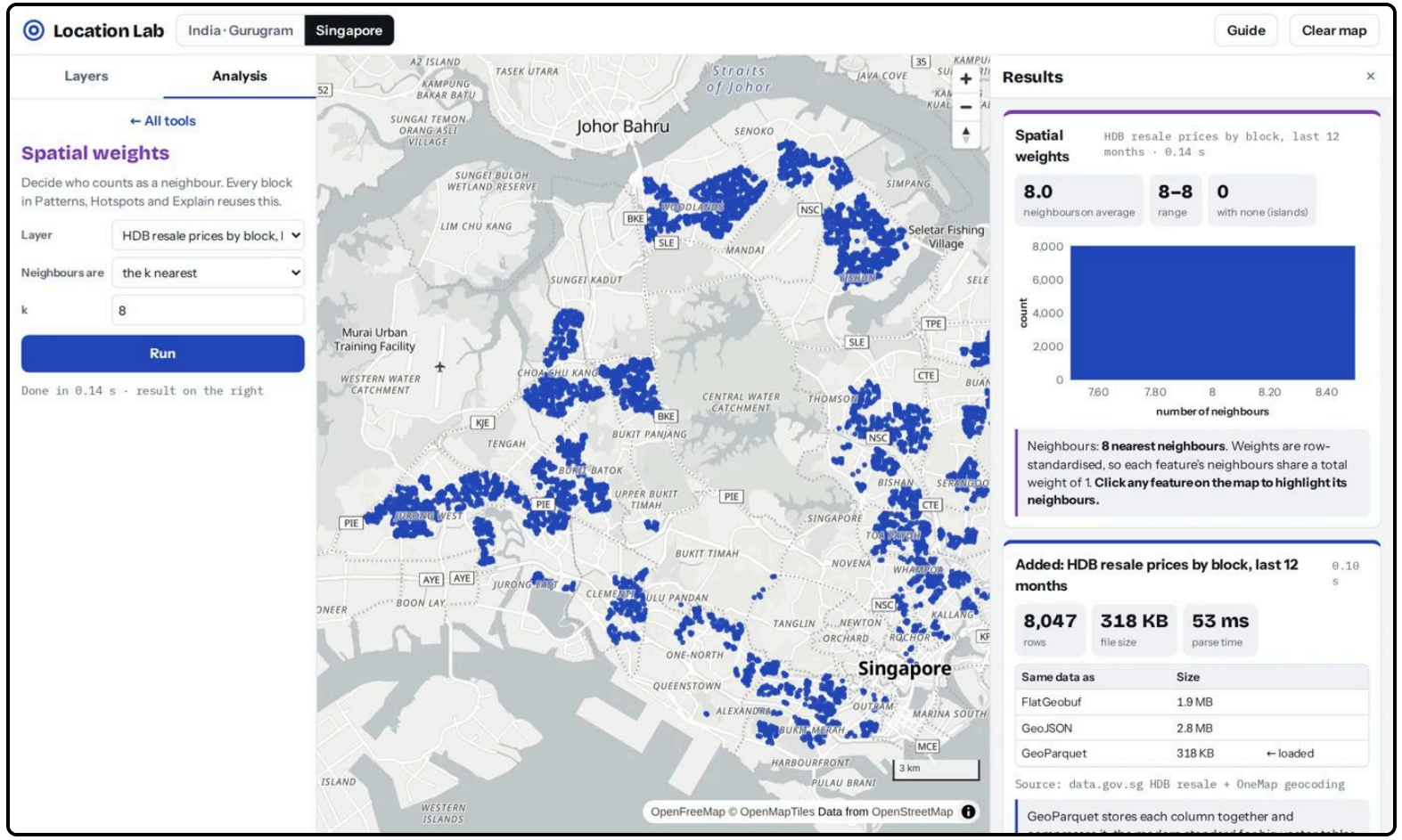
YOU SHOULD SEE

8.0
neighbours on average

8–8
range

0
with none (islands)

> Neighbours: 8 nearest neighbours. Weights are row-standardised, so each feature's neighbours share a total weight of 1. Click any feature on the map to highlight its neighbours.



LAB 2 · STEP 3 OF 3

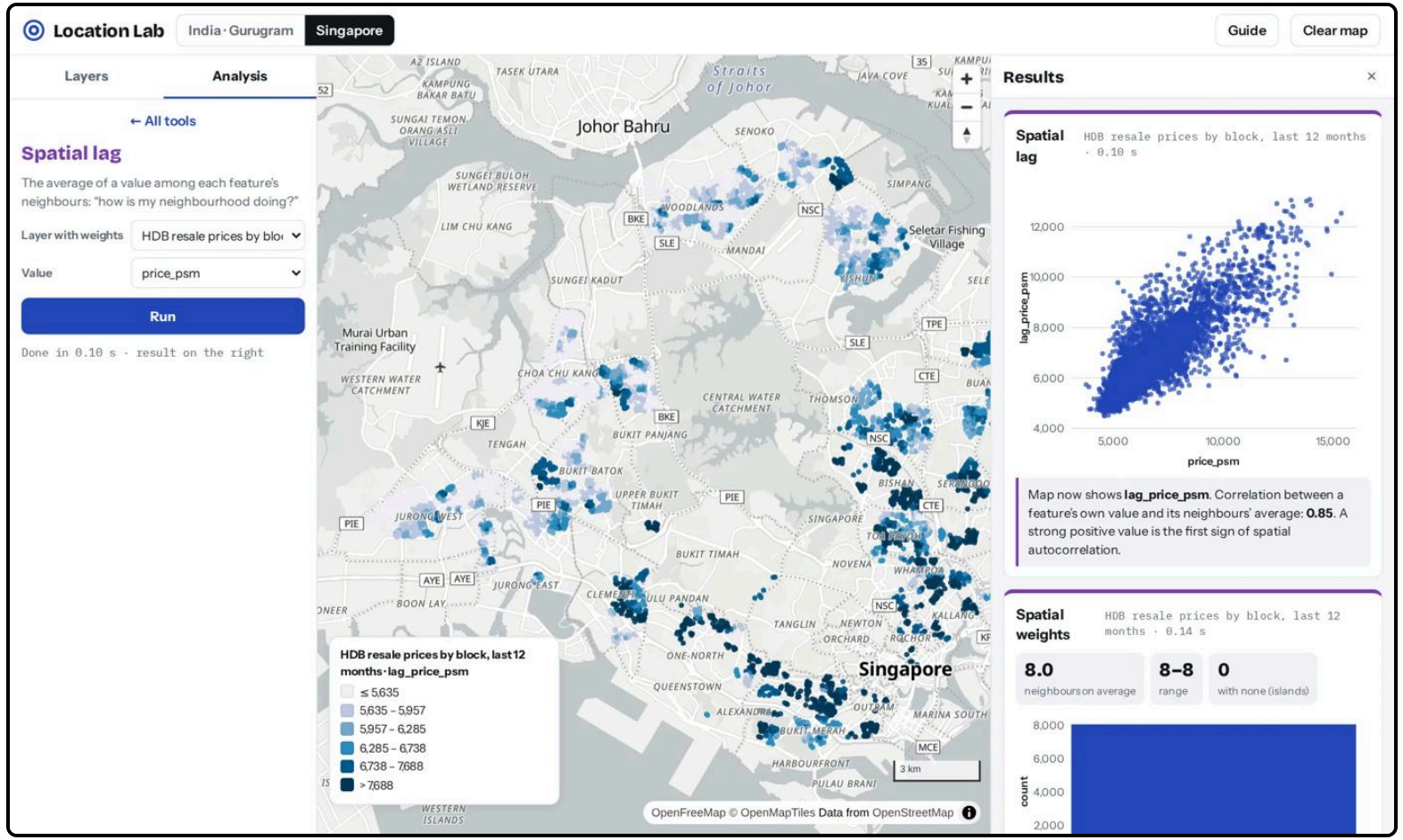
Spatial lag of price per m².

DO THIS

Analysis tab → drag **Spatial lag** onto the map (or click it), check the settings, press **Run**.

Layer with weights	HDB resale prices by block, last 12 months
Value	price_psm

> Map now shows lag_price_psm. Correlation between a feature's own value and its neighbours' average: 0.85. A strong positive value is the first sign of spatial autocorrelation.



Three questions

1. Why use 8 nearest neighbours rather than a 500 m band for HDB blocks?
2. Four blocks A = 10, B = 2, C = 6, D = 8 in a 2 × 2 grid, rook rule. What is the lag of B?
3. What does row-standardising do?

Answers

1. **Why use 8 nearest neighbours rather than a 500 m band for HDB blocks?**
   Density varies; a band gives some blocks dozens of neighbours and some none. k = 8 treats all alike.
2. **Four blocks A = 10, B = 2, C = 6, D = 8 in a 2 × 2 grid, rook rule. What is the lag of B?**
   B touches A and D: (10 + 8) ÷ 2 = 9.
3. **What does row-standardising do?**
   Each row of W sums to 1, so Wy is an average of the neighbours.

PART 3

Point patterns

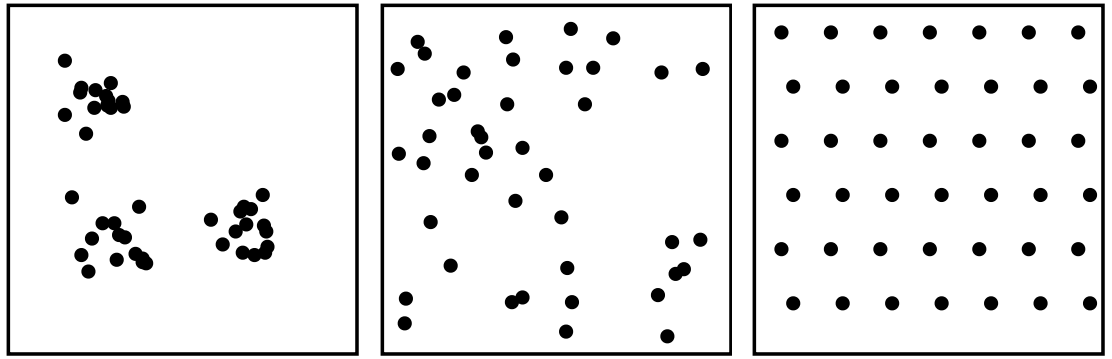
The path we take on the board:

Randomness (CSR) → **Three patterns** → **Nearest-neighbour R** → **Ripley's L + envelope**

What would “random” look like?

Before calling cafés “clustered” we need a benchmark. **Complete spatial randomness (CSR)** means every location in the study area is equally likely and points ignore each other, like rain on a pavement.

Real patterns depart from CSR in two directions. **Clustered**: points attract (food courts, malls, competition for the same footfall). **Dispersed**: points repel (franchise territories, planned services, pharmacies with exclusive catchments).



Clustered R < 1 | Random R ≈ 1 | Dispersed R > 1

Nearest-neighbour ratio R

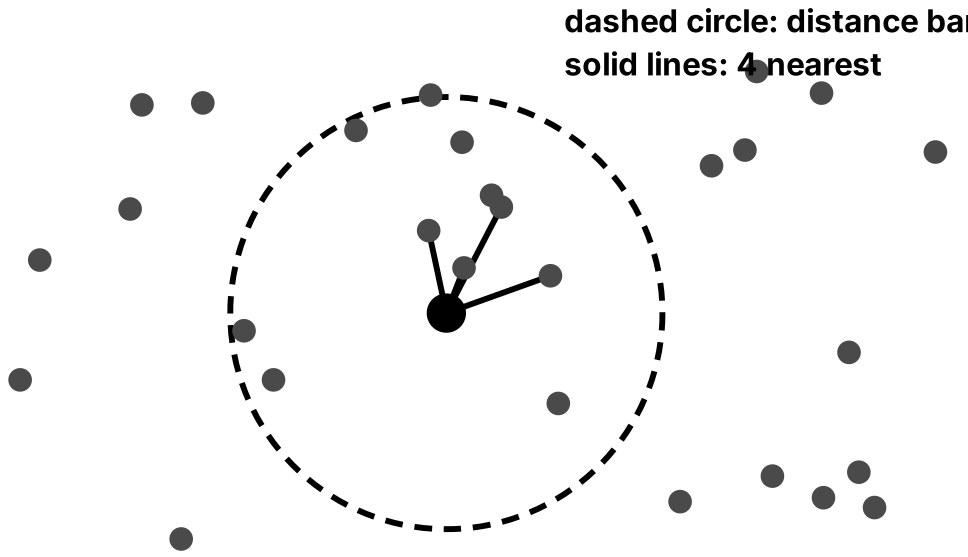
For every point, measure the distance to its nearest neighbour and average. Under CSR with n points in area A, theory gives the expected average:

$$E[d] = 0.5 \div \sqrt{(n / A)}$$

$$R = \text{observed mean } d \div E[d]$$

R < 1: clustered. R ≈ 1: random. R > 1: dispersed. A z-score tells you whether the gap from 1 is larger than chance.

Singapore hawker centres: 803 m observed vs 907 m expected, so **R = 0.89**: only mildly clustered, because hawker centres are planned to serve every estate.



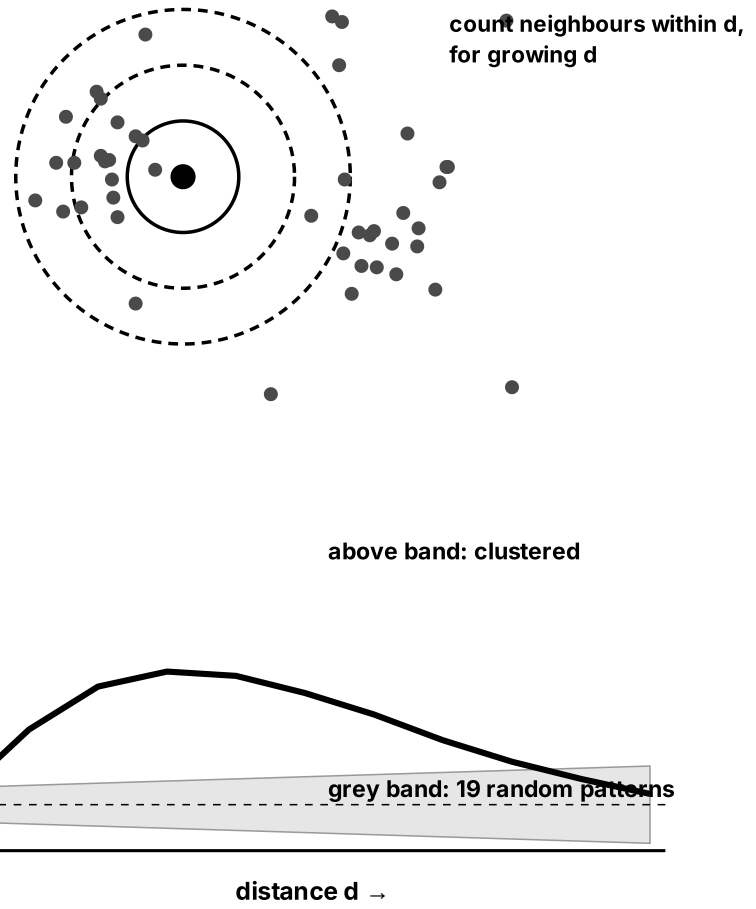
Each point's nearest-neighbour distance, averaged

One scale is not enough: Ripley’s K and L

R looks only at the closest neighbour. A pattern can be clustered at 200 m (shops in a mall) and dispersed at 3 km (malls spaced across the city).

Ripley’s idea: draw circles of growing radius d around every point and count how many others fall inside. Compare with the count CSR would give. The **L function** rescales this so that random is a flat line at zero.

To judge chance, simulate 19 random patterns in the same area and draw their range as a grey band. Where the real line rises above the band, the pattern is clustered at that scale.

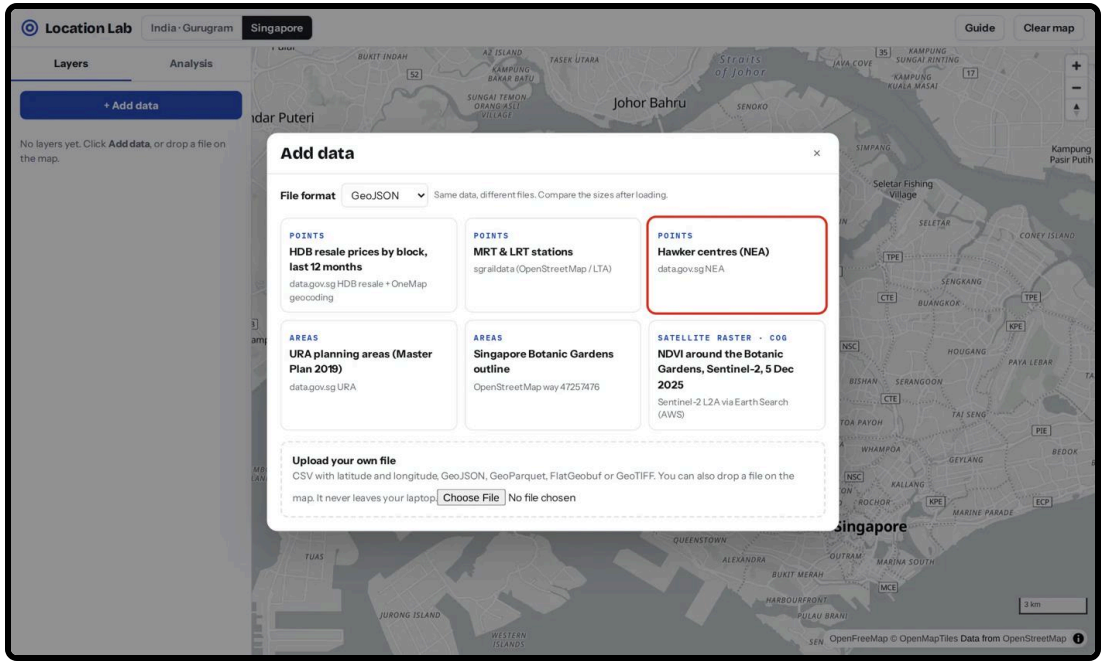


3 · Is the pattern random?

Are hawker centres clustered, spread out, or random across the island?

Open **quantng.github.io/location-lab**, choose **Singapore** in the top bar, then follow the steps.

1. Load the hawker centres.
2. Nearest-neighbour ratio and Ripley's L.



LAB 3 · STEP 1 OF 2

Load the hawker centres.

DO THIS

Layers tab → **+ Add data** → pick the format → click the dataset.

Dataset	Hawker centres (NEA)
File format	GeoJSON

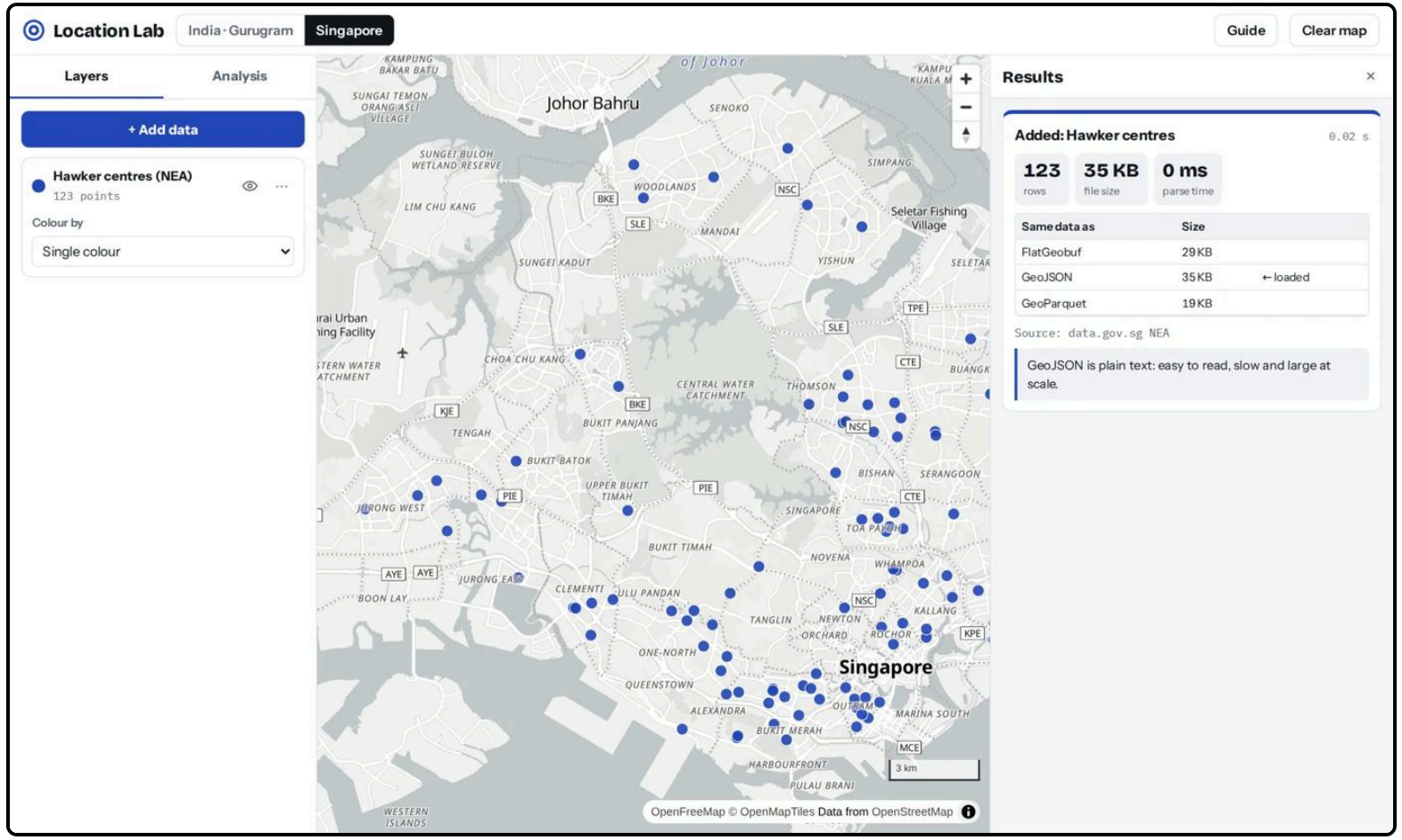
YOU SHOULD SEE

123 rows

35 KB file size

0 ms parse time

> GeoJSON is plain text: easy to read, slow and large at scale.



LAB 3 · STEP 2 OF 2

Nearest-neighbour ratio and Ripley’s L.

DO THIS

Analysis tab → drag **Point pattern test** onto the map (or click it), check the settings, press **Run**.

Points	Hawker centres (NEA)

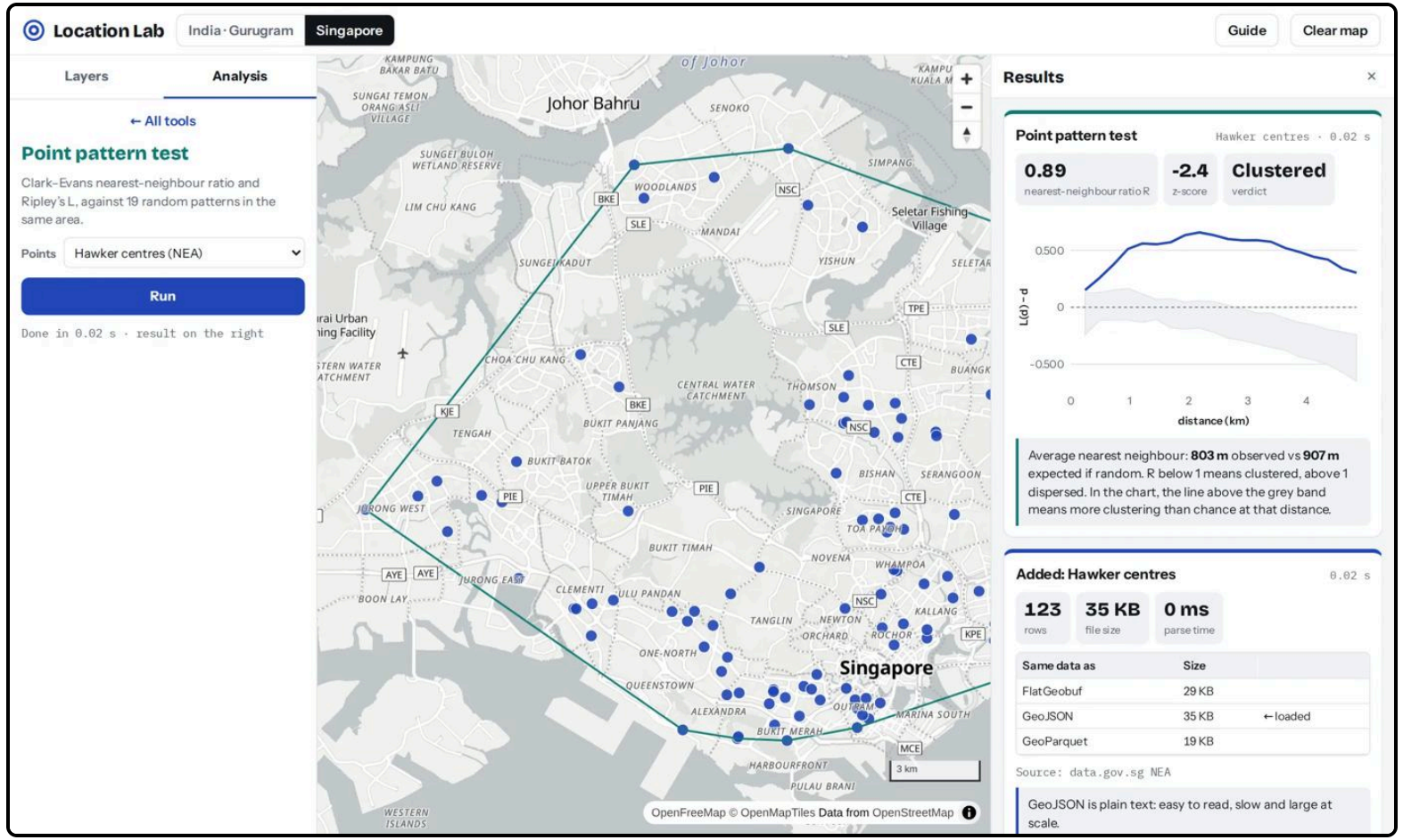
YOU SHOULD SEE

0.89
nearest-neighbour ratio R

-2.4
z-score

Clustered
verdict

> Average nearest neighbour: 803 m observed vs 907 m expected if random. R below 1 means clustered, above 1 dispersed. In the chart, the line above the grey band means more clustering than...



Three questions

1. Hawker centres give R = 0.89. How does that compare with a market-driven pattern?
2. What does Ripley’s L add to R?
3. Why compare with patterns in the same area?

Answers

1. **Hawker centres give R = 0.89. How does that compare with a market-driven pattern?**
   Only mildly clustered: hawker centres are planned to serve every estate.
2. **What does Ripley's L add to R?**
   It tests clustering at many distances, not just the nearest neighbour.
3. **Why compare with patterns in the same area?**
   Clustering only means something relative to chance over the same study area.

PART 4

Autocorrelation and hotspots

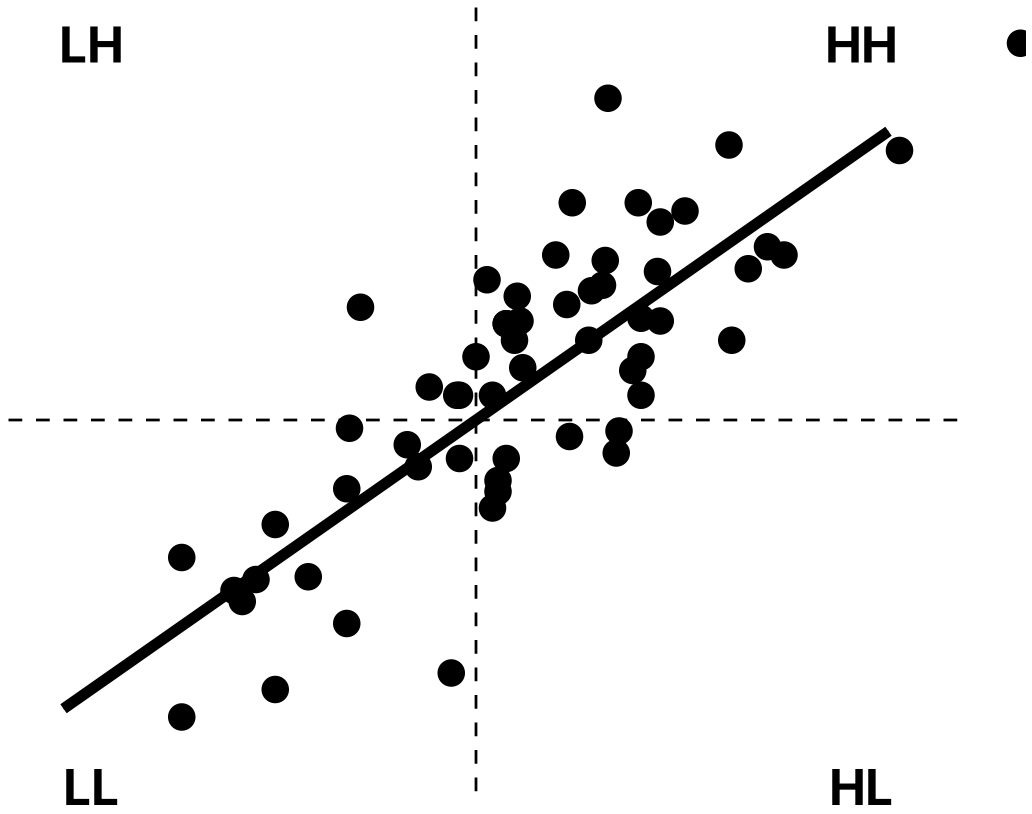
The path we take on the board:

Correlation → **Moran's I** → **Shuffle test** → **LISA** → **Gi***

Recall correlation

Correlation measures whether two variables move together. Standardise each (subtract the mean, divide by the spread), multiply them pairwise, and average. Positive: high goes with high. Negative: high goes with low. Zero: no linear link.

We already saw that a time series can be correlated with its own lag. The obvious next step: correlate a map with **its spatial lag**.



positive correlation: the line rises

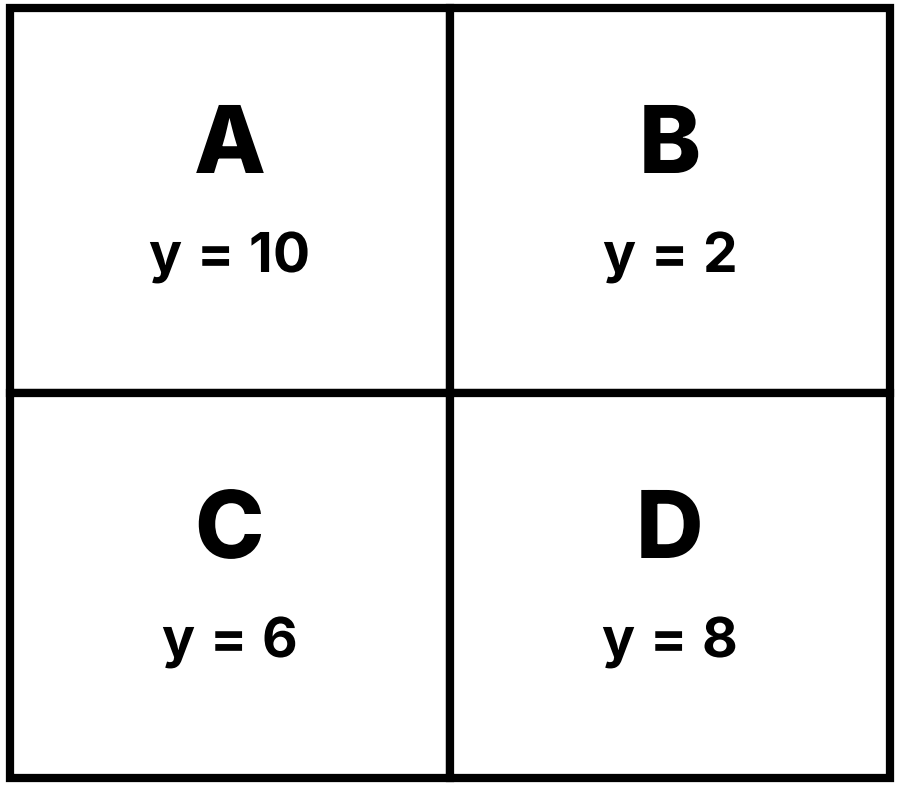
Moran’s I: a map correlated with its own lag

With $z_i = y_i - \bar{y}$ and row-standardised weights:

$$I = \Sigma_i \, z_i \cdot (Wz)_i \div \Sigma_i \, z_i^2$$

It is the slope of the line in the plot of z against its lag. Our 2 × 2 example: mean 6.5, so z = (3.5, −4.5, −0.5, 1.5) and Wz = (−2.5, 2.5, 2.5, −2.5). Numerator −25, denominator 35, so **I = −0.71**: a checkerboard, neighbours unlike each other.

Positive I means similar values cluster (Tobler: near things are more related). Near zero: random.



High A beside low B and C: negative I

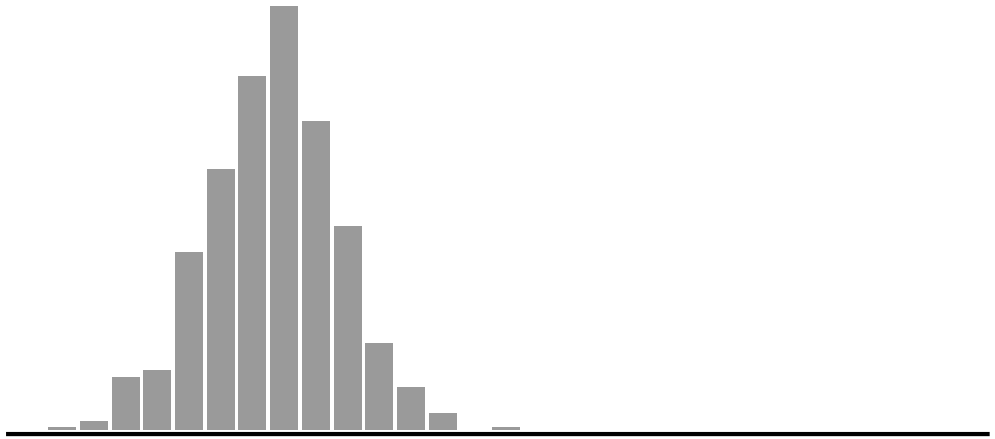
AUTOCORRELATION · STEP 3

Is it chance? Shuffle the map

Keep the values, keep the neighbours, but scatter the values randomly across locations, then recompute I. Do it 999 times. That histogram is what I looks like when location does not matter.

If the real I sits beyond almost every shuffle, the pseudo p-value is (number of shuffles at least as extreme + 1) ÷ 1,000. None out of 999 gives **p = 0.001**, the smallest possible.

HDB price per m²: **I = 0.711**, p = 0.001: price is strongly spatial.



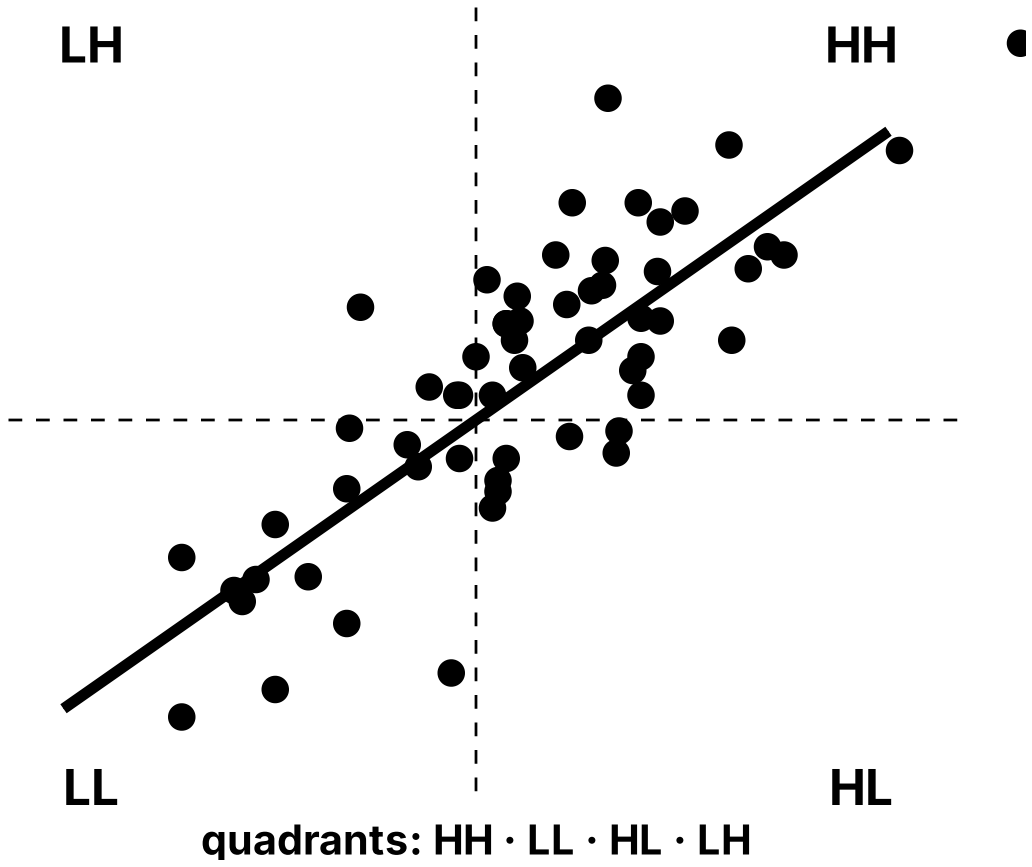
999 shuffled maps

From one number to every place: LISA

Moran's I is an average over the whole map. **Local Moran (LISA)** keeps each place's own term, $I_i = z_i \cdot (Wz)_i$, and tests it by shuffling only that place's neighbours (conditional permutation).

The four quadrants of the Moran scatter give the labels. **High-High**: hotspot. **Low-Low**: coldspot. **High-Low**: a strong performer among weak neighbours. **Low-High**: a weak spot inside a strong area, often an opening or a problem worth checking.

Singapore: 1,358 High-High blocks.



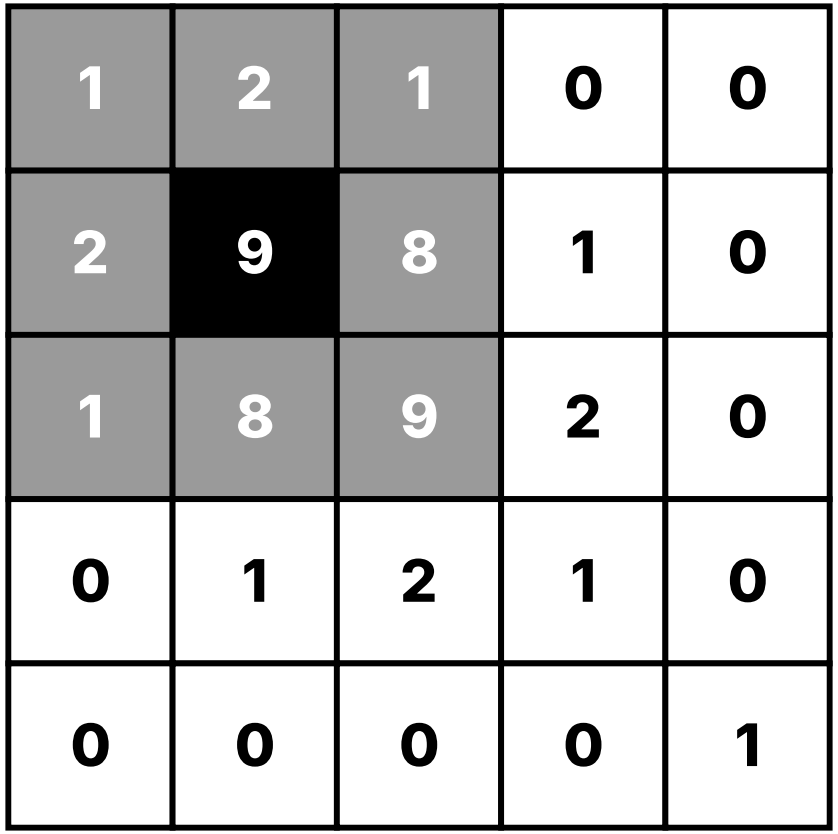
quadrants: HH · LL · HL · LH

Getis-Ord Gi*: is the local total too big?

Gi* draws a window around each place (itself included), adds up the values inside, and compares that local sum with what the window would hold if values were spread at random. The result is a z-score.

In the grid, the black cell's window holds 41 against an expected 17.6 (nine cells × the map mean of 1.96). That is a hotspot.

z ≥ 1.65, 1.96, 2.58 means 90, 95, 99% confidence. A heatmap shows density; Gi* says whether the density is more than chance.



Window of 9 cells around the black cell

Where business already uses this

IN THE NEWS · MALAY MAIL · 7 OCT 2026

Record 209 million-dollar HDB flats sold in September 2026

Toa Payoh (33), Queenstown (28) and Bedok (20) led. Compare with the High-High hotspots the LISA and Gi* blocks find in the HDB price data.

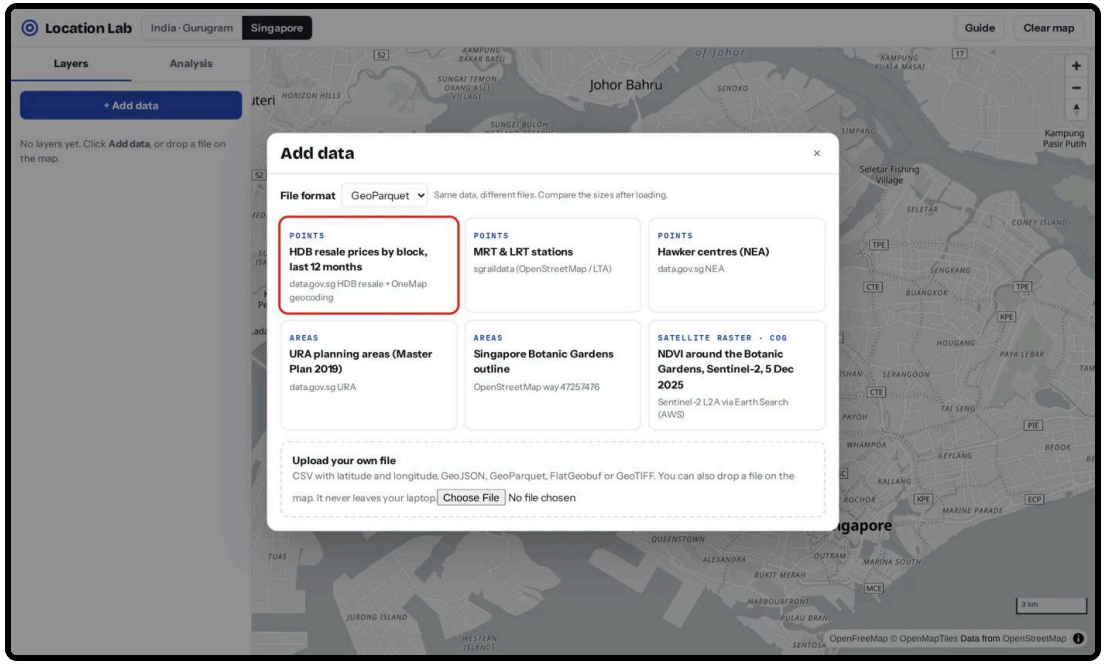
https://www.malaymail.com/news/singapore/2026/10/07/singapores-hdb-market-splits-record-s1m-deals-slower-overall-sales/238098

4 · Autocorrelation and hotspots

Where are prices significantly high or low, given their neighbours?

Open **quantng.github.io/location-lab**, choose **Singapore** in the top bar, then follow the steps.

1. Load HDB resale blocks.
2. 8-nearest-neighbour weights.
3. Global Moran's I.
4. LISA clusters and outliers.
5. Gi* hotspots.



LAB 4 · STEP 1 OF 5

Load HDB resale blocks.

DO THIS

Layers tab → **+ Add data** → pick the format → click the dataset.

Dataset	HDB resale prices by block, last 12 months
File format	GeoParquet

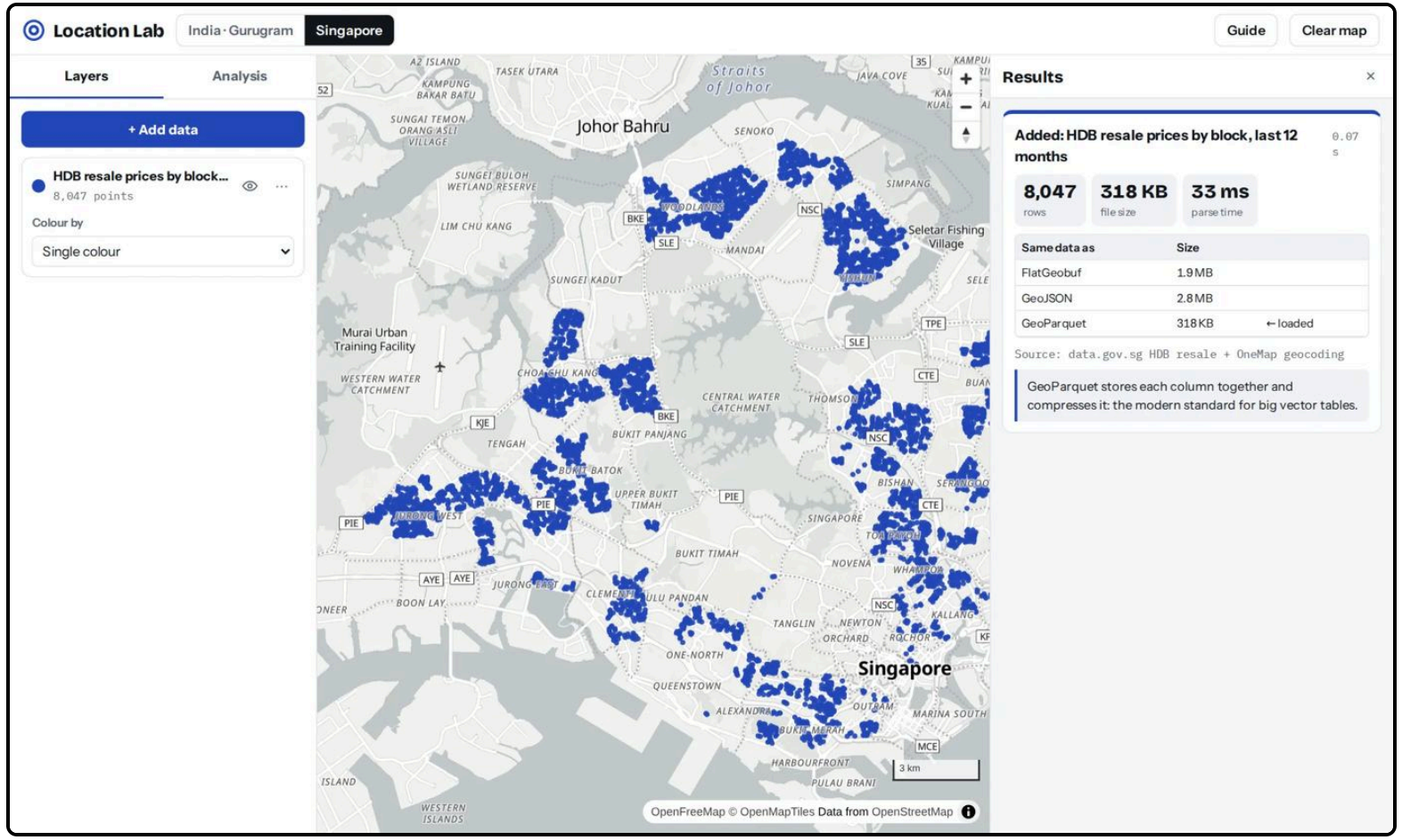
YOU SHOULD SEE

8,047 rows

318 KB file size

33 ms parse time

> GeoParquet stores each column together and compresses it: the modern standard for big vector tables.



8-nearest-neighbour weights.

DO THIS

Analysis tab → drag **Spatial weights** onto the map (or click it), check the settings, press **Run**.

Layer	HDB resale prices by block, last 12 months
Neighbours are	the k nearest
k	8

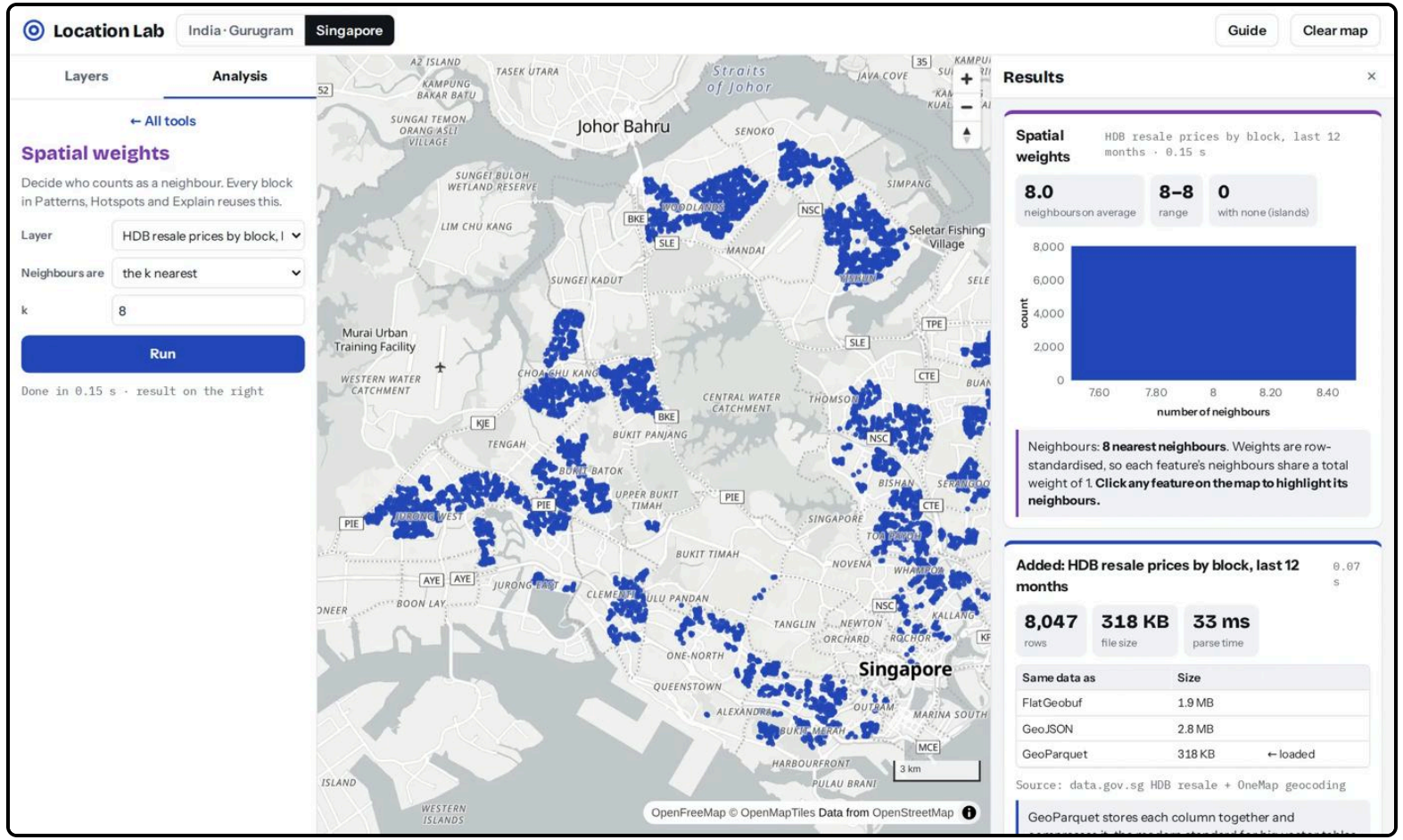
YOU SHOULD SEE

8.0
neighbours on average

8–8
range

0
with none (islands)

> Neighbours: 8 nearest neighbours. Weights are row-standardised, so each feature's neighbours share a total weight of 1. Click any feature on the map to highlight its neighbours.



Global Moran’s I.

DO THIS

Analysis tab → drag **Global Moran’s I** onto the map (or click it), check the settings, press **Run**.

Layer with weights	HDB resale prices by block, last 12 months
Value	price_psm
Shuffles	999

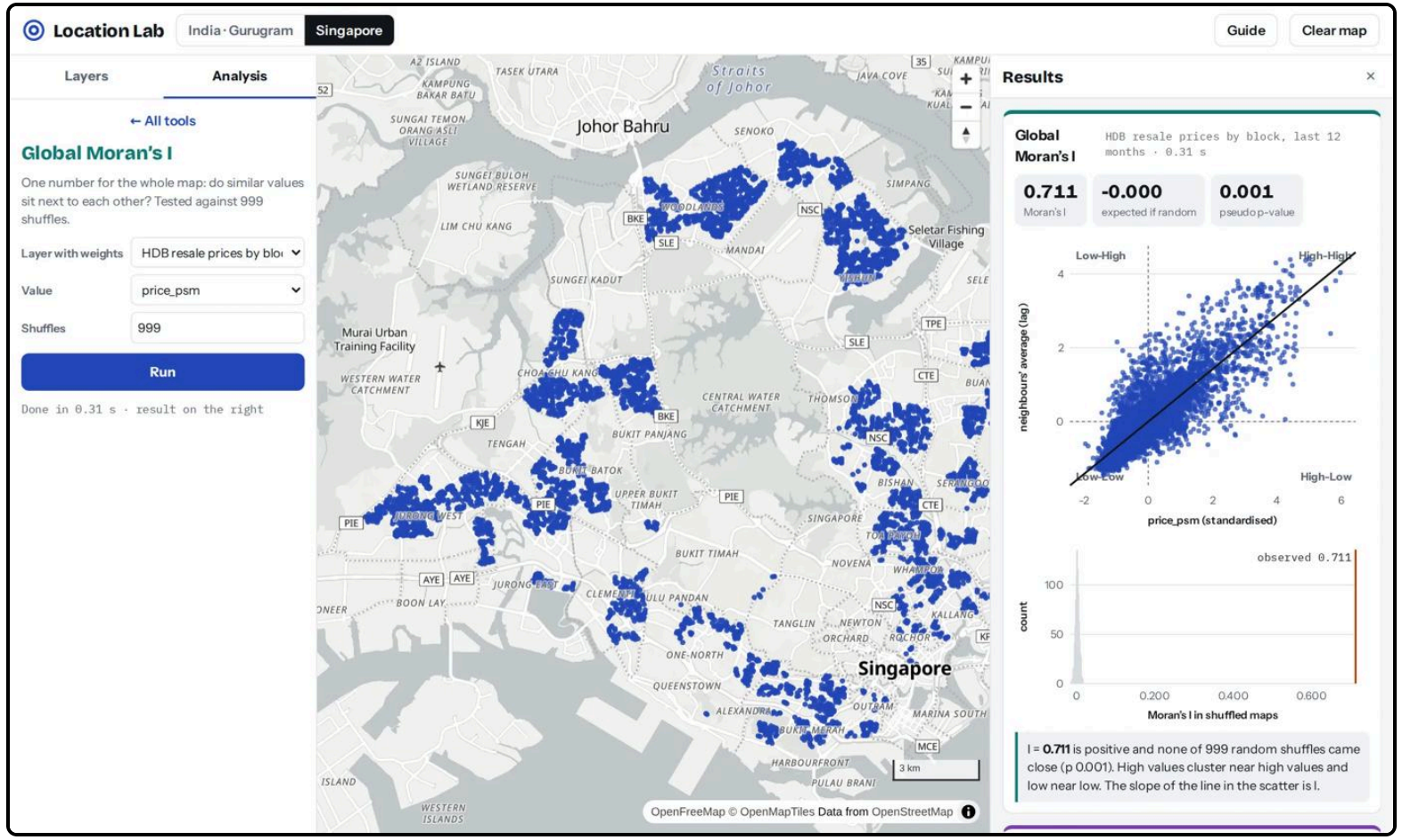
YOU SHOULD SEE

0.711
Moran’s I

-0.000
expected if random

0.001
pseudo p-value

> I = 0.711 is positive and none of 999 random shuffles came close (p 0.001). High values cluster near high values and low near low. The slope of the line in the scatter is I.



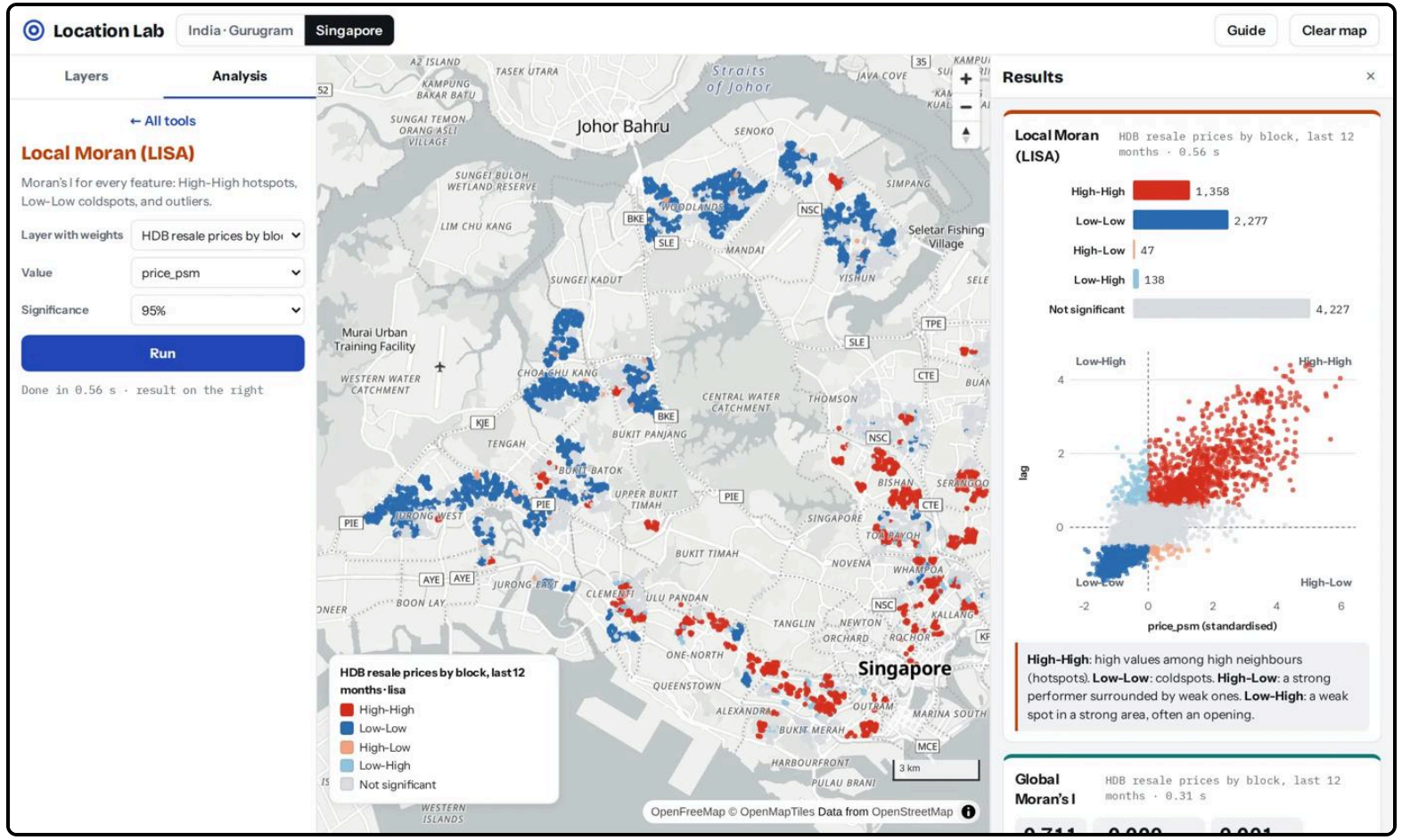
LISA clusters and outliers.

DO THIS

Analysis tab → drag **Local Moran (LISA)** onto the map (or click it), check the settings, press **Run**.

Layer with weights	HDB resale prices by block, last 12 months
Value	price_psm
Significance	95%

> High-High: high values among high neighbours (hotspots). Low-Low: coldspots. High-Low: a strong performer surrounded by weak ones. Low-High: a weak spot in a strong area, often an opening.



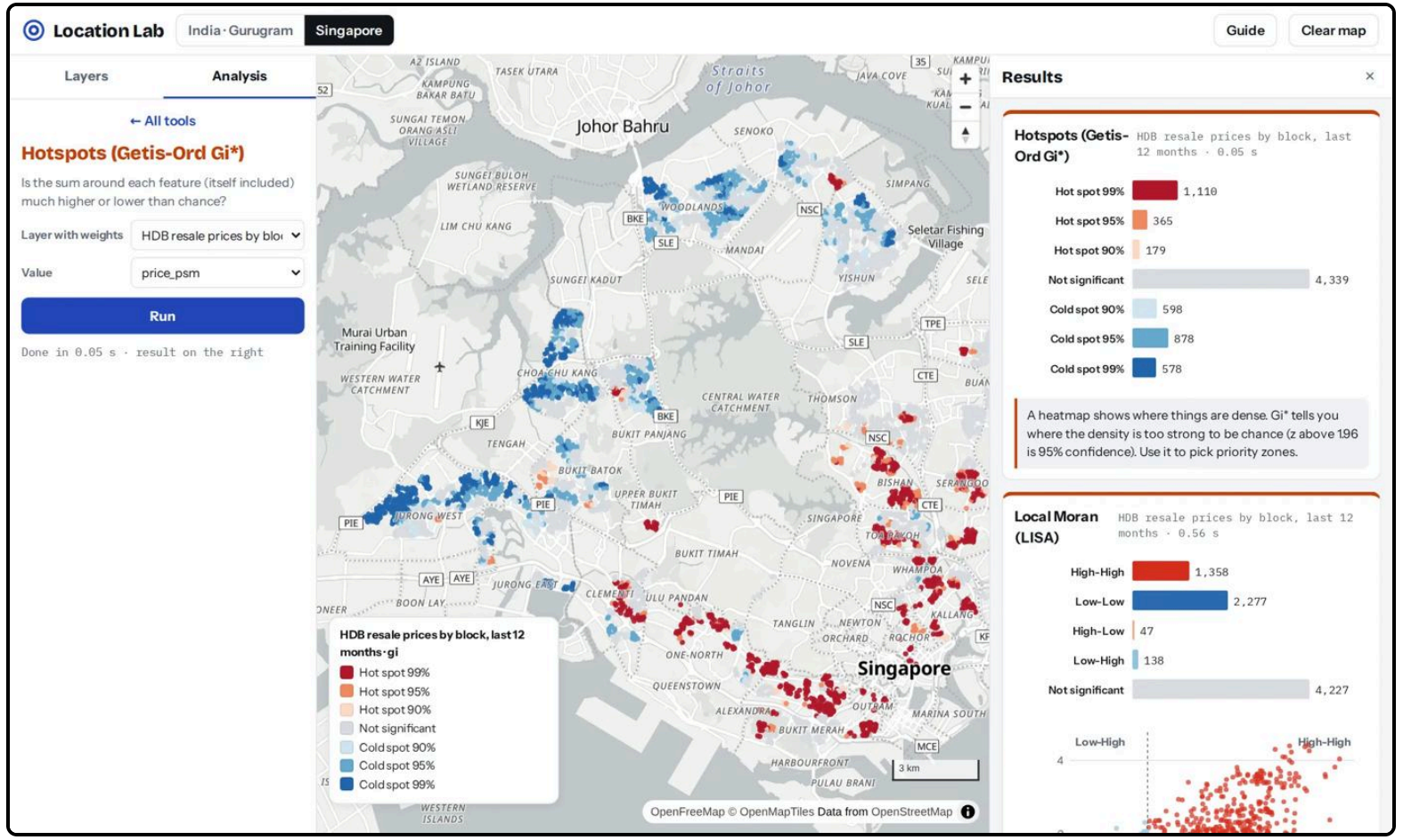
Gi* hotspots.

DO THIS

Analysis tab → drag **Hotspots (Getis-Ord Gi*)** onto the map (or click it), check the settings, press **Run**.

Layer with weights	HDB resale prices by block, last 12 months
Value	price_psm

> A heatmap shows where things are dense. Gi* tells you where the density is too strong to be chance (z above 1.96 is 95% confidence). Use it to pick priority zones.



Three questions

1. Moran's I = 0.711, p = 0.001 for price per m^2. Interpret.
2. A block is Low-High. What could it mean for a buyer?
3. How does Gi* differ from LISA?

Answers

1. **Moran's I = 0.711, p = 0.001 for price per m^2. Interpret.**
   Strong positive autocorrelation: expensive blocks sit near expensive blocks, far beyond chance.
2. **A block is Low-High. What could it mean for a buyer?**
   A cheaper block in an expensive area: possible value, or a block-level issue (lease, age, noise) to check.
3. **How does Gi* differ from LISA?**
   Gi* tests the local total including the place itself; LISA tests similarity between a place and its neighbours and also flags outliers.

PART 5

Clusters for territories

The path we take on the board:

Density → **Core · border · noise** → **ε and minPts**

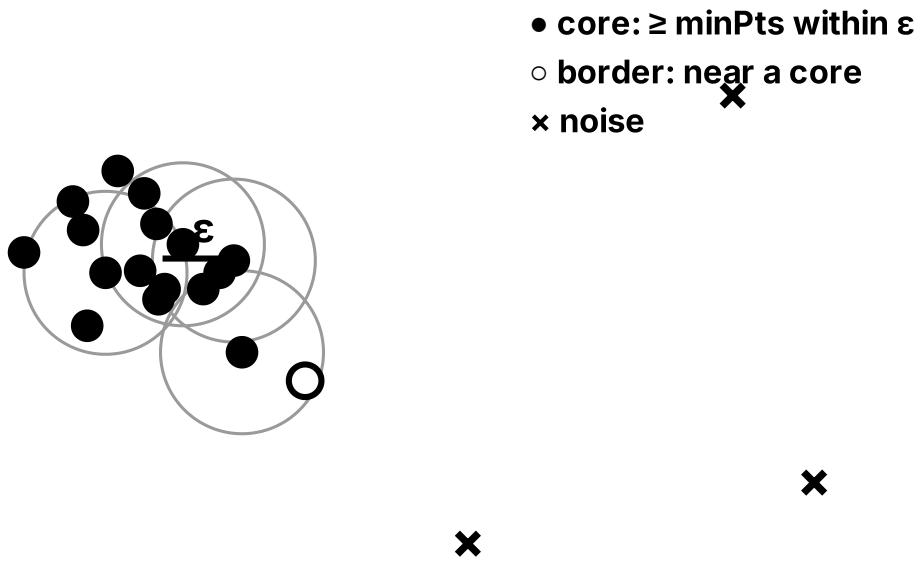
CLUSTERS · STEP 1

Density, not distance to a centre

k-means assumes round clusters and asks you for k in advance. Markets are not round: they follow roads, malls and metro lines.

DBSCAN asks two business questions instead. **ε**: how close is “the same place”? **minPts**: how many points make a market? A point with at least minPts others within ε is a **core**. Points within ε of a core are **border** points. Everything else is **noise**.

Clusters grow by chaining cores together, so they can take any shape, and the number of clusters comes out of the data.



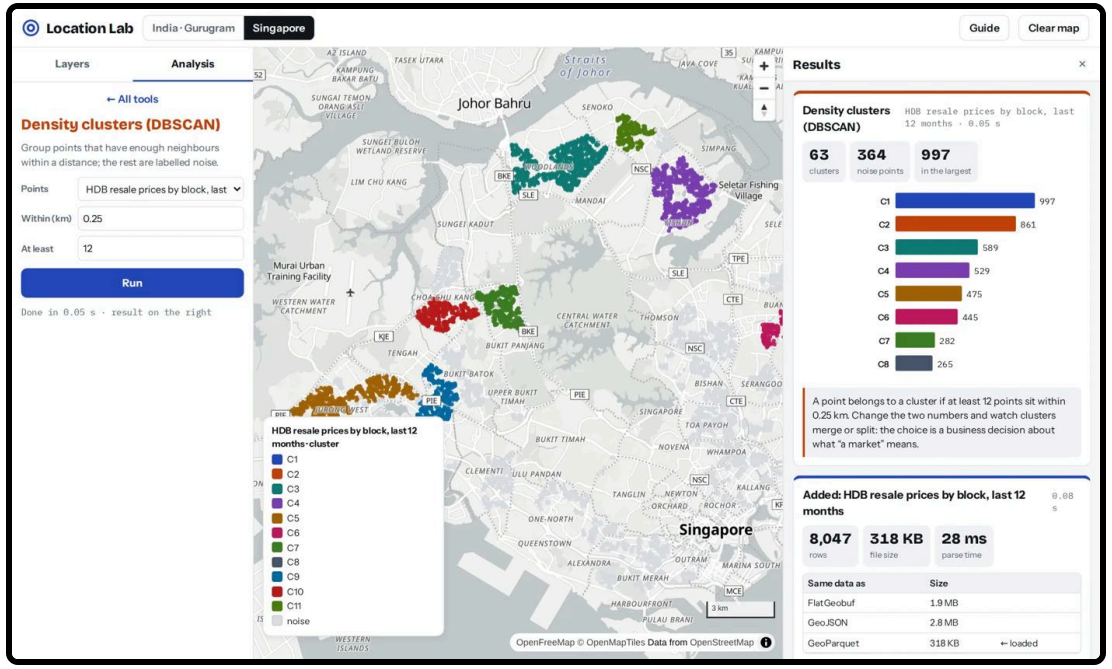
CLUSTERS · STEP 2

Choosing ε and minPts

Set ε from the decision: a delivery radius, a walking distance, a block spacing. Set minPts from what counts as a viable cluster. Larger ε or smaller minPts merges clusters and shrinks noise; the opposite splits them.

Singapore, ε = 250 m and minPts = 12: **63 clusters** that recover HDB estates from block locations alone.

Noise is not useless: an isolated outlet with strong sales may mark an underserved catchment.

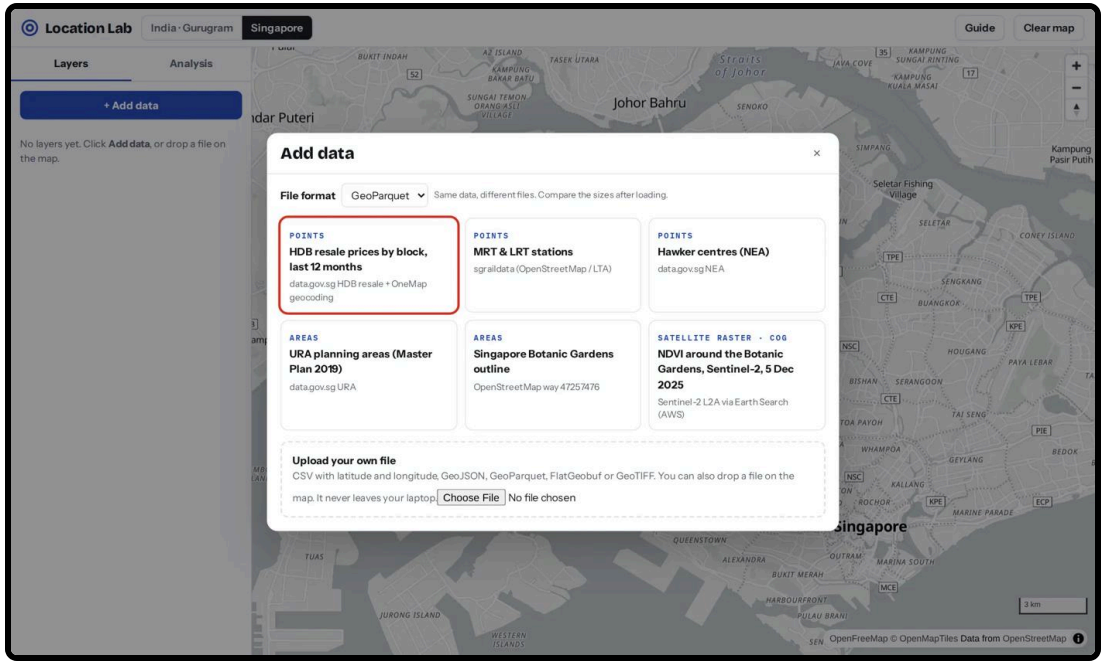


5 · Clusters for territories

DBSCAN finds HDB estates from block locations alone.

Open **quantng.github.io/location-lab**, choose **Singapore** in the top bar, then follow the steps.

1. Load HDB resale blocks.
2. DBSCAN: 12+ blocks within 250 m.



LAB 5 · STEP 1 OF 2

Load HDB resale blocks.

DO THIS

Layers tab → **+ Add data** → pick the format → click the dataset.

Dataset	HDB resale prices by block, last 12 months
File format	GeoParquet

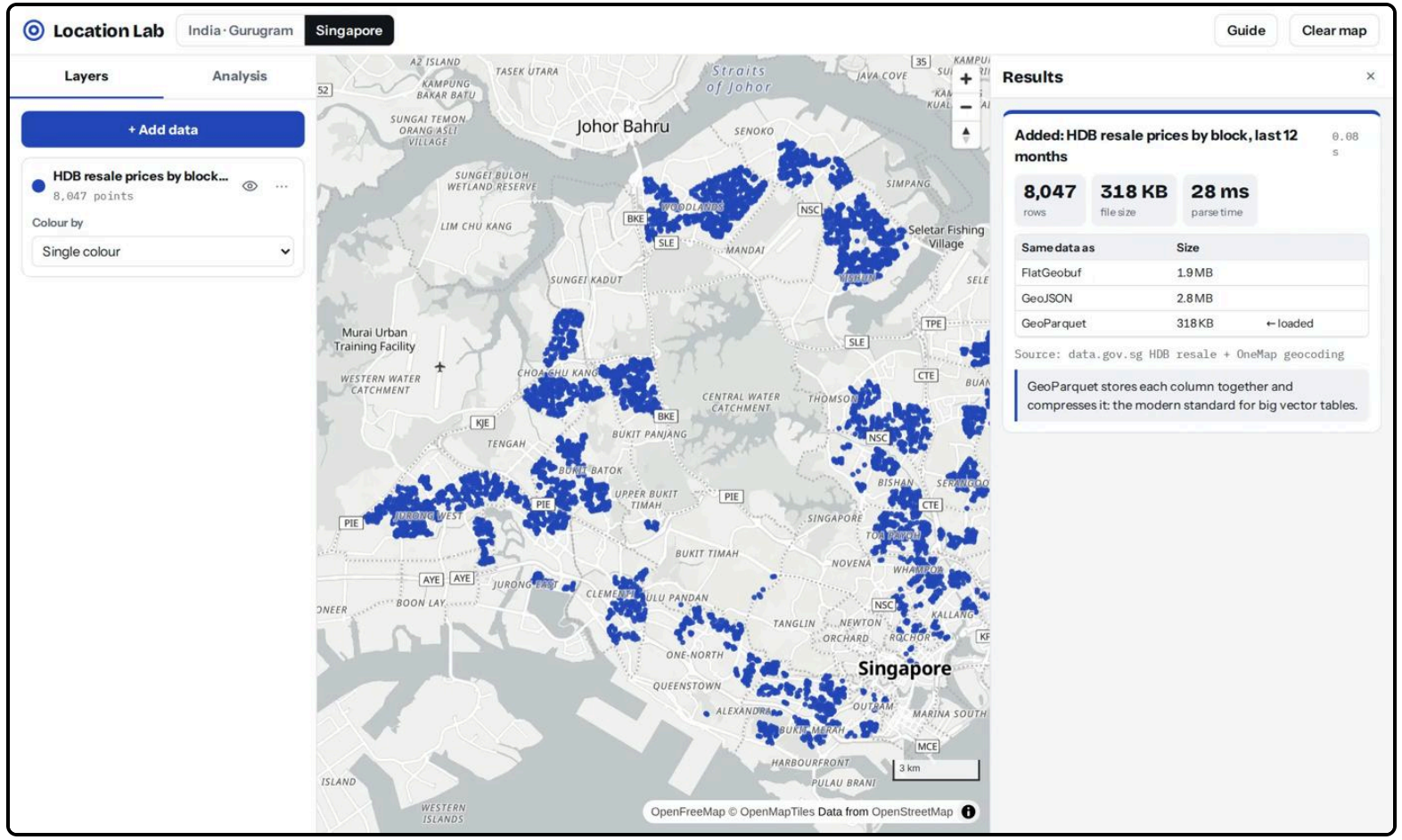
YOU SHOULD SEE

8,047 rows

318 KB file size

28 ms parse time

> GeoParquet stores each column together and compresses it: the modern standard for big vector tables.



DBSCAN: 12+ blocks within 250 m.

DO THIS

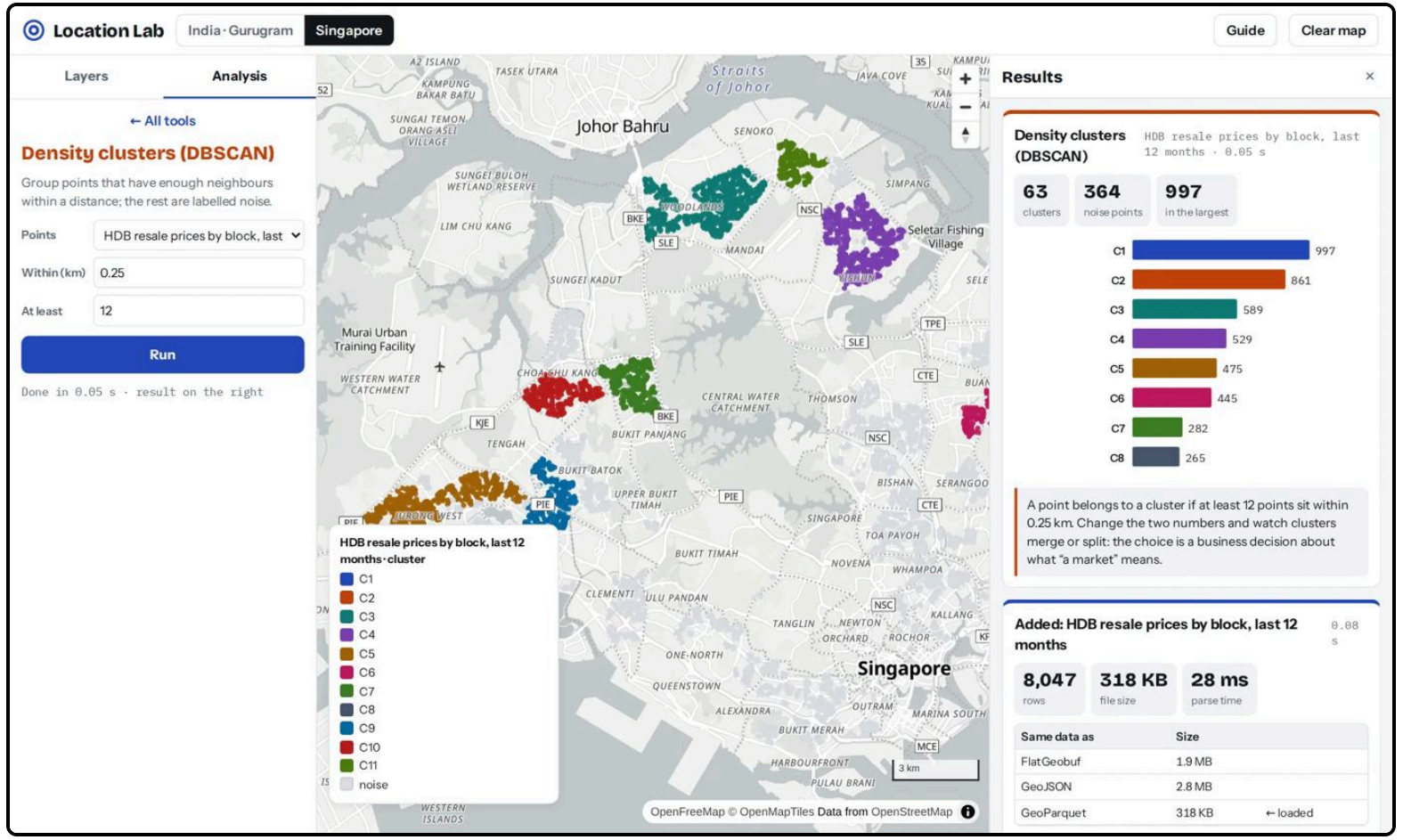
Analysis tab → drag **Density clusters (DBSCAN)** onto the map (or click it), check the settings, press **Run**.

Points	HDB resale prices by block, last 12 months
Within (km)	0.25
At least	12

YOU SHOULD SEE

63	364	997
clusters	noise points	in the largest

> A point belongs to a cluster if at least 12 points sit within 0.25 km. Change the two numbers and watch clusters merge or split: the choice is a business decision about what "a market" means.



Three questions

1. DBSCAN with 12 blocks within 250 m finds 63 clusters. What are they?
2. What happens if ε rises to 500 m?
3. Why is DBSCAN useful for service planning?

Answers

1. **DBSCAN with 12 blocks within 250 m finds 63 clusters. What are they?**
   HDB estates, recovered from block locations alone.
2. **What happens if ε rises to 500 m?**
   Clusters merge and fewer blocks are noise.
3. **Why is DBSCAN useful for service planning?**
   It finds dense groups of any shape and flags isolated points without fixing the number of zones.

PART 6

From samples to surfaces

The path we take on the board:

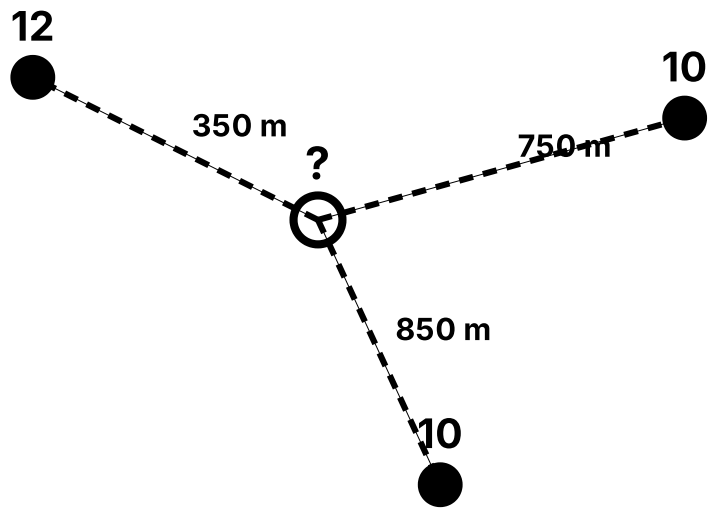
IDW → **Power** → **Variogram** → **Kriging + error** → **Hold-out check**

Guess the value between samples: IDW

You know the value at three places and want it at a fourth. Closer samples should count more. **Inverse distance weighting** gives each sample weight 1 ÷ distance^{p} and takes the weighted average.

Samples 12, 10, 10 at 350, 750, 850 m. With p = 1: (12/350 + 10/750 + 10/850) ÷ (1/350 + 1/750 + 1/850) = **11.1**. With p = 2 the nearest sample dominates more: **11.4**.

IDW is simple and exact at the samples, but it cannot tell you how sure it is.

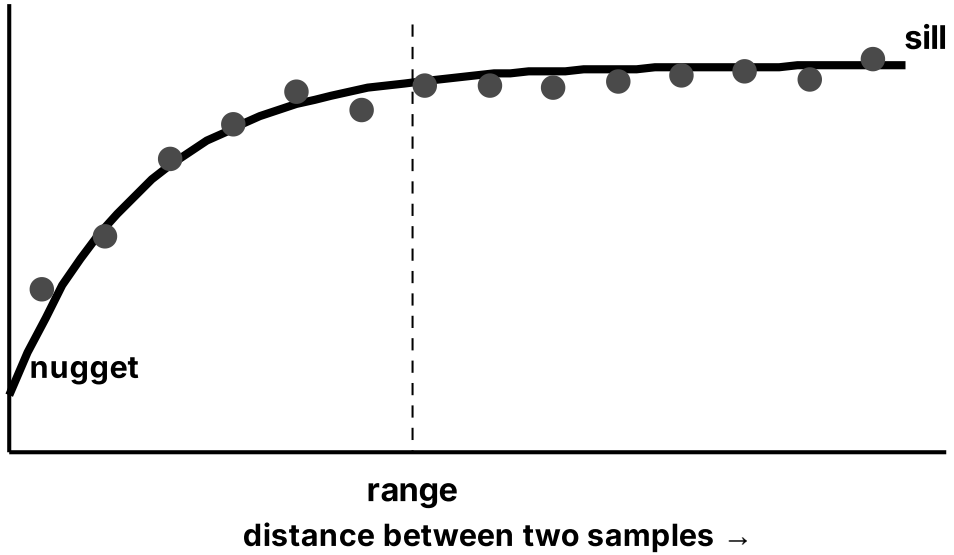


The variogram: how fast similarity fades

Take every pair of samples, compute half the squared difference of their values, and plot it against their distance. Close pairs are similar (low); far pairs are not (high).

Three numbers describe the curve. **Nugget**: the difference at almost zero distance (measurement noise, micro-variation). **Sill**: the plateau, the total variance. **Range**: the distance where the curve flattens, beyond which samples tell you nothing about each other.

HDB price: range ≈ **12.8 km**, a city-wide gradient.

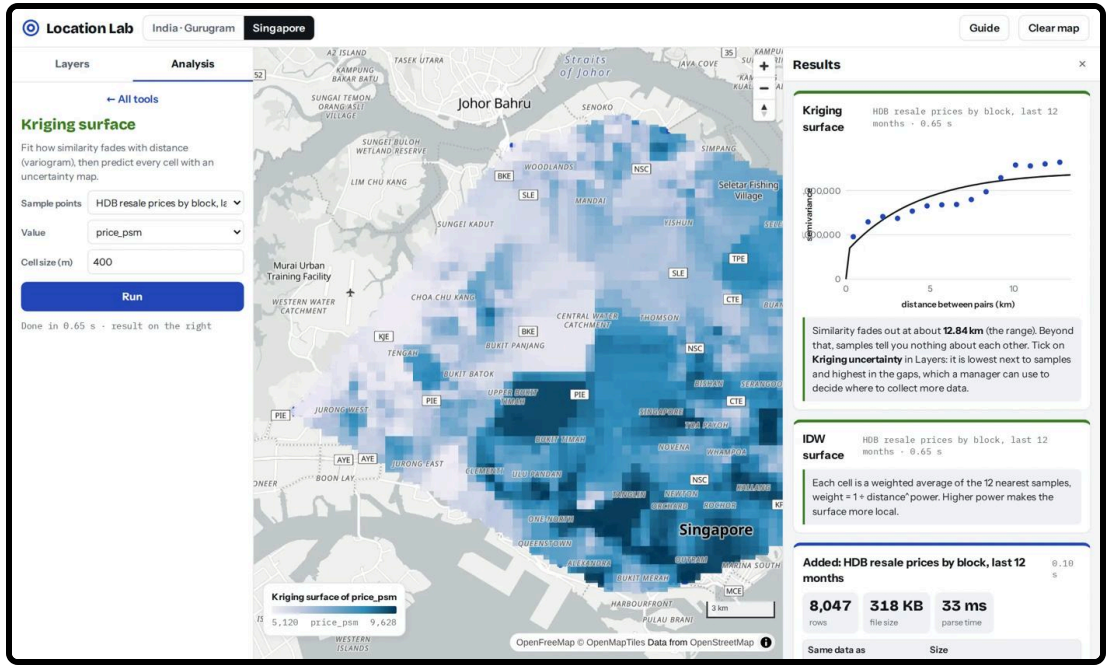


Kriging: the best guess, and how good it is

Kriging uses the variogram to choose the weights instead of a fixed power. Samples that are close and not redundant get more weight. It also returns a **standard error** for every cell: low near samples, high in the gaps.

That second map is a business tool: it tells you where to collect the next sample.

Applied to HDB prices, the kriged surface shows the city-wide gradient and where the estimate is weakest.

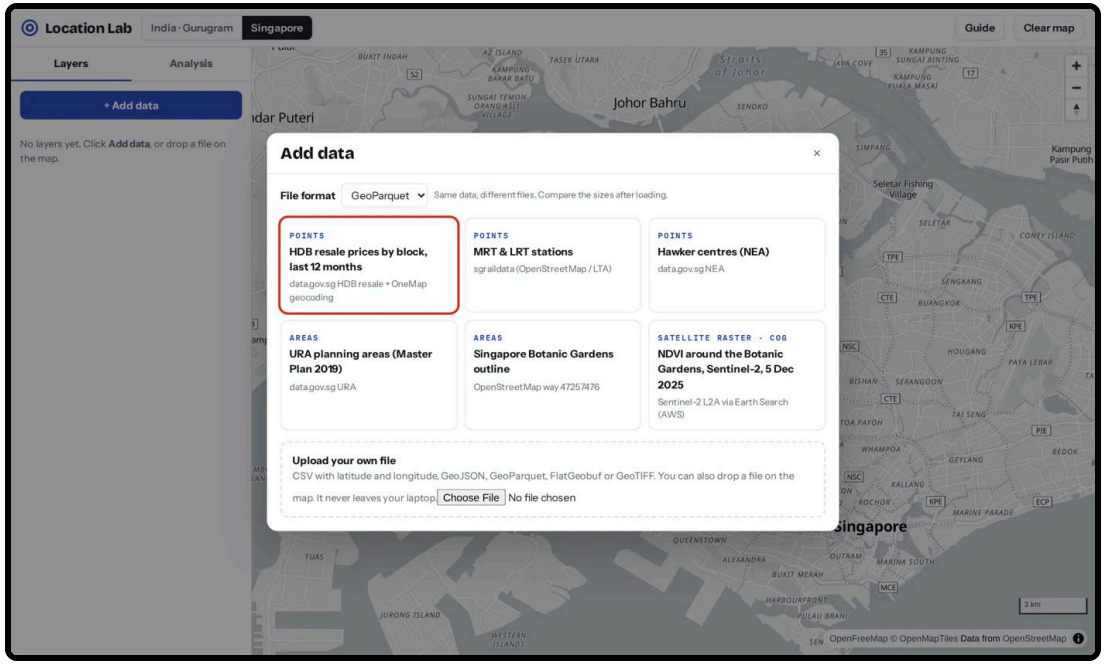


6 · From samples to surfaces

A continuous price surface from block-level sales.

Open **quantng.github.io/location-lab**, choose **Singapore** in the top bar, then follow the steps.

1. Load HDB resale blocks.
2. IDW price surface.
3. Kriging surface and its uncertainty.



LAB 6 · STEP 1 OF 3

Load HDB resale blocks.

DO THIS

Layers tab → **+ Add data** → pick the format → click the dataset.

Dataset	HDB resale prices by block, last 12 months
File format	GeoParquet

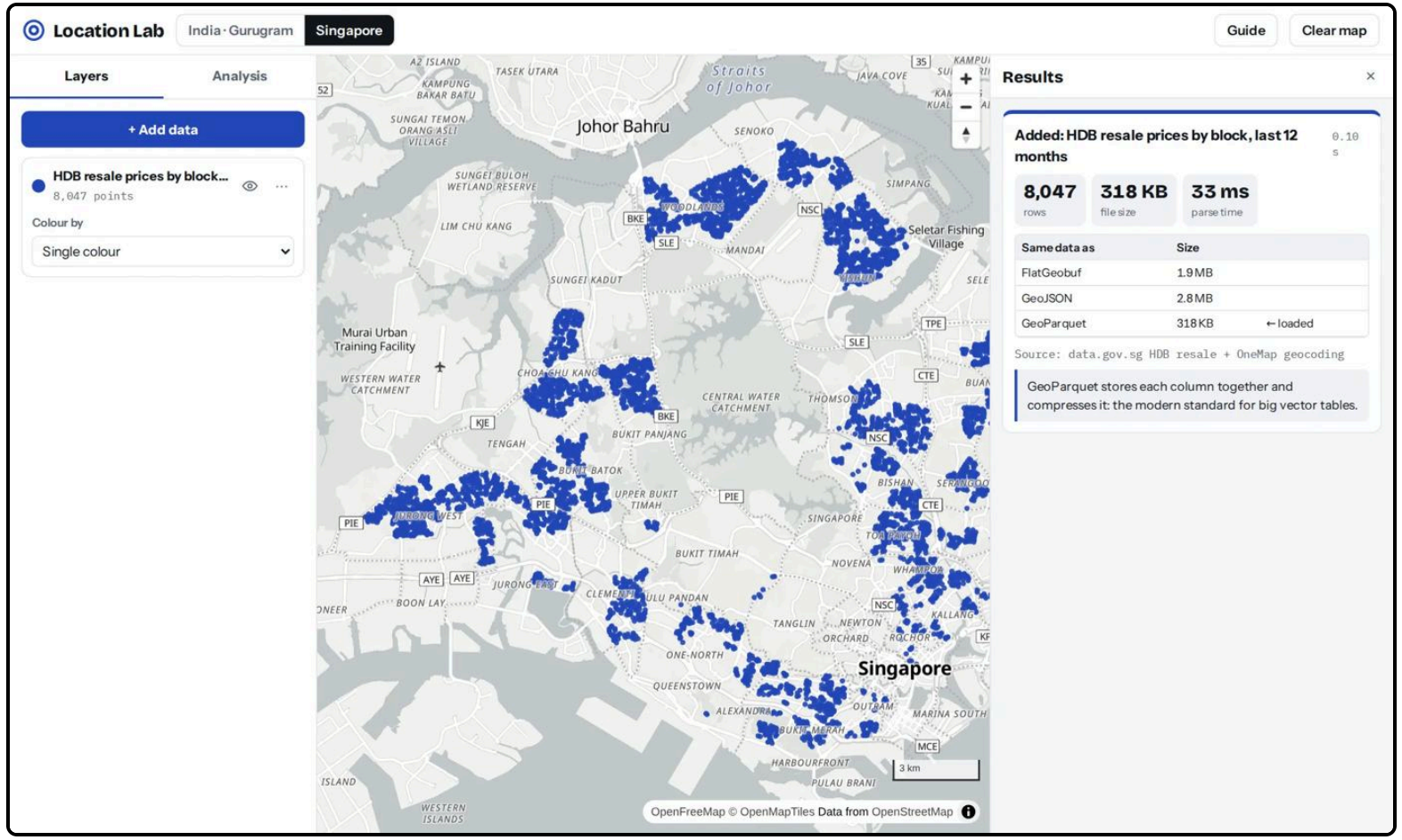
YOU SHOULD SEE

8,047 rows

318 KB file size

33 ms parse time

> GeoParquet stores each column together and compresses it: the modern standard for big vector tables.



LAB 6 · STEP 2 OF 3

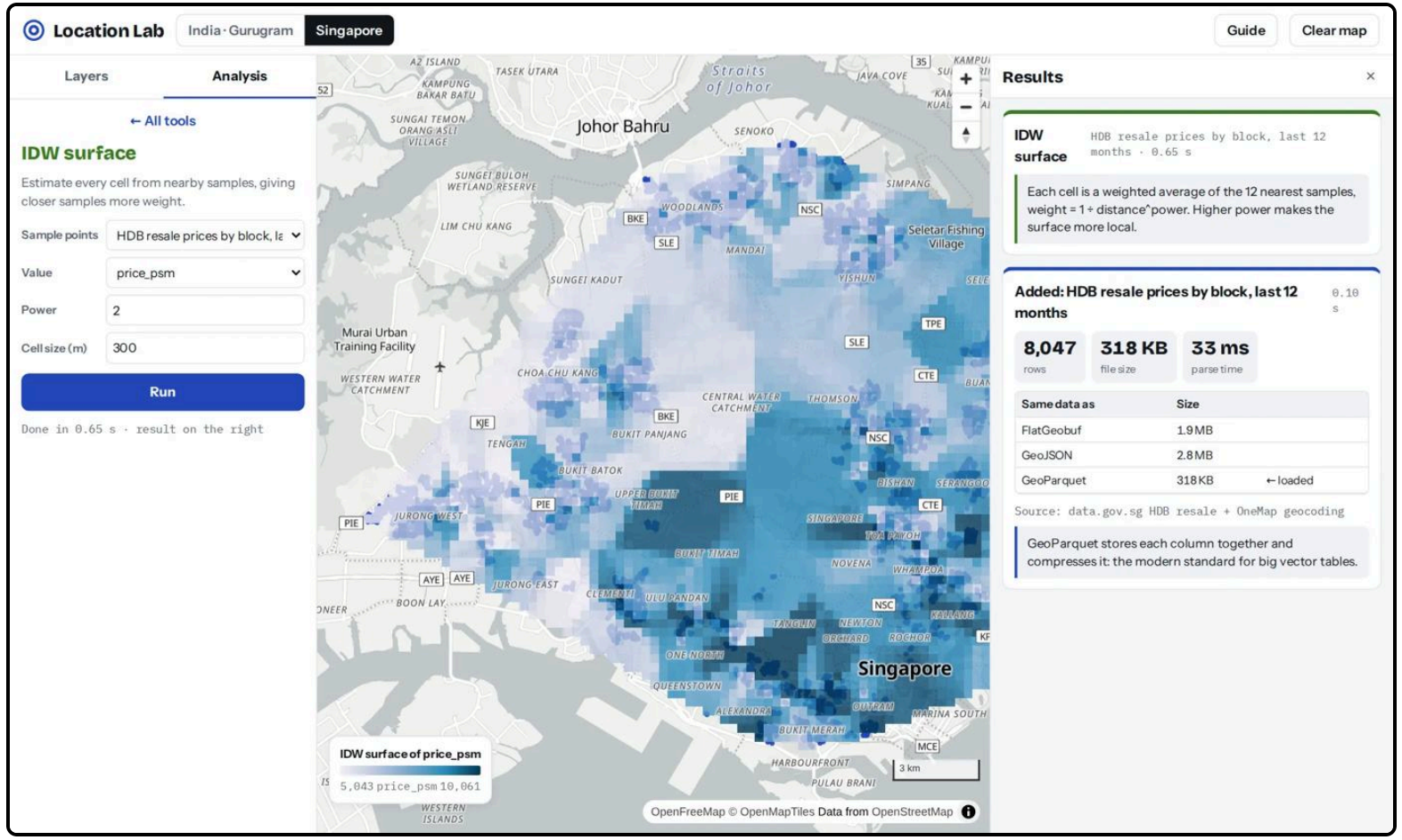
IDW price surface.

DO THIS

Analysis tab → drag **IDW surface** onto the map (or click it), check the settings, press **Run**.

Sample points	HDB resale prices by block, last 12 months
Value	price_psm
Power	2
Cell size (m)	300

> Each cell is a weighted average of the 12 nearest samples, weight = 1 ÷ distance^power. Higher power makes the surface more local.



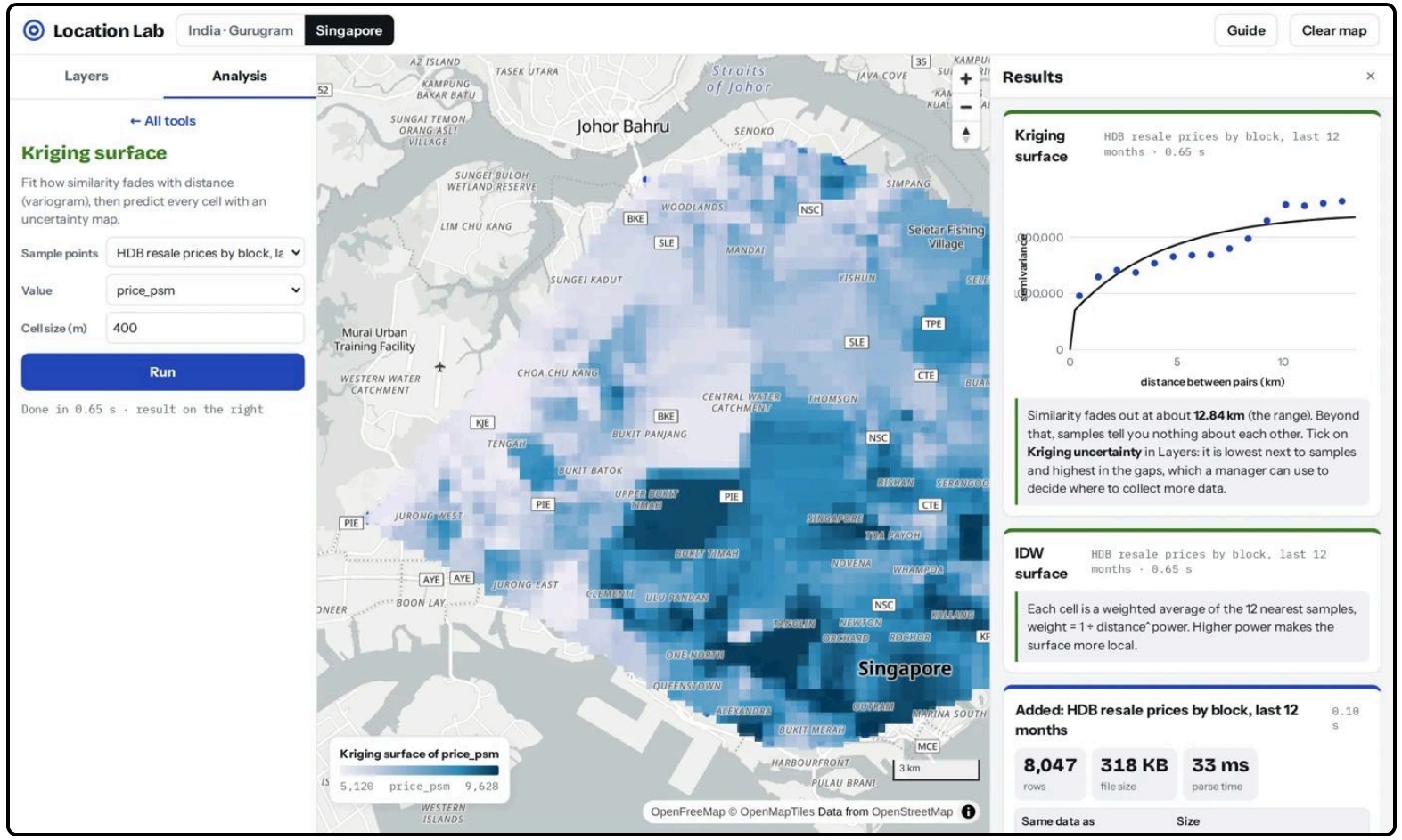
Kriging surface and its uncertainty.

DO THIS

Analysis tab → drag **Kriging surface** onto the map (or click it), check the settings, press **Run**.

Sample points	HDB resale prices by block, last 12 months
Value	price_psm
Cell size (m)	400

> Similarity fades out at about 12.84 km (the range). Beyond that, samples tell you nothing about each other. Tick on Kriging uncertainty in Layers: it is lowest next to samples and highest...



Three questions

1. The price variogram range is about 12.8 km. What does it suggest?
2. Where is the kriging standard error highest?
3. Why does IDW with a high power look patchy?

Answers

1. **The price variogram range is about 12.8 km. What does it suggest?**
   A city-wide gradient: prices relate over long distances, not just within estates.
2. **Where is the kriging standard error highest?**
   In gaps between blocks, such as nature reserves, the airport and industrial land.
3. **Why does IDW with a high power look patchy?**
   The nearest sample dominates, so each cell copies its closest block.

PART 7

Explaining location

The path we take on the board:

OLS → **Independence** → **Residual Moran** → **Lag vs error** → **AIC** → **GWR**

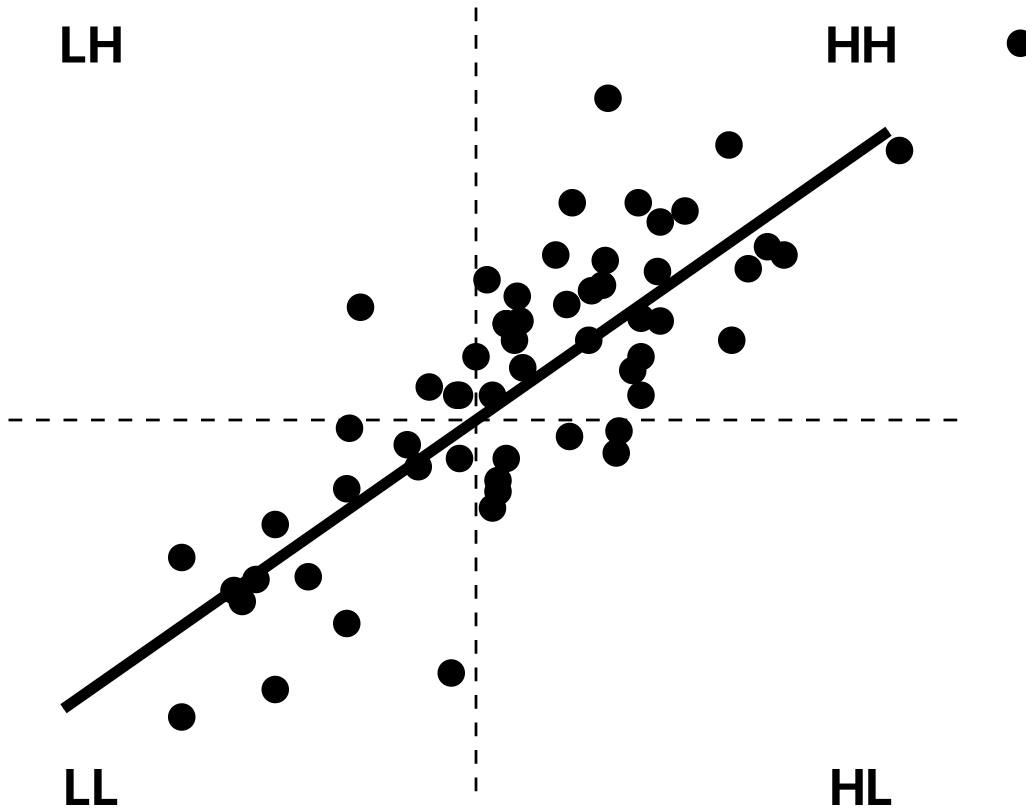
EXPLAIN · STEP 1

OLS: one line for the whole map

Ordinary least squares fits $y = \beta_0 + \beta_1 x_1 + \ldots + \varepsilon$ and reports one coefficient per variable for the whole map. Its inference assumes the errors ε are **independent**.

On a map that assumption is suspect: if an unmeasured factor (prestige, school zones, a new mall) is shared by neighbours, the errors of neighbours move together, standard errors are too small and coefficients may be biased.

Singapore price per m² on distance to MRT, hawker centre and CBD: **$R^2 = 0.69$**.



one slope for all locations

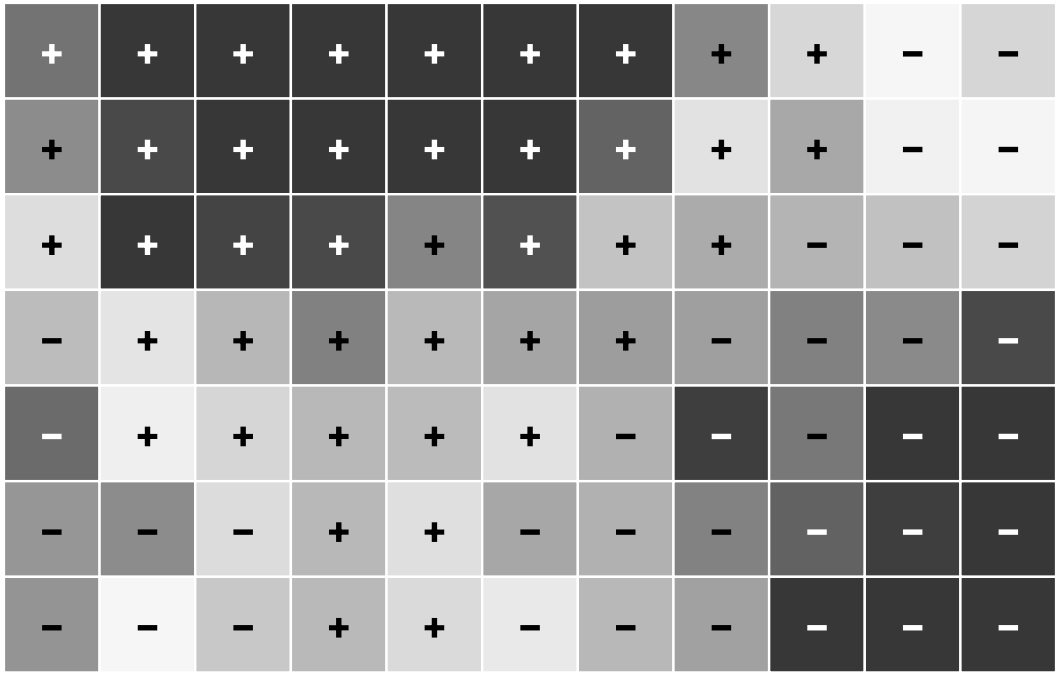
EXPLAIN · STEP 2

Test the errors: map the residuals

Map the residuals (actual – predicted). If OLS has captured everything spatial, the map looks like noise. If plus signs cluster with plus signs, something spatial is missing.

We already have the test: **Moran's I of the residuals**, using the same W.

Singapore: residual **I = 0.653**. OLS is clearly misspecified.



Clustered residuals: the model misses a spatial factor

Two fixes: spillover or shared omissions

Spatial lag model: $y = \rho Wy + X\beta + \varepsilon$. My value depends on my neighbours' values: genuine spillover (a café draws footfall to the next one; a renovated block lifts its neighbours).

Spatial error model: $y = X\beta + u$, $u = \lambda Wu + \varepsilon$. Neighbours share unmeasured factors, but do not cause each other.

Fit both and compare **AIC** (fit penalised for complexity; lower is better). Singapore: $\rho = 0.59$, $\lambda = 0.85$.

$$y = \rho Wy + X\beta + \varepsilon$$

$$y = X\beta + u,\ u = \lambda Wu + \varepsilon$$

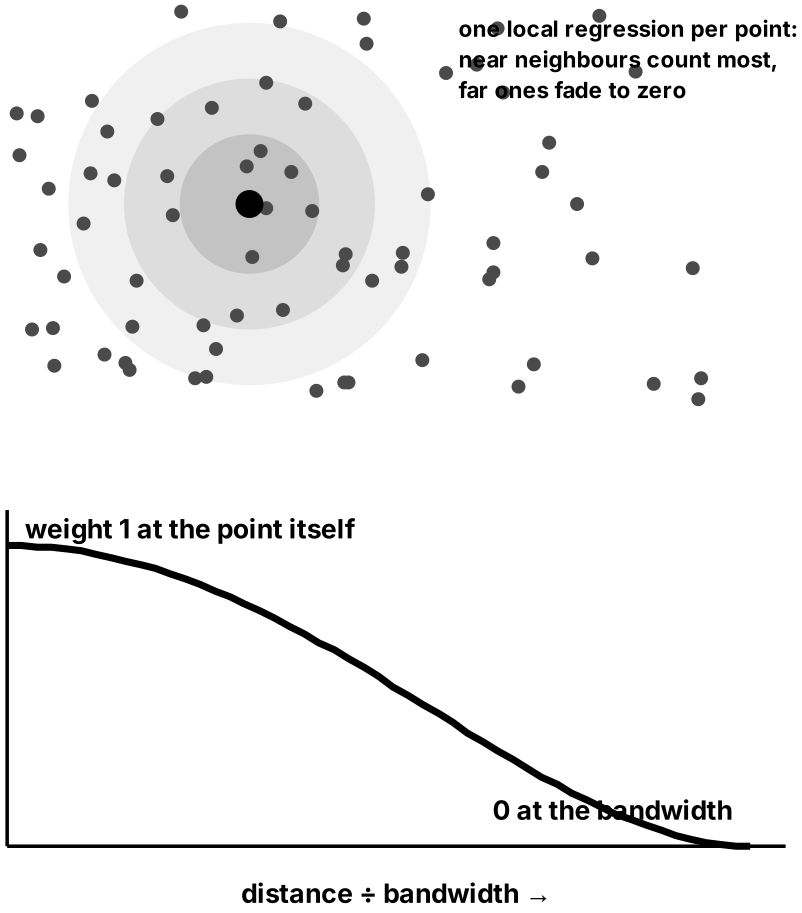
EXPLAIN · STEP 4

GWR: let the coefficients vary

OLS asks “what is the metro effect?” GWR asks “what is the metro effect **here**?” It fits a separate regression at every location, weighting nearby observations with a kernel that falls from 1 to 0 at the bandwidth.

Small bandwidth: very local, noisy. Large: smooth, close to OLS. The bandwidth is chosen automatically by minimising AICc.

Singapore: bandwidth 35 blocks, R^2 rises from 0.69 to **0.92**, and the MRT effect is strong in some towns and absent in others.



Where business already uses this

IN THE NEWS · MINISTRY OF SUSTAINABILITY AND THE ENVIRONMENT · 15 OCT 2024

How Singapore decides where hawker centres go

Population served, nearby food options and land use: the same variables a location model weighs when it explains or predicts demand.

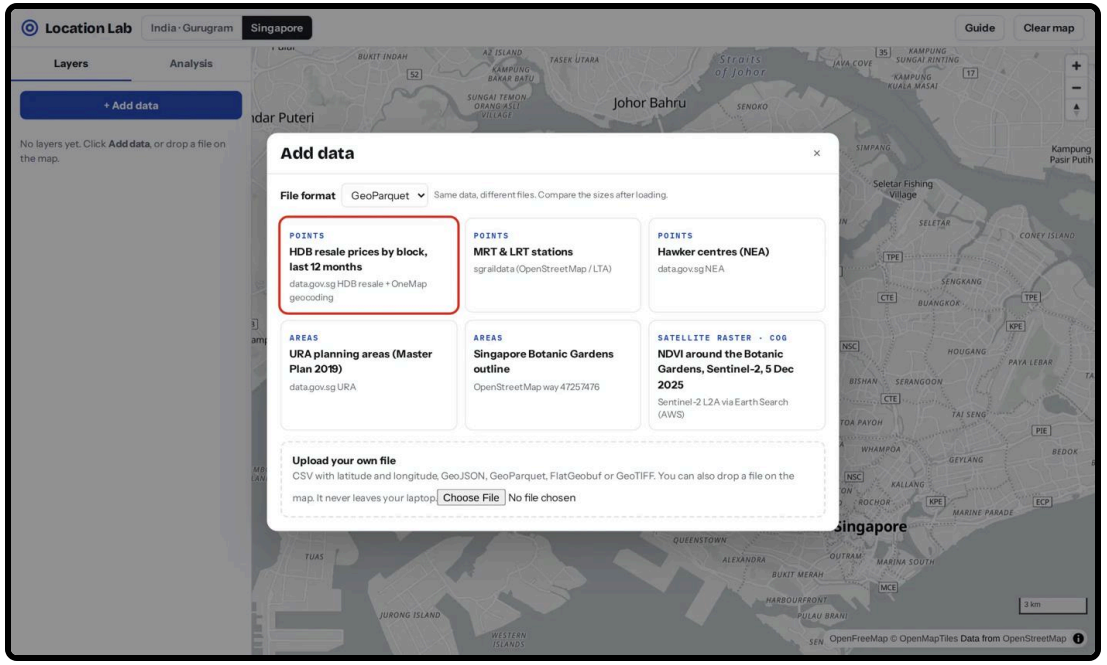
https://www.mse.gov.sg/latest-news/written-reply-to-pq-on-planning-for-hawker-centres/

7 · What explains resale prices?

Does price per m² rise near MRT, hawker centres and the CBD, and is that effect the same island-wide?

Open **quantng.github.io/location-lab**, choose **Singapore** in the top bar, then follow the steps.

1. Load HDB resale blocks.
2. Load MRT & LRT stations.
3. Distance from each block to the nearest station.
4. Load hawker centres.
5. Distance to the nearest hawker centre.
6. Distance to Raffles Place (CBD).
7. 8-nearest-neighbour weights.
8. Ordinary regression and residual Moran's I.
9. Spatial lag model.
10. Spatial error model.
11. GWR: where does MRT access matter most?



LAB 7 · STEP 1 OF 11

Load HDB resale blocks.

DO THIS

Layers tab → **+ Add data** → pick the format → click the dataset.

Dataset	HDB resale prices by block, last 12 months
File format	GeoParquet

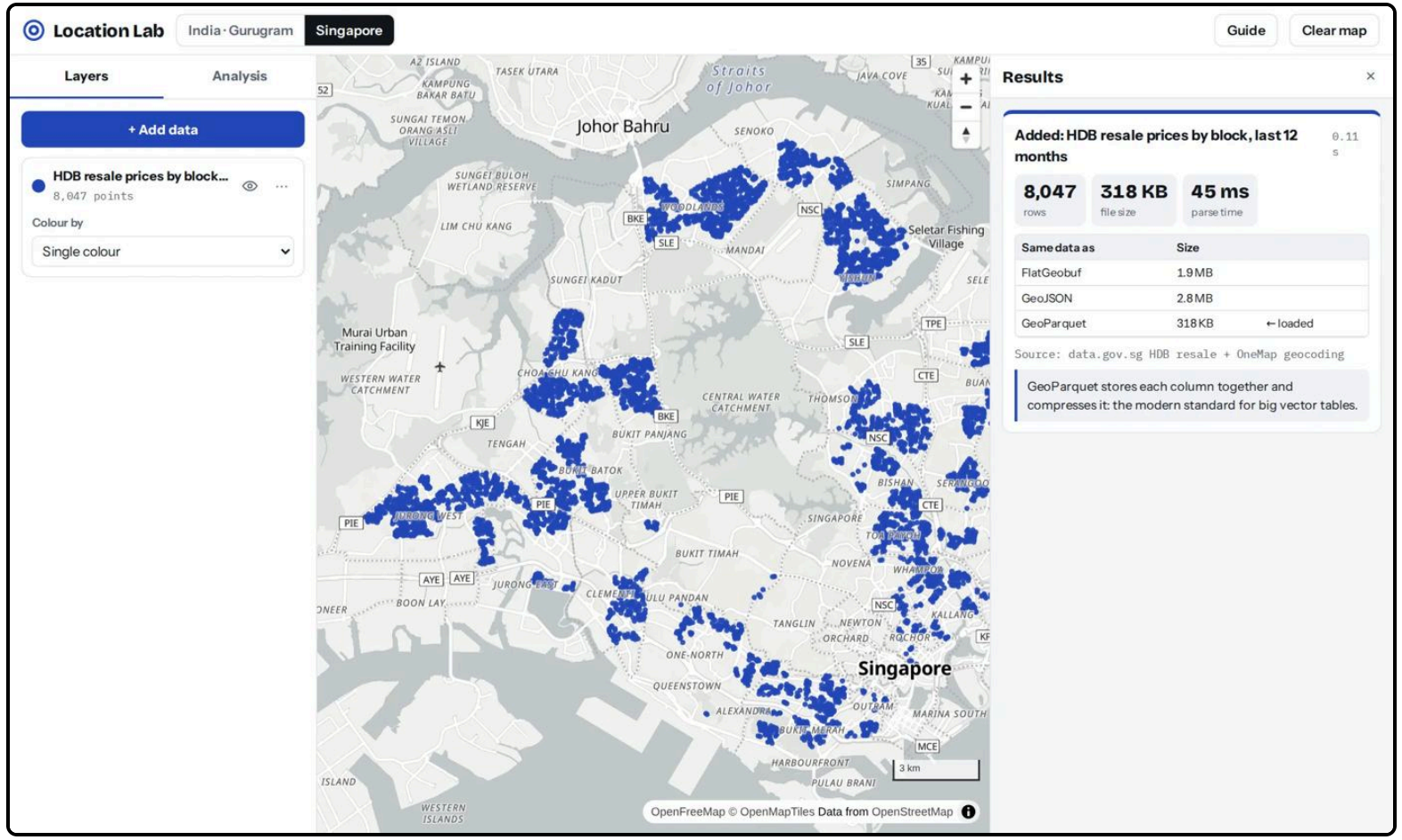
YOU SHOULD SEE

8,047 rows

318 KB file size

45 ms parse time

> GeoParquet stores each column together and compresses it: the modern standard for big vector tables.



Load MRT & LRT stations.

DO THIS

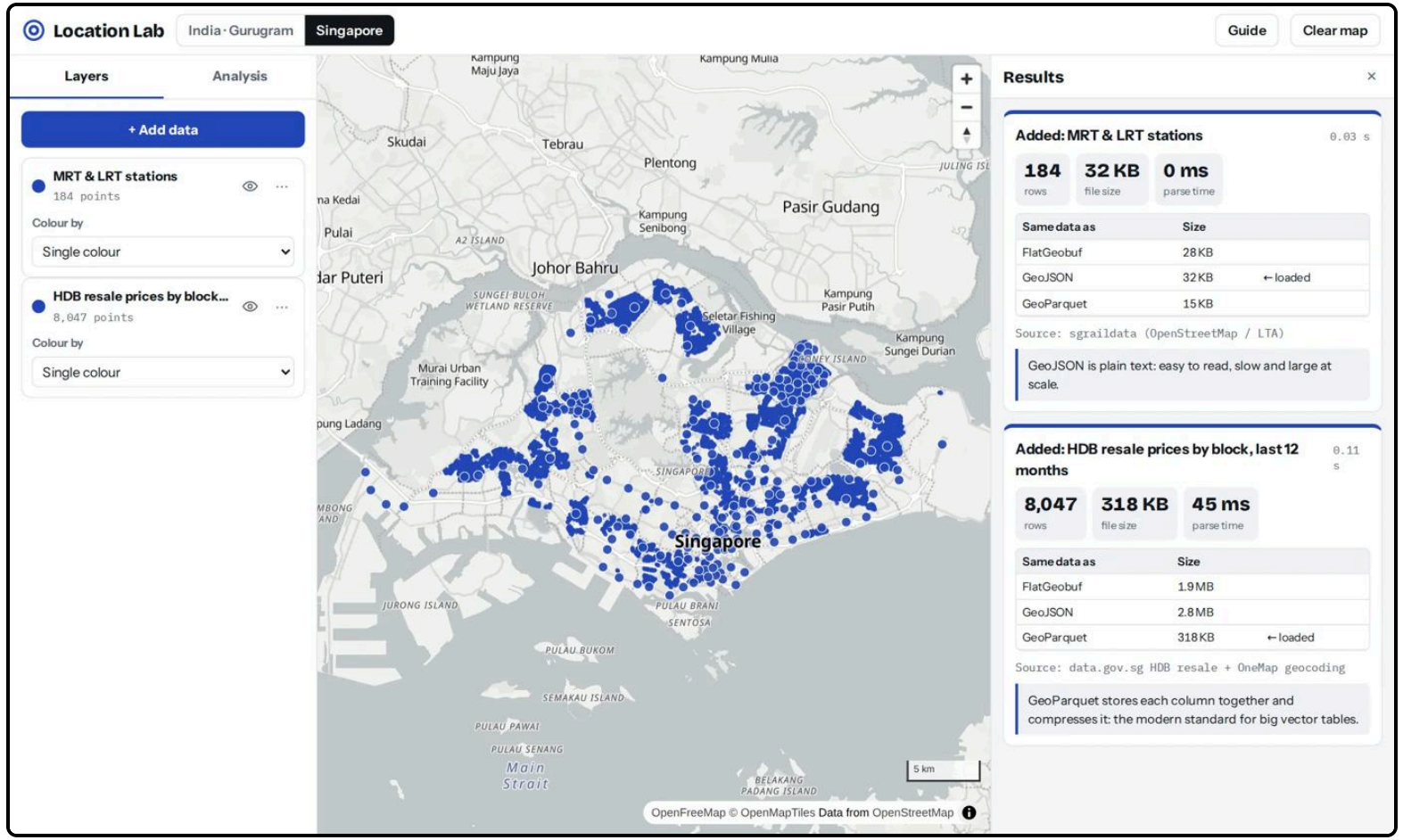
Layers tab → **+ Add data** → pick the format → click the dataset.

Dataset	MRT & LRT stations
File format	GeoJSON

YOU SHOULD SEE

184 rows | **32 KB** file size | **0 ms** parse time

> GeoJSON is plain text: easy to read, slow and large at scale.



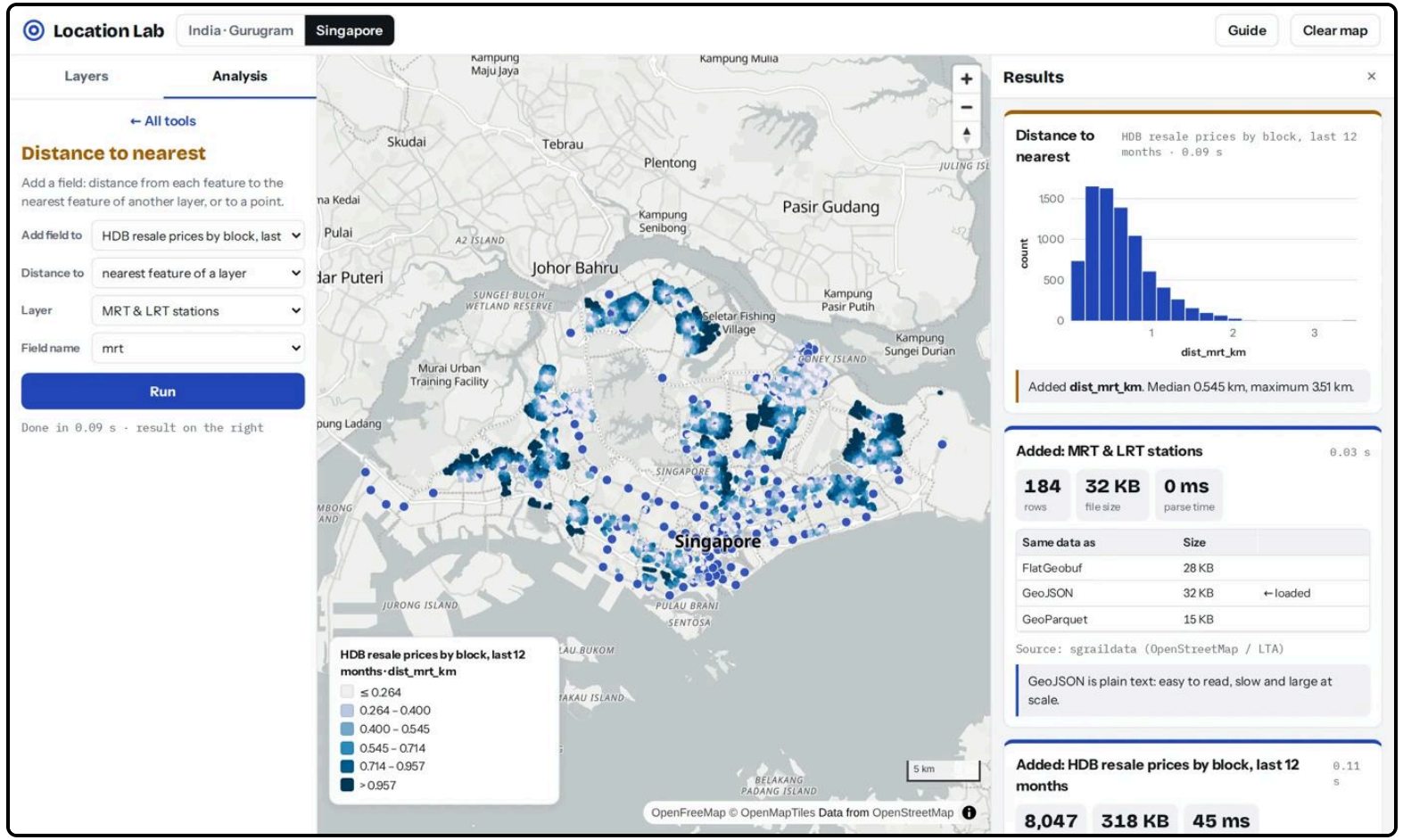
Distance from each block to the nearest station.

DO THIS

Analysis tab → drag **Distance to nearest** onto the map (or click it), check the settings, press **Run**.

Add field to	HDB resale prices by block, last 12 months
Distance to	nearest feature of a layer
Layer	MRT & LRT stations
Field name	mrt

> Added dist_mrt_km. Median 0.545 km, maximum 3.51 km.



Load hawker centres.

DO THIS

Layers tab → **+ Add data** → pick the format → click the dataset.

Dataset	Hawker centres (NEA)
File format	GeoJSON

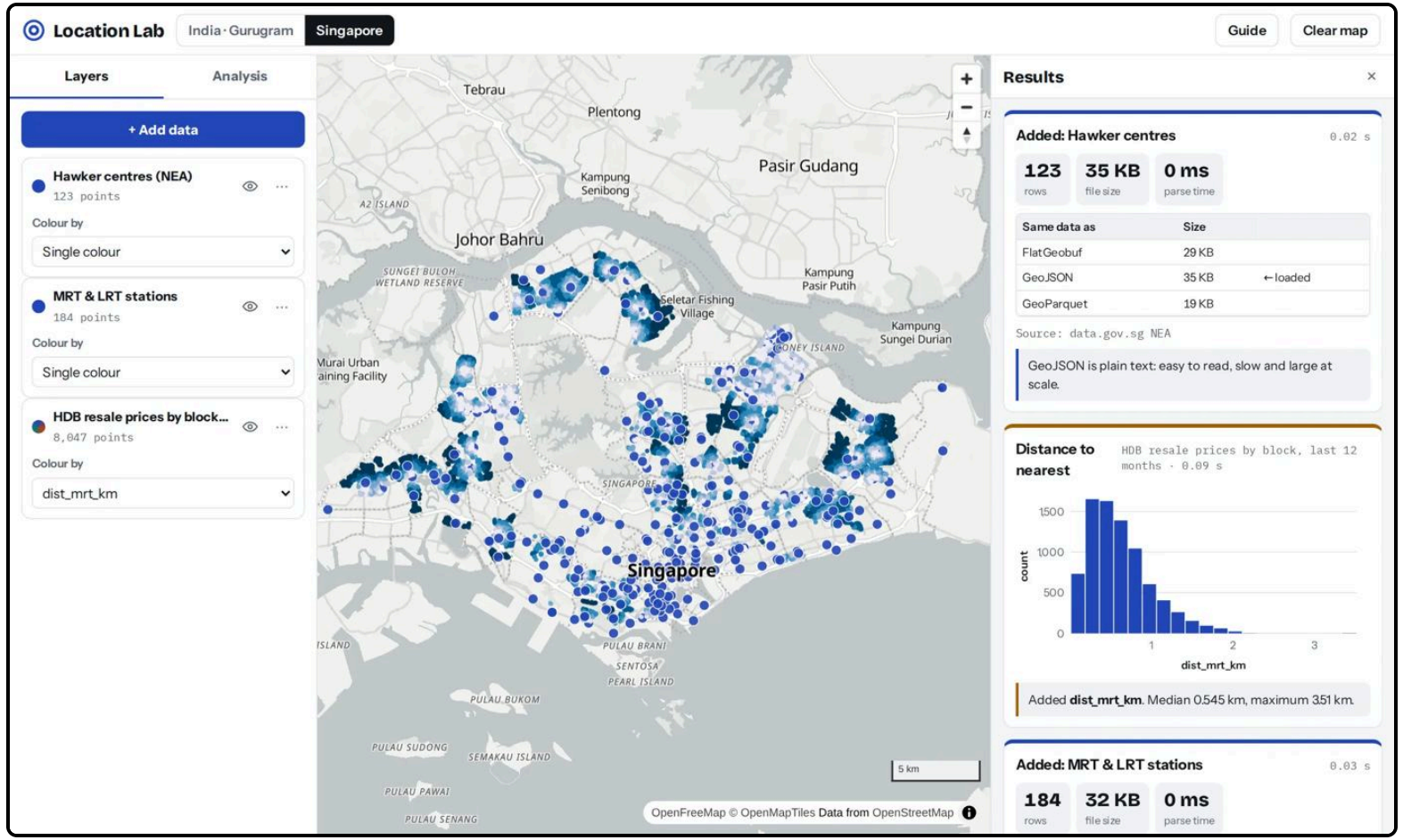
YOU SHOULD SEE

123 rows

35 KB file size

0 ms parse time

> GeoJSON is plain text: easy to read, slow and large at scale.



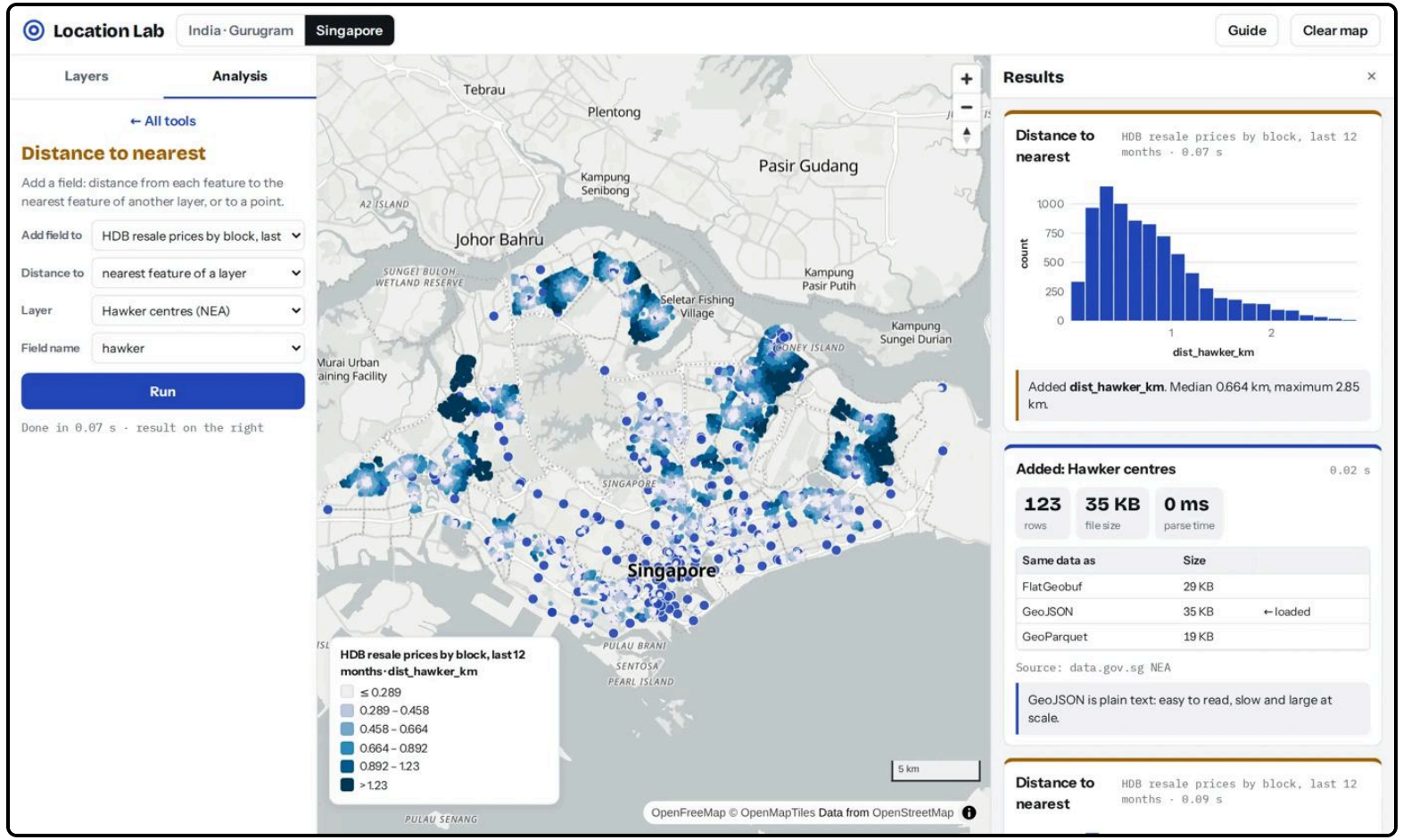
Distance to the nearest hawker centre.

DO THIS

Analysis tab → drag **Distance to nearest** onto the map (or click it), check the settings, press **Run**.

Add field to	HDB resale prices by block, last 12 months
Distance to	nearest feature of a layer
Layer	Hawker centres (NEA)
Field name	hawker

> Added dist_hawker_km. Median 0.664 km, maximum 2.85 km.



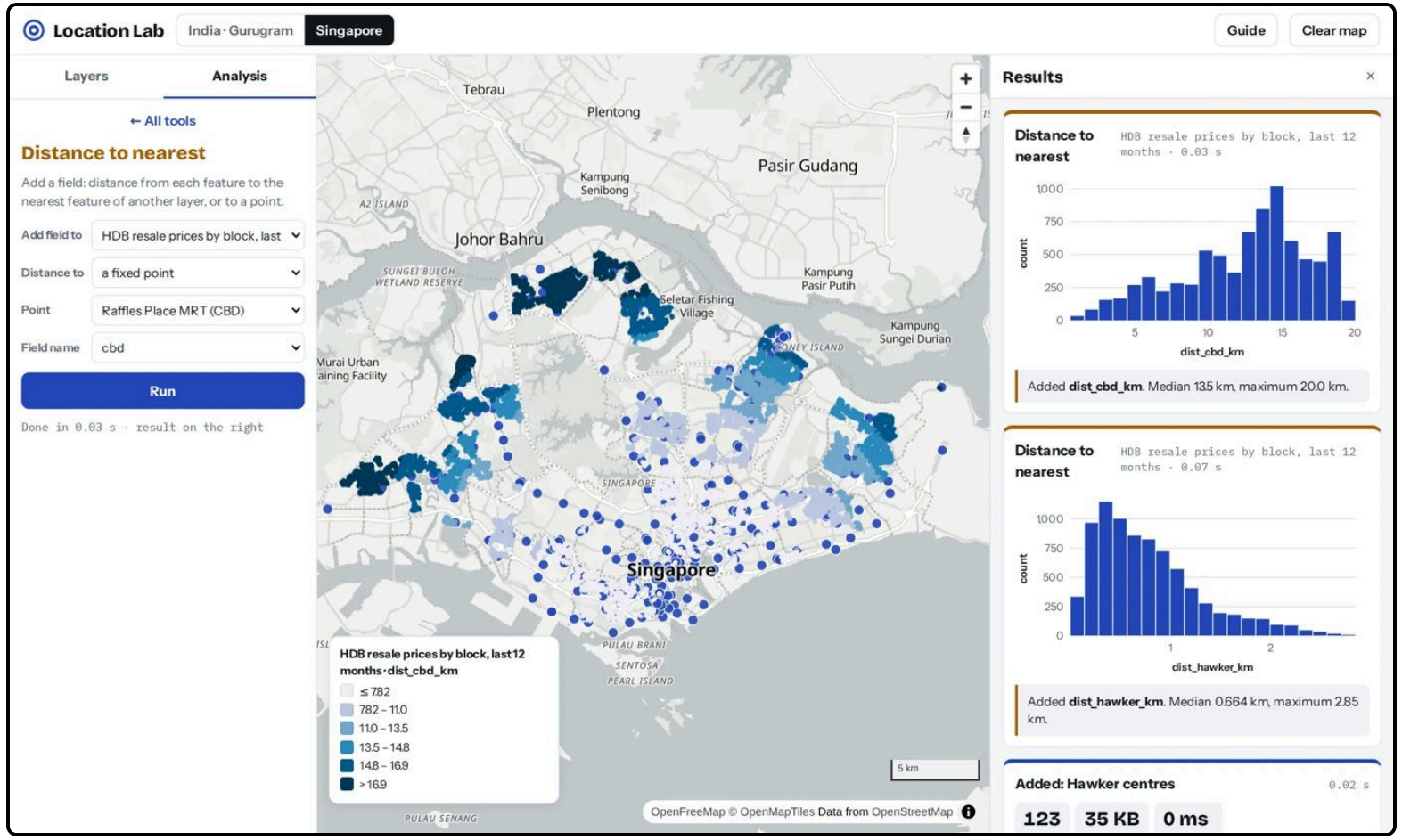
Distance to Raffles Place (CBD).

DO THIS

Analysis tab → drag **Distance to nearest** onto the map (or click it), check the settings, press **Run**.

Add field to	HDB resale prices by block, last 12 months
Distance to	a fixed point
Point	Raffles Place MRT (CBD)
Field name	cbd

> Added dist_cbd_km. Median 13.5 km, maximum 20.0 km.



8-nearest-neighbour weights.

DO THIS

Analysis tab → drag **Spatial weights** onto the map (or click it), check the settings, press **Run**.

Layer	HDB resale prices by block, last 12 months
Neighbours are	the k nearest
k	8

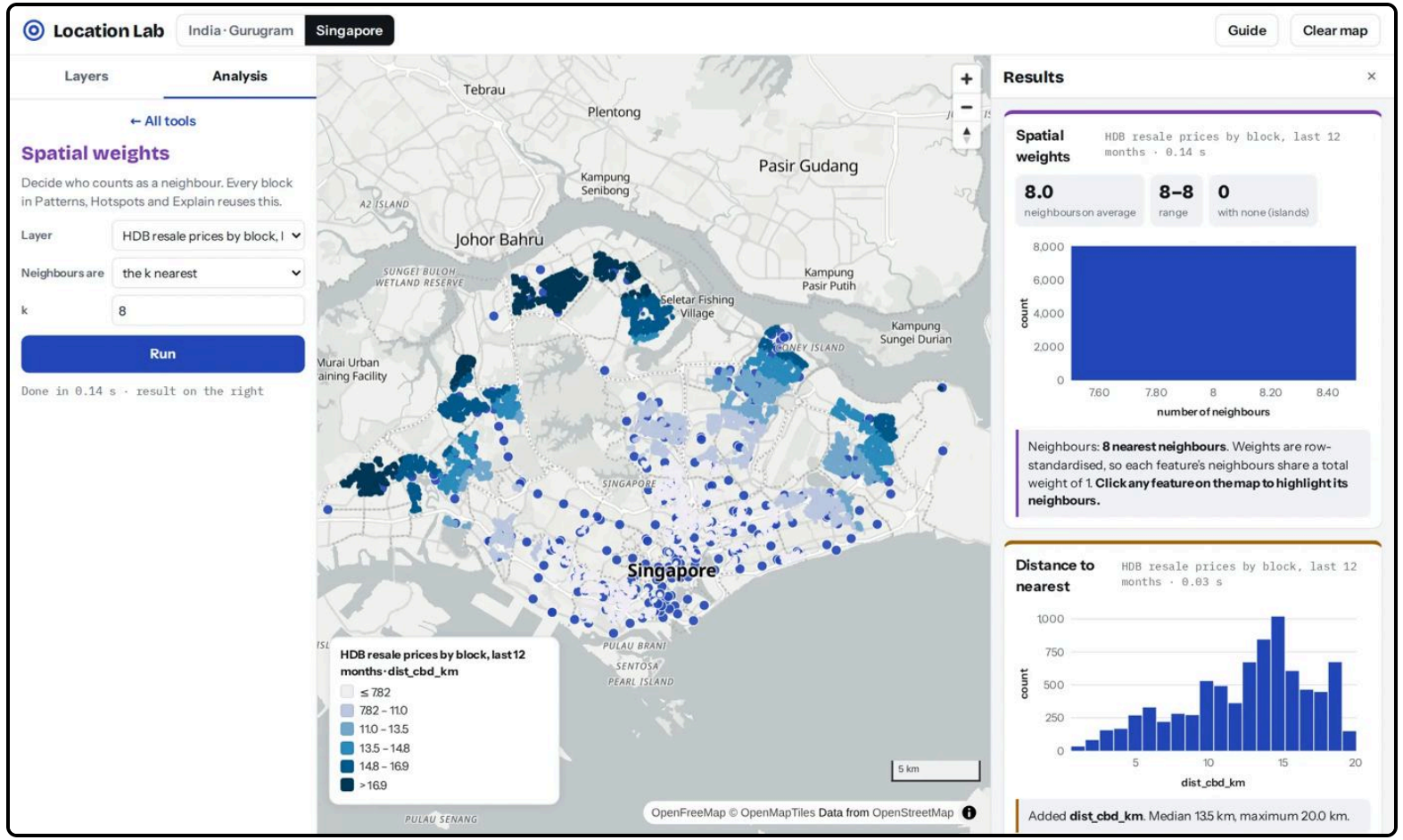
YOU SHOULD SEE

8.0
neighbours on average

8–8
range

0
with none (islands)

> Neighbours: 8 nearest neighbours. Weights are row-standardised, so each feature's neighbours share a total weight of 1. Click any feature on the map to highlight its neighbours.



Ordinary regression and residual Moran’s I.

DO THIS

Analysis tab → drag **Ordinary regression (OLS)** onto the map (or click it), check the settings, press **Run**.

Layer	HDB resale prices by block, last 12 months
Explain (y)	price_psm
Using (x)	storey_mid, lease_start, dist_mrt_km, dist_hawker_km, dist_cbd_km

YOU SHOULD SEE

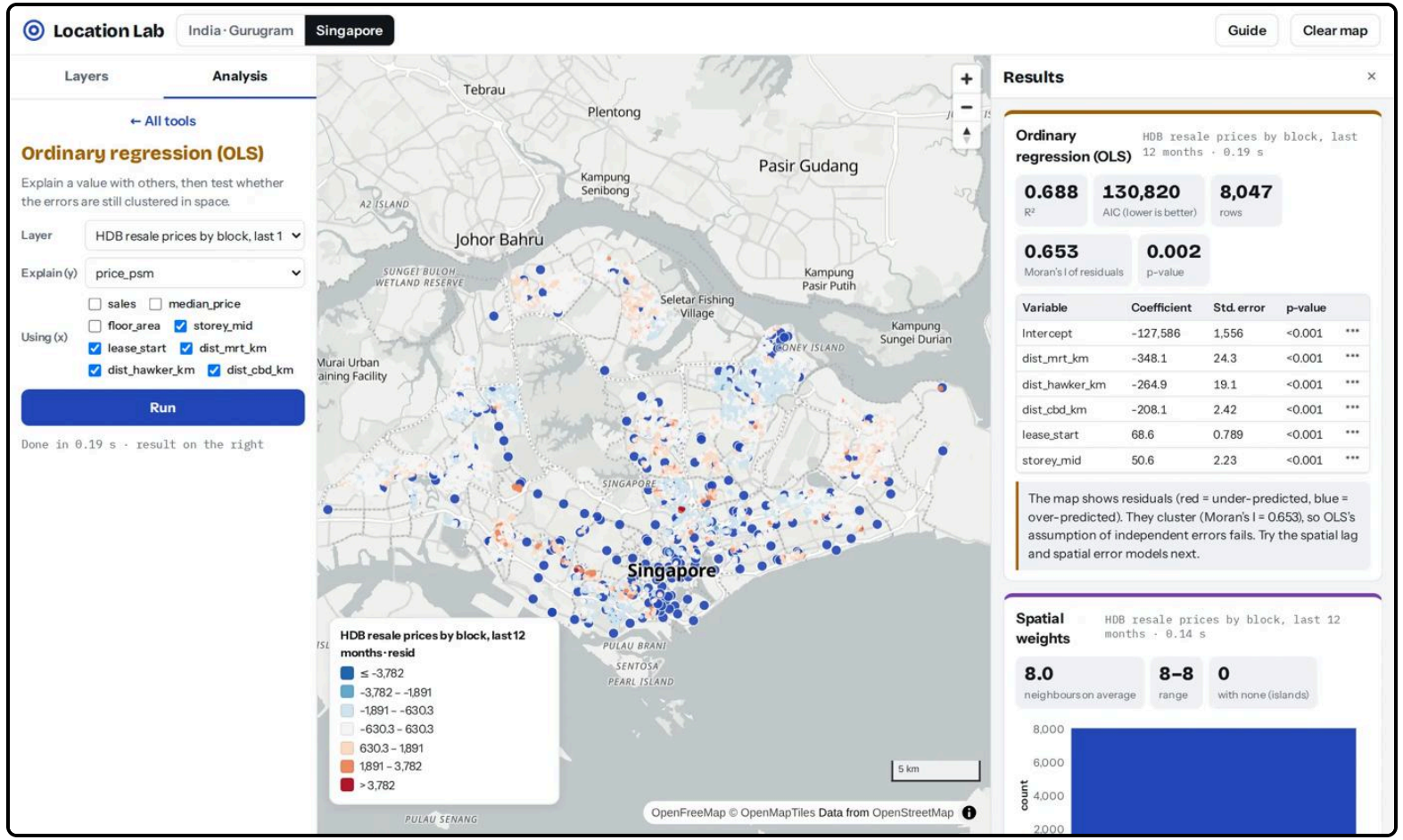
0.688
R^2

130,820
AIC (lower is better)

8,047
rows

0.653
Moran’s I of residuals

> The map shows residuals (red = under-predicted, blue = over-predicted). They cluster (Moran’s I = 0.653), so OLS’s assumption of independent errors fails. Try the spatial lag and spatial...



Spatial lag model.

DO THIS

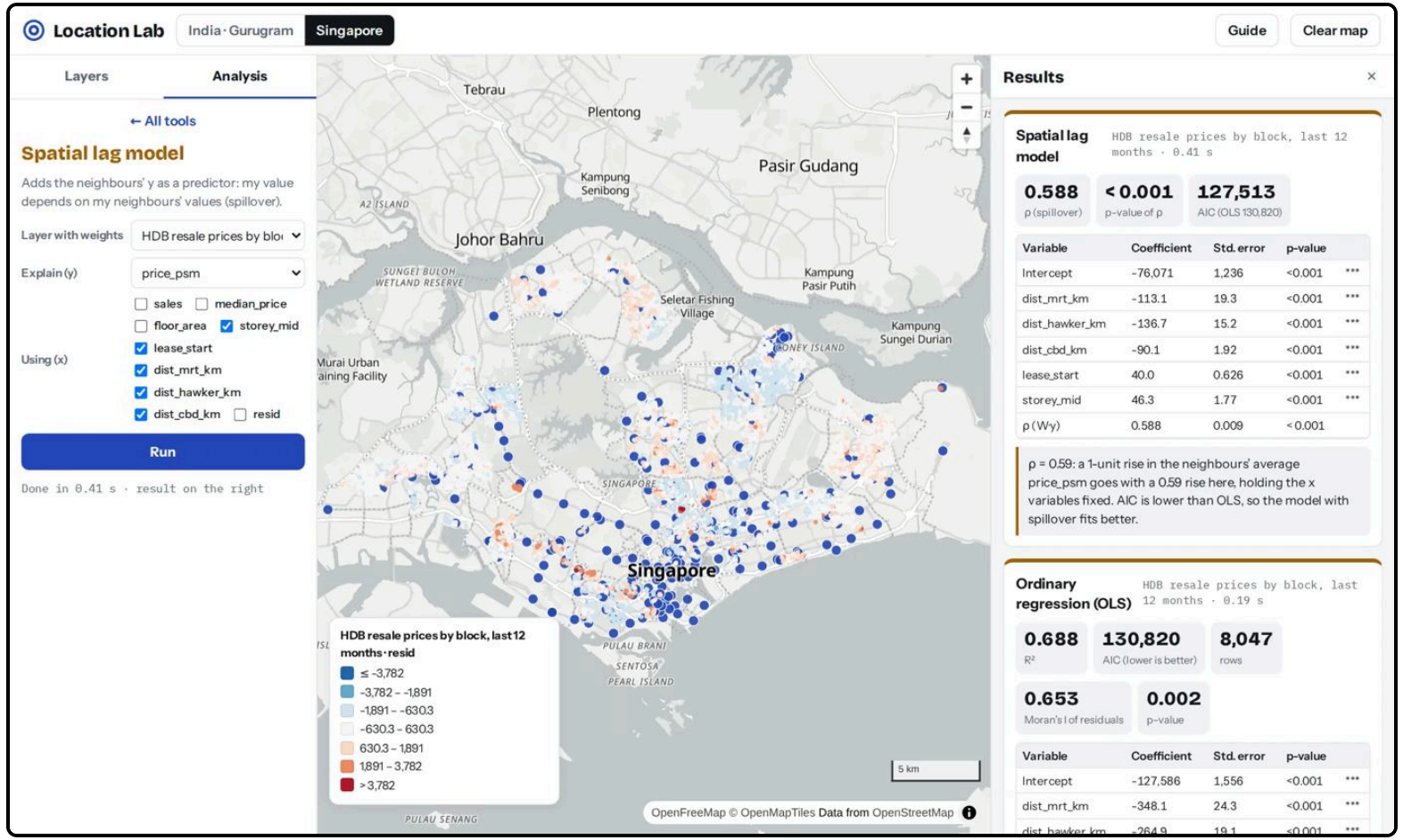
Analysis tab → drag **Spatial lag model** onto the map (or click it), check the settings, press **Run**.

Layer with weights	HDB resale prices by block, last 12 months
Explain (y)	price_psm
Using (x)	storey_mid, lease_start, dist_mrt_km, dist_hawker_km, dist_cbd_km

YOU SHOULD SEE

0.588	< 0.001	127,513
ρ (spillover)	p-value of ρ	AIC (OLS 130,820)

> ρ = 0.59: a 1-unit rise in the neighbours' average price_psm goes with a 0.59 rise here, holding the x variables fixed. AIC is lower than OLS, so the model with spillover fits better.



Spatial error model.

DO THIS

Analysis tab → drag **Spatial error model** onto the map (or click it), check the settings, press **Run**.

Layer with weights	HDB resale prices by block, last 12 months
Explain (y)	price_psm
Using (x)	storey_mid, lease_start, dist_mrt_km, dist_hawker_km, dist_cbd_km

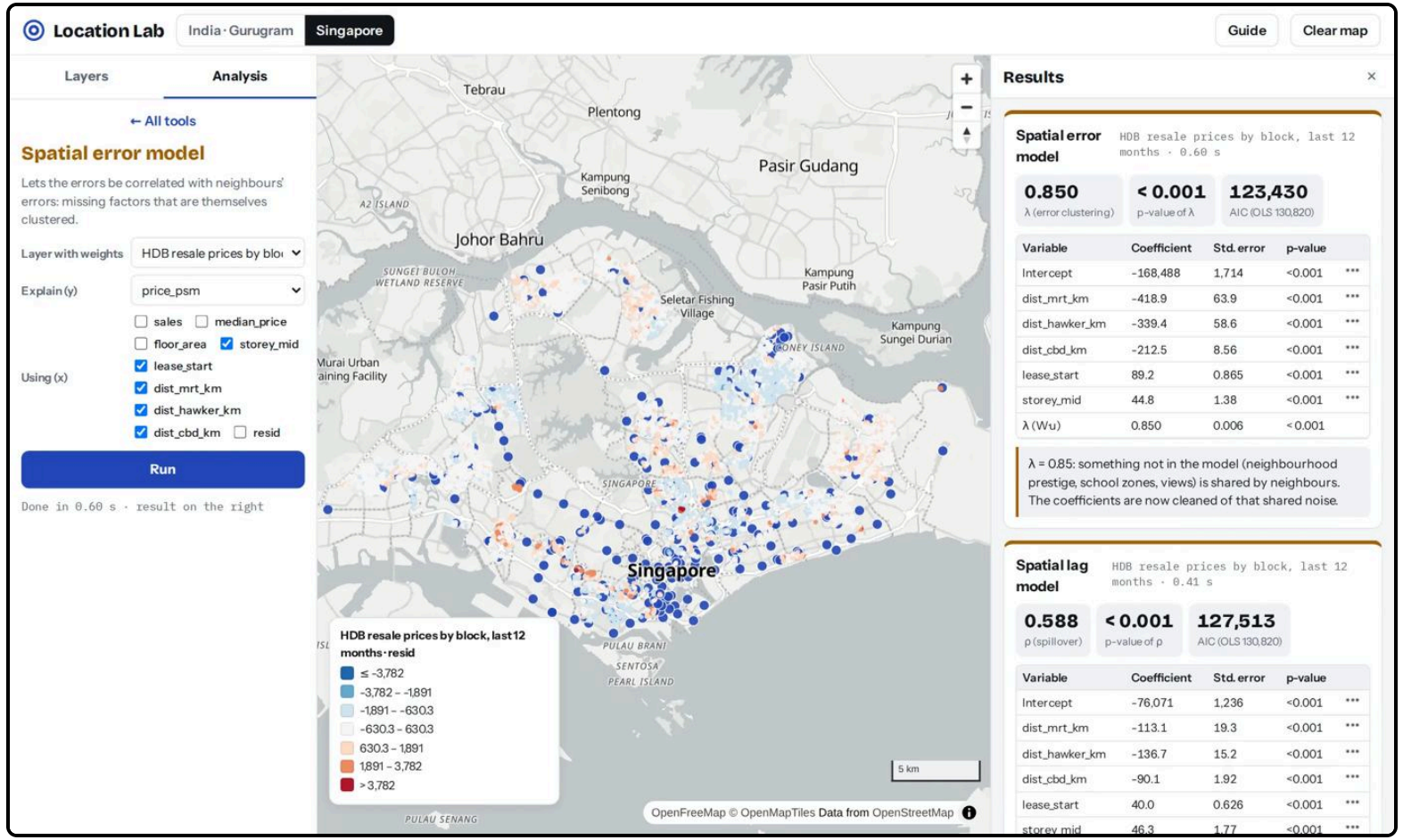
YOU SHOULD SEE

0.850
λ (error clustering)

< 0.001
p-value of λ

123,430
AIC (OLS 130,820)

> λ = 0.85: something not in the model (neighbourhood prestige, school zones, views) is shared by neighbours. The coefficients are now cleaned of that shared noise.



GWR: where does MRT access matter most?

DO THIS

Analysis tab → drag **Geographically weighted regression** onto the map (or click it), check the settings, press **Run**.

Layer	HDB resale prices by block, last 12 months
Explain (y)	price_psm
Using (x)	lease_start, dist_mrt_km, dist_cbd_km
Neighbours per fit (0 = auto)	0
Max rows (speed)	1500

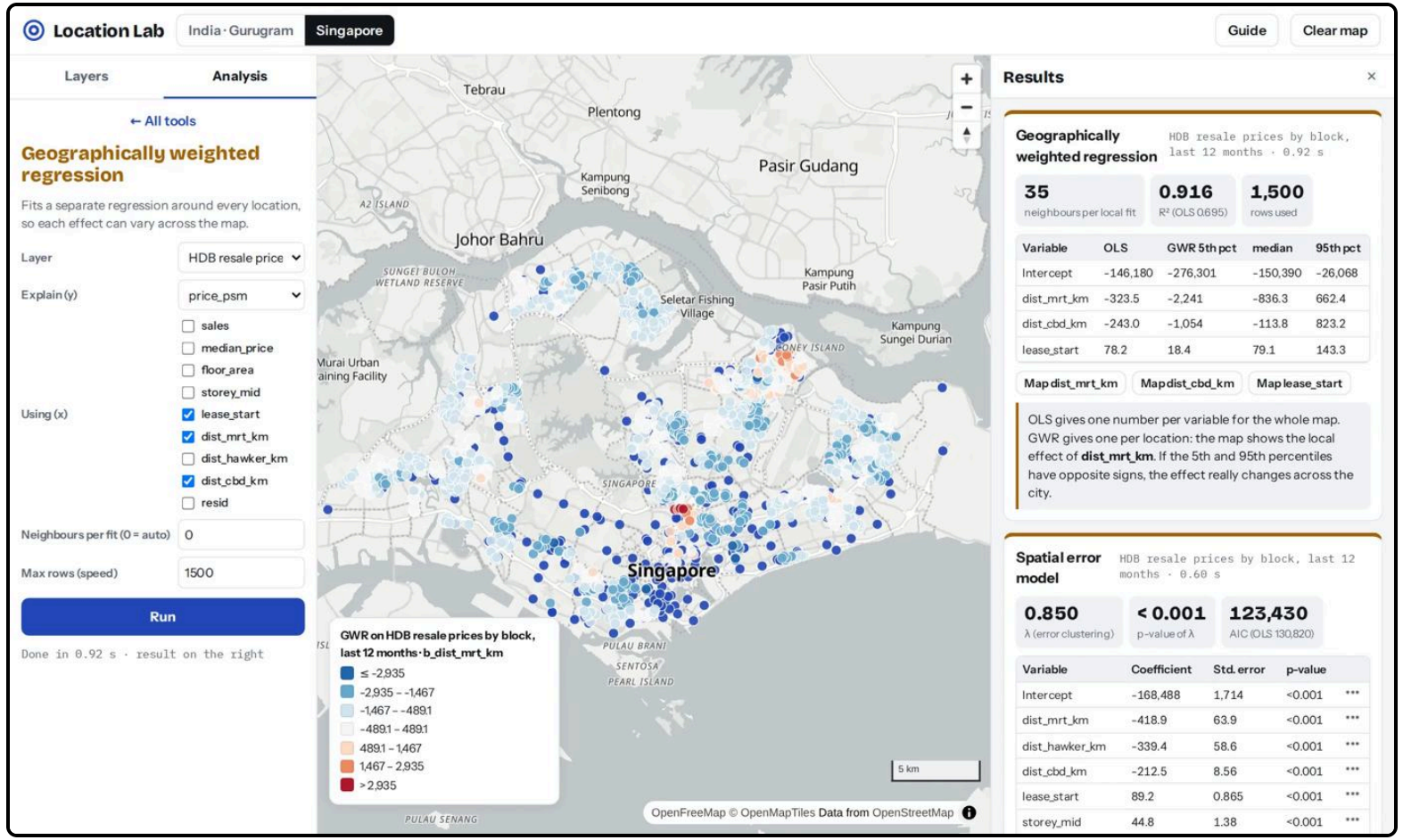
YOU SHOULD SEE

35 neighbours per local fit

0.916 R^2 (OLS 0.695)

1,500 rows used

> OLS gives one number per variable for the whole map. GWR gives one per location: the map shows the local effect of dist_mrt_km. If the 5th and 95th percentiles have opposite signs, the…



Three questions

1. OLS residuals have Moran's I = 0.653. What should you do?
2. $\rho = 0.59$ and $\lambda = 0.85$. Interpret each.
3. GWR R^2 is 0.92 with bandwidth 35. What does the MRT coefficient map show?

Answers

1. **OLS residuals have Moran's I = 0.653. What should you do?**
   Errors cluster, so OLS is misspecified: fit spatial lag and spatial error models and compare AIC.

2. **ρ = 0.59 and λ = 0.85. Interpret each.**
   ρ: neighbours' prices spill over into a block's own price. λ: unmeasured factors shared by neighbours (school zones, estate age) are strong.

3. **GWR R^2 is 0.92 with bandwidth 35. What does the MRT coefficient map show?**
   MRT distance lowers prices strongly in some towns and barely in others; one island-wide number hides this.

FINAL PRODUCT

A one-page decision brief

Question. A buyer with a fixed budget wants the best-value HDB block.

Evidence. Price per m^2 is strongly spatial (I = 0.711); OLS on MRT, hawker and CBD distance fits well (R^2 0.69) but leaves clustered residuals (I = 0.653). The spatial error model (λ = 0.85) beats OLS on AIC; GWR reaches R^2 0.92 with an MRT effect that changes by town.

Decision. Look at Low-High blocks inside High-High estates, in towns where GWR shows MRT distance matters least, and check lease remaining before shortlisting.

Your deliverable

One page: the question, three maps or charts exported from the tool (layer card → download, or a screenshot), the numbers that matter, and a decision with its main risk. Use the same structure as the example on the left.

CLOSING

Glossary

Lag (time)	The series shifted by one period	**LISA**	Local Moran: hotspots, coldspots, outliers
Rook / queen	Neighbours share an edge / an edge or corner	**Gi***	z-score of the local total vs chance
Weights W	Who is a neighbour, row-standardised to sum to 1	**DBSCAN**	Density clusters from ε and minPts; noise flagged
Spatial lag Wy	Neighbours' weighted average	**IDW**	Weighted average, weight 1 ÷ distance^p
CSR · R	Randomness benchmark · observed ÷ expected NN distance	**Variogram**	Nugget, sill, range of fading similarity
Ripley's L	Clustering at many distances, with envelope	**Kriging**	Variogram-weighted prediction with standard error
Moran's I	Slope of z against its spatial lag	**SLM ρ · SEM λ**	Spillover · shared unmeasured factors
Permutation test	Compare with 999 shuffled maps	**GWR**	Local regressions; bandwidth by AICc